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MIL-STD-3006D
17 April 2026

SUPERSEDING
MIL-STD-3006C
1 June 2008

# DEPARTMENT OF WAR STANDARD PRACTICE

## FOOD PROTECTION REQUIREMENTS FOR COMMERCIAL FOOD ESTABLISHMENTS

AMSC N/A

FSC 89GP

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# FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of War (DoW).

2. The DoW is committed to the acquisition of food products produced, handled and stored in clean, sanitary food establishments in order to prevent the transmission of contaminants and foodborne disease to members of the Armed Forces. The requirements contained herein are based on available national food and drug regulatory requirements, industry and trade standards, and scientific guidelines published by various professional societies. These requirement documents are selectively applied herein to provide a scientific basis upon which to evaluate the safety of commercially produced food purchased by the United States Armed Forces.

3. In keeping with DoW policy to align food protection requirements with other federal agencies, the requirements for food establishments contained in this standard are based upon the requirements set forth in the Food Safety Modernization Act (FSMA).

4. This standard establishes the enforcement of Current Good Manufacturing Practices (CGMPs), as accepted industry food safety practices, as provided in the United States Code of Federal Regulations (CFR), Title 21, Parts 112, 117, and 121, for food establishments requiring listing in the Worldwide Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement (hereafter referred to as the Worldwide Directory). This standard identifies parts of the FSMA that are enforced by the DoW.

5. This standard is applicable to all commercial food establishments that require food protection audits and supply subsistence to DoW, whether in the Continental United States (CONUS) or outside the Continental United States (OCONUS). Food and Water Risk Assessments (FWRA) are conducted on eateries and certain OCONUS caterers under specific guidelines and requirements of Military Standard 3041A (MIL-STD-3041A). The sanitation inspection of military food facilities is conducted utilizing the Technical Bulletin Medical (TB MED) 530, Tri-Service Food Code.

6. This standard is not used to determine an establishment's capabilities to comply with product specifications or other technical purchase requirements. In cases where CGMPs are provided for specific subsistence items (e.g., Low-acid canned foods), the specific requirements for that item will be applied in addition to those found in the primary CFRs (Parts 112, 117, and 121 as applicable). Additional CGMP documents provided by industry and recognized by the U.S. Food and Drug Administration (FDA), the United States Department of Agriculture (USDA) or the United States Department of Commerce, may be used in conjunction with 21 CFR Part 117, with Directory listing authority approval.

7. Comments, suggestions, or questions on this document shall be addressed to the Defense Health Agency Veterinary Service (DHA VS), 7700 Arlington Boulevard, Falls Church, VA 22042.



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## 1. SCOPE

1.1 Scope. This standard establishes food protection requirements for commercial establishments that produce, process, or store various types of food products before or after final acceptance by an element of the DoW.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF WAR HANDBOOKS

MIL-HDBK-3006D - Procedures for Auditing Food Establishments.

(Copies of this document are available online at https://assist.dla.mil/online/start/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 21, Part 112 - Standards for the Growing, Harvesting, Packing, And Holding of Produce for Human Consumption

Title 21, Part 117 - Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food

Title 21, Part 121 - Mitigation Strategies to Protect Food Against Intentional Adulteration



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(Copies of these are available online at: https://www.govinfo.gov/.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act, 1938, as amended by Public Law 107-188, Public Health Security and Bioterrorism Preparedness and Response Act of 2002.

(Copies of this document are available online at: http://www.fda.gov or https://www.govinfo.gov/app/details/PLAW-107publ188.)

U.S. DEPARTMENT OF AGRICULTURE

The National Advisory Committee on Microbiological Criteria for Foods (NACMCF):

"Requisite Scientific parameters for Establishing the Equivalence of Alternative Methods of Pasteurization", Aug 2004.

(Copies of this document are available on-line at: https://downloads.regulations.gov/FDA-2011-N-0920-0001/attachment_177.pdf.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

WORLD HEALTH ORGANIZATION

CODEX ALIMENTARIUS, General Principles of Food Hygiene, current version.

(Copies of this document are available online at: https://www.fao.org/fao-who-codexalimentarius/codex-texts/codes-of-practice/en/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. DEFINITIONS

3.1 General. The definitions found in 21 CFR, Parts 112, 117, and 121 are applicable to this standard.



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3.2 Acceptable audit rating. An audit rating generated when no critical findings are identified at the time of the audit and all CARs are adequately addressed by the establishment within established suspense date(s).

3.3 Adequate. That which is needed to accomplish the intended purpose, in keeping with good public health practices.

3.4 Adulterated. Within the meaning of the Federal Food, Drug and Cosmetic Act, Section 402, a food has been manufactured under such conditions that it is unfit for food, or, it has been prepared, packed, or held under insanitary conditions whereby it may have become contaminated with filth, or whereby it may have been rendered injurious to health and in violation of Section 361 of the Public Health Service Act (42 U.S.C. 264).

3.5 Agricultural water. Water used in farms handling produce that must meet the requirements of 21 CFR 112 where water is intended to, or is likely to, contact produce or food contact surfaces, including water used in growing activities (including irrigation water applied using direct water application methods, water used for preparing crop sprays, and water used for growing sprouts) and in harvesting, packing, and holding activities (including water used for washing or cooling harvested produce and water used for preventing dehydration).

3.6 Bioterrorism. Bioterrorism is the use of biological agents, such as pathogenic organisms or agricultural pests, for terrorist purposes against a civilian or military population.

3.7 Caterers. Establishments that produce meals for an event or to order for immediate consumption.

3.8 Correction. An action that identifies and corrects a nonconformance without implementing further actions that reduce the likelihood the problem will recur.

3.9 Corrective action. An action that eliminates the root cause of a detected problem or nonconformance.

3.10 Corrective action request (CAR). A written request from the auditor to management addressing Critical and Major findings that require a response from management as to the root cause of the finding, as well as the action(s) taken to correct and prevent recurrence of the finding(s).

3.11 Critical finding. A condition, practice, procedure, or loss of food protection control indicating an imminent health hazard (IHH).

3.12 Directed audit. An audit conducted at the direction of the DoW for establishments listed in the Worldwide Directory when public health concerns indicate a need for further food protection verification. The focus of this audit is normally limited to the areas of concern.

3.13 Directory listing authority. Refers to the headquarters or staff office within the U.S. Army Veterinary Service that is responsible for maintaining the Worldwide Directory.



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3.14 Food defense. The effort to protect food from acts of intentional adulteration.

3.15 Food defense finding. Any condition, practice, step, or procedure noted relating to the risk of intentional food tampering or increased food vulnerability. Food defense findings can occur at any stage during receipt, storage, processing, packaging, packing, warehousing, or distribution.

3.16 Food defense plan. A written document (or set of documents) that details the food defense program: vulnerability assessments; actionable process steps; mitigation strategies; monitoring procedures; corrective actions; and verification process.

3.17 Food defense program. A program developed by an establishment to assess and mitigate the vulnerabilities within the food system or infrastructure to an attack from deliberate or intentional acts of food destruction, contamination, or tampering.

3.18 Food protection. Overarching term that includes both food safety and food defense. Food safety is implementing protection measures necessary to identify, mitigate, or eliminate the risk from unintentional contamination. A food is deemed safe when it has been prepared, packed, stored and distributed under sanitary conditions whereby it has not been rendered injurious to health. Food defense is invoking protection measures necessary to identify and mitigate the threat of intentional contamination of food.

3.19 Food protection audit. The systematic, independent and documented examination (through observation, investigation, discussions, records review, and as appropriate, sampling and laboratory analysis) to assess compliance to applicable food protection requirements.

3.20 Food safety plan. A collection of written documents, prepared, or its preparation is overseen, by a qualified individual, that details the approach for identifying and controlling hazards within a food safety system.

3.21 Food safety program (FSP). Food establishment practices (similar to HACCP) that are implemented and practiced by establishment personnel and are designed to ensure the safe production of food.

3.22 Imminent health hazard (IHH). A significant threat or danger to health considered to exist when there is evidence sufficient to show that a product, practice, circumstance, or event creates a situation that requires immediate correction or cessation of operation to prevent illness or injury based on: (1) the number of potential illnesses and injuries, and (2) the nature, severity, and duration of the anticipated illness or injury.

3.23 Initial Audit. A comprehensive evaluation of a commercial food establishment to determine the food protection status for the first time before listing in the Worldwide Directory. This audit is used to approve or disapprove the establishment as a source for Armed Forces procurement. Initial food protection audits include a full assessment of the facility’s food safety system and associated controls.



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3.24 Low-acid canned food (LACF). Any food (other than alcoholic beverages) with a finished equilibrium pH greater than 4.6 and a water activity greater than 0.85, excluding tomatoes and tomato products having a finished equilibrium pH less than 4.7.

3.25 Major finding. Condition, practice, or procedure that adversely affects food protection or could lead to an IHH if not mitigated or corrected.

3.26 Minor finding. Condition, practice, or procedure that deviates from food protection requirements, and does not meet the criteria of a Critical or Major finding. Minor findings do not require CARs.

3.27 Potable Water. Water that is safe and satisfactory for drinking and cooking.

3.28 Preventive controls. Preventive controls means those risk based, reasonably appropriate procedures, practices, and processes that a person knowledgeable about the safe manufacturing, processing, packing, or holding of food would employ to significantly minimize or prevent the hazards identified under the hazard analysis that are consistent with the current scientific understanding of safe food manufacturing, processing, packing, or holding at the time of the analysis.

3.29 Root cause analysis. The research and investigative process to identify the fundamental reason for the occurrence of a finding.

3.30 Routine audit. An evaluation to determine the current food protection status of a commercial food establishment listed in the Worldwide Directory.

3.31 Special audit. An evaluation to determine whether the establishment will remain an approved source for the procurement of subsistence for the Armed Forces. This audit is performed after an establishment has been rated unacceptable after a Routine or Directed Audit.

3.32 Unacceptable audit rating. An audit rating generated when one or more critical findings are identified at the time of the audit or failure of an establishment to respond and satisfactorily address all CARs generated within an established suspense date.

3.33 Validation. The evaluation of scientific and technical information designed to determine whether the control measures or food safety plan, when properly implemented, effectively controls the identified food hazards.

3.34 Verification. The application of methods, procedures, tests and other evaluations, in addition to monitoring, to determine whether a control measure or combination of control measures is, or has been, operating as intended. This process is also designed to establish the validity of the food safety plan.

3.35 Veterinary audit rating. A rating based upon the results of the food protection audit, rated as either acceptable or unacceptable.



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3.36 Veterinary audit report (VAR). A written record of the results from a DoW food protection audit.

3.37 Veterinary Corps Officer (VCO). U.S. Army Veterinarian (Doctor of Veterinary Medicine (DVM)/Veterinary Medicine Doctor (VMD) - 64 series AOC - and Warrant Officers (Veterinary Corps Food Safety Officer, military occupational specialty (MOS) 640A). These officers are described as auditors in the publication.

3.38 Vulnerability. A point, step, or procedure in a food system that is susceptible to intentional adulteration or to disruption by a natural hazard.

3.39 Working papers. Papers maintained by the auditor, which may include written notes and instructions, checklists, questionnaires, matrixes, draft reports, etc.

## 4. GENERAL REQUIREMENTS

4.1 General. Commercial food establishments will be audited in accordance with the list of CFR Parts and Subparts located in Appendix A and B, as well as the applicable product Appendices. The U.S. Army Veterinary Services Lethality Calculator may be utilized in applicable establishment audits (pasteurized products, etc.).

## 5. DETAILED REQUIREMENTS

5.1 Food protection audits. One of four types of audits (Initial, Routine, Directed, or Special), as defined in this document, may be conducted. Each establishment must receive an acceptable audit rating in order to be added, or remain listed in, the Worldwide Directory. Establishments that receive an unacceptable audit rating will be suspended from, denied listing in, or removed from the Worldwide Directory. Findings identified during the audit are scored Critical (C), Major (M), or Minor (m). All Routine audits are normally announced and scheduled annually; however, unannounced or more frequent routine audits may occur.

5.1.1 Verification of compliance. Compliance with the requirements of this standard and the applicable appendices will be verified through an audit of the food establishment. The auditor(s) shall examine and evaluate the adequacy of an establishment's food safety, food defense, and other applicable control systems to include an examination of the methods used to receive, handle, process, and store and transport food. The food protection audit will be performed in the presence of management or a designated representative. Results of the audit will be documented. The DoW reserves the right to audit against all or part of the requirements during the audit. Establishments shall provide all documents pertinent to the scope of the audit.

5.1.2. Audit results. Upon completion, the audit will receive either an acceptable or unacceptable rating.

5.1.3 Acceptable rating. Rating awarded to an audit without critical findings and after the establishment adequately addresses all other findings within established suspense date(s).



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5.1.4 Unacceptable rating. Rating awarded to an audit with at least one critical finding or when an establishment fails to respond and satisfactorily address Critical or Major findings documented in a CAR's suspense date.

5.1.5 Corrective action requests. CARs will be generated and issued for all Critical and Major findings with a suspense date of no more than 90 days from the audit date. Establishments receiving CARs must define the root cause of the finding as well as the action(s) taken to correct and prevent recurrence of the finding(s) within the established suspense date. Failure to reply to a CAR within the specified suspense date will result in an unacceptable rating and either suspension, denial of listing, or removal from the Worldwide Directory. Future audits will not be scheduled or conducted until the establishment has provided an actionable response to all CARs, which adequately addresses the root cause of the finding.

5.2 Harassment or abuse. Under no circumstances shall an auditor be forced to endure harassment or abuse from establishment representatives. Auditors who feel they are being verbally abused, harassed, or otherwise treated improperly have the option of terminating the audit at their discretion. This may include, but is not limited to, discrimination based on race, color, religion, national origin, and gender (including sexual harassment). Under these circumstances, the auditor should excuse themselves from the establishment and consult telephonically with the chain of command. Similarly, auditors shall neither harass nor abuse establishment personnel. Establishment representatives experiencing abusive behavior from a DoW auditor shall contact the contracting agency representative to resolve these issues.

5.3 Appendices. The items in the appendices found in Military Standard 3006D are derived from the CFRs or industry standards applicable to specific products and shall apply to the appropriate food establishment.

5.4 Denial of access to auditors. Establishments may be removed from or denied listing in the Worldwide Directory for refusing auditor entry into the establishment, refusal to schedule the audit, or refusal to provide access to applicable processes and documentation.

5.5 Laboratory testing. All food products and the environment in the establishment are subject to laboratory testing. If samples are selected for laboratory testing, the audit report will not be considered final until laboratory results are received. Results of the establishment's internal/contract laboratory tests will be made available to DoW auditors upon request. DoW Food Safety and Quality Assurance Laboratory Action Levels, Appendix O, are available at https://sph.health.mil/.

5.6 Water potability.

5.6.1 Introduction. Commercial food establishments must adhere to the water potability requirements listed in this standard. Microbiological, chemical, and radiological water testing is required for commercial food establishments that process food requiring the declaration of water as an ingredient on their product label(s) in accordance with 21 CFR 101.4. All other establishments that process food not requiring the declaration of water as an ingredient on their product label(s) only require microbiological testing. Food storage facilities that do not



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process exposed food are exempt from water potability testing. The audited commercial food establishment is responsible for laboratory testing and the associated costs. Source water testing for bottled water does not apply to this policy and is required in accordance with 21 CFR 129 and 165.

5.6.2 Microbiological quality. Microbiological quality will be assessed annually for municipal sources and quarterly for wells.

5.6.2.1 Sampling. Microbiological samples must be drawn from within the establishment. Microbiological testing performed outside the establishment by a municipality does not satisfy this requirement. The establishment shall draw microbiological samples from the potable water supply, as close as possible to the primary point of use, to account for possible contaminant sources within the establishment. The following limits apply:

a. Coliforms: NEG/100 mL.

b. *Escherichia coli (E. coli)* or fecal coliforms: NEG/100 mL (testing for *E. coli* and speciation required when coliforms are present).

5.6.3 Chemical and radiological quality. Chemical testing is required annually, and radiological testing is required every three years. Chemical and radiological results must comply with the maximum contaminant levels (MCLs) of all chemical and radiological contaminants listed in the US Environmental Protection Agency (EPA) National Primary Drinking Water Regulations (40 CFR 141). Establishments may reduce the frequency of their chemical and radiological testing in accordance with 40 CFR 141.

5.6.3.1 Sampling. Water samples for chemical and radiological testing can be drawn at the municipality source or at the establishment. Municipal or external laboratory reports for chemical and radiological testing satisfies this requirement. When municipal water quality reports are used to demonstrate chemical and radiological compliance, the report can be electronic or printed, and each required parameter shall show a result (or below detectable limit). If a well is supplying the source water, samples must be drawn at the well. If multiple wells are supplying the source water, samples must be drawn separately from each well.

5.6.4 Authorized testing laboratories. Establishments must use external and internal accredited laboratories (e.g., EPA, state-certified laboratories, and ISO 17025 accredited laboratories) with verifiable certification, either by the website of the accrediting organization or a current accreditation certificate. Internal laboratories will be evaluated by the DOW auditor for process validation of acceptable laboratory practices (only applies to microbiological testing).

5.6.5 Threat assessment. Audited food establishments will have a process in place to identify known or suspected threats to water quality, as an auditable program. Identification of such a threat requires an immediate re-assessment of microbiological, chemical, or radiological potability as described in the previous paragraphs. Testing is only required for parameters appropriate to the threat. This process will include final testing upon identification and resolution of the event that initiated the retesting. Sites that use well sources will have a similar process,



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and shall include monitoring the nearest municipality warning system as an indicator that a natural or manmade event has occurred in the locality that could affect their water table. Examples (not all inclusive) of threats are listed below.

a. Acts of Nature, such as storms, flooding, etc. (requires microbiological retesting).

b. Alerts/notices provided by local water authorities that are accompanied by a water warning or boil water advisory (requires testing appropriate to the Alert).

c. Indications that infrastructure that may be carrying water has degraded or become contaminated in some way, such as lead leaching from water mains (requires testing appropriate to the indicator).

d. Major plumbing work of any kind conducted on the premises of the audited establishment (requires testing of microbiological, chemical, and radiological).

5.7 Other water systems. Many establishments have a second or dedicated water system for fire suppression (sprinklers). This system is normally segregated from the main water supply line and is protected by a reduced pressure zone (RPZ) device. Boiler make-up tanks and boiler chemicals (water softeners and injection of descaling chemicals) shall also be isolated from the potable water supply; this is normally accomplished using an air-gap, RPZ or double check valve assembly (DCVA), or a combination of these devices.

5.8 Sanitizers. Chemicals and sanitizers shall be approved for use in contact with food in accordance with Title 21, Part 173, Secondary Direct Food Additives Permitted in Food for Human Consumption, Subpart D, and Title 21, Part 178.1010, Sanitizing Solutions.

5.9 Films, packages, containers and lids. These items shall be approved for contact with food in accordance with Title 21, Part 177, Indirect Food Additives - Polymers. Establishments shall have a letter of guaranty or certificate of compliance from their film or package supplier indicating compliance.

5.10 Safety of films. Additionally, the oxygen transmission rate (OTR), or permeability of packaging films shall be considered when *Clostridium botulinum* is reasonably likely to occur (fresh fruits and vegetables, seafood, etc.). If the films are not oxygen permeable, a vacuum packaged or gas flushed (Carbon Dioxide or Nitrogen) environment can create anaerobic conditions conducive to the growth and subsequent toxin production associated with *C. botulinum*. The food safety plan shall address these hazards and implement appropriate controls to mitigate the risks. Additionally, the establishment shall have film studies or Certificates of Analysis or Compliance from their film suppliers containing technical data (oxygen transfer rates - OTRs) and indicating compliance.

5.11 Nondisclosure of audit information and results. U.S. Army Medical Command Veterinary Service (MEDCOM VS) is the custodian for the records associated with food protection audits. Audit information and reports will not be shared outside of the normal chain of custody without approval of MEDCOM VS.



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5.11.1 Protection of audit information. Audit notes, supporting documents, draft reports, final reports, facility proprietary information, and any other information or material collected during the course of a food protection audit is proprietary and sensitive in nature and therefore shall be protected in accordance with 18 USC § 1905.

5.11.2 Nondisclosure agreements. Due to the preexisting information protection requirements set forth in 18 USC § 1905, nondisclosure agreements shall not be signed by U.S. Army Veterinary Service auditors.

5.11.3 Proprietary information. After the audit, auditors shall limit their discussion of proprietary information to parties that have a need to know as part of their official duties (audit review chain) and properly dispose of (shred) audit working papers. Failure to protect proprietary information may be a violation of the U.S. Code and may be punishable under the Uniform Code of Military Justice or applicable service regulations.

## 6. NOTES

6.1 Intended use. This standard is intended to ensure that food products procured by DoW for use by Armed Forces personnel are safe.

6.2 Acquisition requirements. Acquisition documents should specify the title, number, and date of this standard.

6.3 Tailoring guidance. To ensure proper application of this standard, invitations for bids, requests for proposals, and contractual statements of work should tailor the requirements in section 5 of this standard to exclude any unnecessary requirements. For example, if the requirement applies to a cheese establishment, only the requirements contained in this standard and in the applicable appendices should be mandated.

6.4 Audit guidelines. The DoW will perform audits of establishments based on guidelines contained in MIL-HDBK-3006D, Guidelines for Auditing Commercial Food Establishments, and the associated product or process appendices contained therein. Copies of this document are available online at: https://assist.dla.mil/online/start/.

6.5 Listing of establishments. Establishments in compliance with this standard may be listed in the Worldwide Directory. Requests for listing in the Worldwide Directory should be directed to the appropriate procurement agency. Information related to the directory's use should be directed to the Directory listing authority.

6.6 Subject term (key word) listing.

Auditing personnel
Corrective action request
Directed audit
Food defense



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Food safety
Initial audit
Laboratory audit
Methodology
Routine audit
Special audit
Veterinary food protection audit report
Worldwide Directory

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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# GENERAL REQUIREMENTS

## A.1 SCOPE

A.1.1 Scope. This appendix establishes the general food safety requirements for auditing food processing and food storage establishments. This appendix is a mandatory part of this standard. The requirements contained herein are intended for compliance.

## A.2 GENERAL REQUIREMENTS

A.2.1 General. Commercial food establishments will be audited in accordance with 21 CFR 117, 121, and the applicable food Appendix. Some CFR Parts and Subparts will apply without an associated scored item; some are not adopted entirely.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 117 | - | Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food |

(Copies of these are available online at: https://www.govinfo.gov/.)

## A.3 CHECKLISTS

A.3.1 Checklists. Guidance for auditing food production establishments is contained in the following Tables A-I through A-IV and in the applicable product or process Appendix.

A.3.2 HACCP and Preventive Controls checklists. Items A-30 through A-61 are not applicable to Appendices H, J, M, and O, as each of the Appendices have HACCP items built into their respective requirements.



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B.

| ITEM | REQUIREMENT |
|---|---|
| A-1 | Personnel - Disease Control. Management must take reasonable measures and precautions to ensure disease control so that any person found medically unfit IAW this section is excluded from any operations when there is a reasonable possibility of food, food-contact surfaces, or food-packaging materials becoming contaminated until the condition is corrected, unless conditions such as open lesions, boils, and infected wounds are adequately covered (e.g., by an impermeable cover). Personnel must be instructed to report such health conditions to their supervisors. 117.10(a) |
| A-2 | Personnel - Cleanliness. Management must take reasonable measures and precautions to ensure cleanliness whereby all persons working in direct contact with food, food-contact surfaces, and food-packaging materials conform to hygienic practices while on duty to the extent necessary to protect against allergen cross-contact and against contamination of food. Methods for maintaining cleanliness:<br>(1) Wearing outer garments appropriate to the operation in a manner that protects against allergen cross-contact and against the contamination of food, food-contact surfaces, or food-packaging materials.<br>(2) Maintaining adequate personal cleanliness.<br>(3) Washing hands thoroughly (and sanitizing if necessary to protect against contamination with undesirable microorganisms) in an adequate hand-washing facility before starting work, after each absence from the work station, and at any other time when the hands may have become soiled or contaminated.<br>(4) Unsecured jewelry removed. If hand jewelry cannot be removed, it may be adequately covered.<br>(5) Maintaining gloves, if they are used in food handling, in an intact, clean, and sanitary condition.<br>(6) Wearing, where appropriate, in an effective manner, hair nets, headbands, caps, beard covers, or other effective hair restraints.<br>(7) Storing clothing or other personal belongings in areas other than where food is exposed or where equipment or utensils are washed.<br>(8) Confining the following to areas other than where food may be exposed or where equipment or utensils are washed: eating food, chewing gum, drinking beverages, or using tobacco.<br>(9) Taking any other necessary precautions to protect against allergen cross-contact and against contamination of food, food-contact surfaces, or food-packaging materials with microorganisms or foreign substances (including perspiration, hair, cosmetics, tobacco, chemicals, and medicines applied to the skin). 117.10(b) |
| A-3 | Plant and Grounds - Grounds. The grounds about a food plant under the control of the operator must be kept in a condition that will protect against the contamination of food. The methods for adequate maintenance of grounds must include:<br>(1) Properly storing equipment, removing litter and waste, and cutting weeds or grass within the immediate vicinity of the plant that may constitute an attractant, breeding place, or harborage for pests; |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-3 cont. | (2) Maintaining roads, yards, and parking lots.<br>(3) Adequately draining areas that may contribute contamination to food by seepage, foot-borne filth, or providing a breeding place for pests.<br>(4) Operating systems for waste treatment and disposal in an adequate manner so that they do not constitute a source of contamination in areas where food is exposed.<br>(5) If the plant grounds are bordered by grounds not under the operator's control and not maintained in the manner described in the paragraphs above in this section, care must be exercised in the plant by inspection, extermination, or other means to exclude pests, dirt, and filth that may be a source of food contamination. 117.20(a) |
| A-4 | Plants and Grounds - Plant Construction and Design. The plant must be suitable in size, construction, and design to facilitate maintenance and sanitary operations for food-production purposes (i.e., manufacturing, processing, packing, and holding). The plant must:<br>(1) Provide adequate space for such placement of equipment and storage of materials as is necessary for maintenance, sanitary operations, and the production of safe food.<br>(2) Permit the taking of adequate precautions to reduce the potential for allergen cross-contact and for contamination of food, food-contact surfaces, or food-packaging materials with microorganisms, chemicals, filth, and other extraneous material. The potential for allergen cross-contact and for contamination may be reduced by adequate food safety controls and operating practices or effective design, including the separation of operations in which allergen cross-contact and contamination are likely to occur, by one or more of the following means: location, time, partition, air flow systems, dust control systems, enclosed systems, or other effective means.<br>(3) Permit the taking of adequate precautions to protect food in installed outdoor bulk vessels by any effective means, including:<br>(i) Using protective coverings.<br>(ii) Controlling areas over and around the vessels to eliminate harborages for pests.<br>(iii) Checking on a regular basis for pests and pest infestation.<br>(iv) Skimming fermentation vessels, as necessary.<br>(4) Be constructed in such a manner that floors, walls, and ceilings may be adequately cleaned and kept clean and kept in good repair; that drip or condensate from fixtures, ducts and pipes does not contaminate food, food-contact surfaces, or food-packaging materials; and that aisles or working spaces are provided between equipment and walls and are adequately unobstructed and of adequate width to permit employees to perform their duties and to protect against contaminating food, food-contact surfaces, or food-packaging materials with clothing or personal contact.<br>(5) Provide adequate lighting in hand-washing areas, dressing and locker rooms, and toilet rooms and in all areas where food is examined, manufactured, processed, packed, or held and where equipment or utensils are cleaned; and provide shatter-resistant light |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-4 cont. | bulbs, fixtures, skylights, or other glass suspended over exposed food in any step of preparation or otherwise protect against food contamination in case of glass breakage. (6) Provide adequate ventilation or control equipment to minimize dust, odors and vapors (including steam and noxious fumes) in areas where they may cause allergen cross-contact or contaminate food; and locate and operate fans and other air-blowing equipment in a manner that minimizes the potential for allergen cross-contact and for contaminating food, food-packaging materials, and food-contact surfaces. (7) Provide, where necessary, adequate screening or other protection against pests. 117.20(b) |
| A-5 | Sanitary Operations - General Maintenance. Buildings, fixtures, and other physical facilities of the plant maintained in clean, sanitary condition and repair adequate to prevent food from becoming adulterated. Cleaning and sanitizing of utensils and equipment must be conducted in a manner that protects against allergen cross-contact and against contamination of food, food-contact surfaces, or food-packaging materials. 117.35(a) |
| A-6 | Sanitary Operations - Substances Used in Cleaning and Sanitizing; Storage of Toxic Materials. Cleaning compounds and sanitizing agents used in cleaning and sanitizing procedures must be free from undesirable microorganisms and must be safe and adequate under the conditions of use. Compliance must be verified. Limited toxic materials may be used or stored in a plant where food is processed or exposed (see 117.35 for restrictions). 117.35(b) |
| A-7 | Sanitary Operations - Pest Control. Pests are not allowed in any area of a food plant (see authorized exceptions). Effective measures are taken to exclude pests from the manufacturing, processing, packing, and holding areas and to protect against the contamination of food on the premises by pests. Proper and limited use of pesticides to control pests in the plant is permitted. 117.35(c) |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-8 | Sanitary Operations - Sanitation of Food-Contact Surfaces. All food-contact surfaces, including utensils and food-contact surfaces of equipment, must be cleaned as frequently as necessary to protect against allergen cross-contact and against contamination of food.<br>(1) Food-contact surfaces used for manufacturing/processing, packing, or holding low-moisture food must be in a clean, dry, sanitary condition before use. When the surfaces are wet-cleaned, they must, when necessary, be sanitized and thoroughly dried before subsequent use.<br>(2) In wet processing, when cleaning is necessary to protect against allergen cross-contact or the introduction of microorganisms into food, all food-contact surfaces must be cleaned and sanitized before use and after any interruption during which the food-contact surfaces may have become contaminated. Where equipment and utensils are used in a continuous production operation, the utensils and food-contact surfaces of the equipment must be cleaned and sanitized as necessary.<br>(3) Single-service articles (such as utensils intended for one-time use, paper cups, and paper towels) stored, handled, and disposed of properly. 117.35(d) |
| A-9 | Sanitary Operations - Sanitation of Non-Food-Contact Surfaces. Non-food-contact surfaces of equipment used in the operation of a food plant must be cleaned to protect against allergen cross-contact and against contamination of food, food-contact surfaces, and food-packaging materials. 117.35(e) |
| A-10 | Sanitary Operations - Storage and Handling of Cleaned Portable Equipment and Utensils. Cleaned and sanitized portable equipment with food-contact surfaces and utensils stored in a location and manner that protects food-contact surfaces from allergen cross-contact and from contamination. 117.35(f) |
| A-11 | Sanitary Facilities and Controls - Water Supply. Water supply is adequate for the operations intended, derived from an adequate source, suitable temperature and pressure, and of adequate sanitary quality. Any water that contacts food, food-contact surfaces, or food-packaging materials must be safe and of adequate sanitary quality. Running water at a suitable temperature, and under pressure as needed, must be provided in all areas where required for the processing of food, for the cleaning of equipment, utensils, and food-packaging materials, or for employee sanitary facilities. 117.37(a) |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-12 | Sanitary Facilities and Controls - Plumbing. Plumbing must be of adequate size and design and adequately installed and maintained to:<br>(1) Carry adequate quantities of water to required locations throughout the plant.<br>(2) Properly convey sewage and liquid disposable waste from the plant.<br>(3) Avoid constituting a source of contamination to food, water supplies, equipment, or utensils or creating an unsanitary condition.<br>(4) Provide adequate floor drainage in all areas where floors are subject to flooding-type cleaning or where normal operations release or discharge water or other liquid waste on the floor.<br>(5) Provide that there is not backflow from, or cross-connection between, piping systems that discharge wastewater or sewage and piping systems that carry water for food or food manufacturing. 117.37(b) |
| A-13 | Sanitary Facilities and Controls - Sewage Disposal. Sewage must be disposed of into an adequate sewerage system or disposed of through other adequate means. 117.37(c) |
| A-14 | Sanitary Facilities and Controls - Toilet Facilities. Each plant has adequate, readily accessible toilet facilities. Toilet facilities must be kept clean and must not be a potential source of contamination of food, food-contact surfaces, or food-packaging materials. 117.37(d) |
| A-15 | Sanitary Facilities and Controls - Hand Washing Facilities. Each plant provides hand-washing facilities that are adequate, convenient, and furnish running water at a suitable temperature. 117.37(e) |
| A-16 | Sanitary Facilities and Controls - Rubbish and Offal Disposal. Rubbish and any offal is conveyed, stored, and disposed of properly. 117.37(f) |
| A-17 | Equipment and Utensils.<br>(1) All plant equipment and utensils used in manufacturing, processing, packing, or holding food so designed and of such material and workmanship as to be adequately cleanable, and must be adequately maintained to protect against allergen cross-contact and contamination.<br>(2) Equipment and utensils designed, constructed, and used appropriately to avoid adulteration of food with lubricants, fuel, metal fragments, contaminated water, or any other contaminants.<br>(3) Equipment installed to facilitate the cleaning and maintenance of the equipment and of adjacent spaces.<br>(4) Food-contact surfaces corrosion-resistant when in contact with food.<br>(5) Food-contact surfaces made of nontoxic materials and designed to withstand the environment of their intended use and the action of food, and, if applicable, cleaning compounds, sanitizing agents, and cleaning procedures.<br>(6) Food-contact surfaces maintained to protect food from allergen cross-contact and from being contaminated by any source, including unlawful indirect food additives. 117.40(a) |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-18 | Equipment and Utensils. Seams on food-contact surfaces smoothly bonded or maintained. 117.40(b) |
| A-19 | Equipment and Utensils. Equipment in areas where food is manufactured, processed, packed, or held and that does not come into contact with food is constructed that it can be kept in a clean and sanitary condition. 117.40(c) |
| A-20 | Equipment and Utensils. Holding, conveying, and manufacturing systems, including gravimetric, pneumatic, closed, and automated systems, of a design and construction that enables them to be maintained in an appropriate clean and sanitary condition. 117.40(d) |
| A-21 | Equipment and Utensils. Each freezer and cold storage compartment used to store and hold food capable of supporting growth of microorganisms is fitted with an accurate indicating thermometer, temperature-measuring device, or temperature-recording device. 117.40(e) |
| A-22 | Equipment and Utensils. Instruments and controls used for measuring, regulating, or recording temperatures, pH, acidity, water activity, or other conditions that control or prevent the growth of undesirable microorganisms in food are accurate and precise and adequately maintained, and adequate in number for designated uses. 117.40(f) |
| A-23 | Equipment and Utensils. Compressed air or other gases mechanically introduced into food or used to clean food-contact surfaces or equipment are treated in such a way that food is not contaminated with unlawful indirect food additives. 117.40(g) |
| A-24 | Processes and Controls - General.<br>(1) All operations conducted in accordance with adequate sanitation principles.<br>(2) Appropriate quality control operations ensure that food is suitable for human consumption and that food-packaging materials are safe and suitable.<br>(3) Overall sanitation of the plant is under the supervision of one or more competent individuals assigned responsibility for this function.<br>(4) Adequate precautions taken to ensure that production procedures do not contribute to allergen cross-contact and to contamination from any source.<br>(5) Chemical, microbial, or extraneous-material testing procedures are used where necessary to identify sanitation failures or possible allergen cross-contact and food contamination.<br>(6) All adulterated food is rejected, or if appropriate, treated or processed to eliminate the contamination. 117.80(a) |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-25 | Processes and Controls - Raw Materials and Other Ingredients.<br>(1) Raw materials and other ingredients inspected and segregated or otherwise handled to ensure they are clean and suitable for processing into food; are stored under conditions protecting against allergen cross-contact and against contamination and minimize deterioration. Raw materials are washed or cleaned as necessary to remove soil or other contamination. Water used for washing, rinsing, or conveying food is safe and of adequate sanitary quality. Reused water is safe and does not cause cross-allergen or other contamination.<br>(2) Raw materials and other ingredients must either not contain levels of microorganisms that may render the food injurious to the health of humans, or they must be pasteurized or otherwise treated during manufacturing operations so that they no longer contain levels that would cause the product to be adulterated.<br>(3) Raw materials and other ingredients susceptible to contamination with aflatoxin or other natural toxins comply with FDA regulations for poisonous or deleterious substances before these raw materials or other ingredients are incorporated into finished food.<br>(4) Raw materials, other ingredients, and rework susceptible to contamination with pests, undesirable microorganisms, or extraneous material comply with FDA regulations for natural or unavoidable defects if these materials are used in manufacturing food.<br>(5) Raw materials, other ingredients, and rework must be held in bulk, or in containers designed and constructed so as to protect against allergen cross-contact and against contamination and must be held at such temperature and relative humidity and in such a manner as to prevent the food from becoming adulterated. Material scheduled for rework must be identified as such.<br>(6) Frozen raw materials and other ingredients kept frozen. Thawing is done in a manner that prevents the raw materials and other ingredients from becoming adulterated.<br>(7) Liquid or dry raw materials and other ingredients received and stored in bulk form held in a manner that protects against allergen cross-contact and against contamination.<br>(8) Raw materials and other ingredients that are food allergens, and rework that contains food allergens, are identified and held in a manner that prevents allergen cross-contact. 117.80(b) |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-26 | Processes and Controls - Manufacturing Operations.<br>(1) Equipment and utensils and food containers must be maintained in an adequate condition through appropriate cleaning and sanitizing, as necessary. Insofar as necessary, equipment must be taken apart for thorough cleaning.<br>(2) All food manufacturing, processing, packing, and holding must be conducted under such conditions and controls as are necessary to minimize the potential for the growth of microorganisms, allergen cross-contact, contamination of food, and deterioration of food.<br>(3) Food that can support the rapid growth of undesirable microorganisms must be held at temperatures that will prevent the food from becoming adulterated during manufacturing, processing, packing, and holding.<br>(4) Measures such as sterilizing, irradiating, pasteurizing, cooking, freezing, refrigerating, controlling pH, or controlling aw that are taken to destroy or prevent the growth of undesirable microorganisms must be adequate under the conditions of manufacture, handling, and distribution to prevent food from being adulterated.<br>(5) Work-in-process and rework must be handled in a manner that protects against allergen cross-contact, contamination, and growth of undesirable microorganisms.<br>(6) Effective measures must be taken to protect finished food from allergen cross-contact and from contamination by raw materials, other ingredients, or refuse. When raw materials, other ingredients, or refuse are unprotected, they must not be handled simultaneously in a receiving, loading, or shipping area if that handling could result in allergen cross-contact or contaminated food. Food transported by conveyor must be protected against allergen cross-contact and against contamination as necessary.<br>(7) Equipment, containers, and utensils used to convey, hold, or store raw materials and other ingredients, work-in-process, rework, or other food must be constructed, handled, and maintained during manufacturing, processing, packing, and holding in a manner that protects against allergen cross-contact and against contamination.<br>(8) Adequate measures must be taken to protect against the inclusion of metal or other extraneous material in food.<br>(9) Food, raw materials, and other ingredients that are adulterated must be disposed of in a manner that protects against the contamination of other food; or, if the adulterated food is capable of being reconditioned, it must be: (a) Reconditioned (if appropriate) using a method that has been proven to be effective; or (b) Reconditioned (if appropriate) and reexamined and subsequently found not to be adulterated within the meaning of the Federal Food, Drug, and Cosmetic Act before being incorporated into other food.<br>(10) Washing, peeling, trimming, cutting, sorting and inspecting, mashing, dewatering, cooling, shredding, extruding, drying, whipping, defatting, and forming are performed so as to protect food against allergen cross-contact and against contamination. Food is protected from contaminants that may drip, drain, or be drawn into the food. |



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TABLE A-I. Current Good Manufacturing Practice - 21 CFR 117 Subpart B – Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-26 (cont.) | (11) Heat blanching is conducted properly, and contamination by thermophilic microorganisms in blanchers is minimized by the use of adequate operating temperatures and periodic cleaning and sanitizing as necessary.<br>(12) Batters, breading, sauces, gravies, dressings, dipping solutions, and other similar preparations that are held and used repeatedly over time treated or maintained in such a manner that they are protected against allergen cross-contact and against contamination, and minimizing the potential for the growth of undesirable microorganisms.<br>(13) Filling, assembling, packaging, and other operations ensure food is protected against allergen cross-contact, contamination and growth of undesirable microorganisms.<br>(14) Food, such as dry mixes, nuts, intermediate moisture food, and dehydrated food that relies principally on the control of aw for preventing the growth of undesirable microorganisms, processed to and maintained at a safe moisture level.<br>(15) Food, such as acid and acidified food that relies principally on the control of pH for preventing the growth of undesirable microorganisms, is monitored and maintained at a pH of 4.6 or below.<br>(16) When ice is used in contact with food, it is made from water that is safe and of adequate sanitary quality in accordance with §117.37(a), and is used only if it has been manufactured in accordance with current good manufacturing practice as outlined in this part. 117.80(c) |
| A-27 | Warehousing and Distribution. Storage and transportation of food is under conditions that protect against allergen cross-contact and against biological, chemical (including radiological), and physical contamination of food, as well as against deterioration of the food and the container. 117.93 |
| A-28 | Defect Action Levels. Quality control operations are used at all times that reduce natural or unavoidable defects to the lowest level currently feasible. 117.110(a) |
| A-29 | Defect Action Levels. Mixing of a food containing defects at levels that render that food adulterated with another lot of food is not permitted and renders the final food adulterated, regardless of the defect level of the final food. 117.110(b) |



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TABLE A-II. Food Safety Plan - 21 CFR 117 Subpart C.

| ITEM | REQUIREMENT |
|---|---|
| A-30 | Food Safety Plan - Preparation.<br>Prepare, or have prepared, and implemented a written Food Safety Plan. 117.126(a) |
| A-31 | Food Safety Plan - Contents. Written food safety plan includes:<br>(1) A written hazard analysis;<br>(2) A written supply-chain program as required by Table A-IV (Supply-Chain Program) of this document.<br>(3) A written recall plan as required by Table A-II, item A-36 (Recall Plan – Included Procedures) of this document;<br>(4) Record keeping and documentation procedures a required by Table A-III (Requirements applying to records that must be established and maintained). 117.126(b) |
| A-32 | Food Safety Plan – Record Keeping. The food safety plan required by this section is a record that is subject to the requirements of Table A-III of this document (Requirements Applying to Records That Must Be Established and Maintained). 117.126(c) |
| A-33 | Hazard Analysis - Requirements for a Hazard Analysis.<br>(1) A hazard analysis must be conducted to identify and evaluate, based on experience, illness data, scientific reports, and other information, known or reasonably foreseeable hazards for each type of food manufactured, processed, packed, or held at the facility.<br>(2) Hazard analysis is written regardless of its outcome. 117.130(a) |
| A-34 | Hazard Analysis - Hazard Identification. Hazard identification considers known or reasonably foreseeable hazards that include: biological hazards, including microbiological hazards such as parasites, environmental pathogens, and other pathogens; chemical hazards, including radiological hazards, substances such as pesticide and drug residues, natural toxins, decomposition, unapproved food or color additives, and food allergens; and physical hazards (such as stones, glass, and metal fragments); and the hazard identification considers known or reasonably foreseeable hazards that may be present in food for any of the following reasons: the hazard occurs naturally; the hazard may be unintentionally introduced; or the hazard may be intentionally introduced for purposes of economic gain. 117.130(b) |
| A-35 | Hazard Analysis - Hazard Evaluation.<br>(1) Hazard analysis includes an evaluation of the hazards identified Item A-34 of this document to assess the severity of the illness or injury if the hazard were to occur and the probability that the hazard will occur.<br>(2) The hazard evaluation required by paragraph (1) of this section must include an evaluation of environmental pathogens whenever a ready-to-eat food is exposed to the environment prior to packaging and the packaged food does not receive a treatment or otherwise include a control measure (such as a formulation lethal to the pathogen) that would significantly minimize the pathogen. 117.130(c) |
| A-36 | Supply-Chain Controls. Supply-chain controls. Supply-chain controls include the supply-chain program as required by Table A-V of these General Provisions. 117.135(c)(4)MOD |



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TABLE A-II. Food Safety Plan - 21 CFR 117 Subpart C - Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-37 | Provision of assurances required under §117.136(a) (2), (3), and (4).<br>A facility that provides a written assurance under §117.136(a) (2), (3), or (4) must act consistently with the assurance and document its actions taken to satisfy the written assurance. 117.137 |
| A-38 | Recall Plan - Included Procedures (Applies only for food requiring a preventive control). Recall plan must include procedures that describe the steps to be taken, and assign responsibility for taking those steps, to perform the following actions as appropriate to the facility:<br>(1) Directly notify the direct consignees of the food being recalled, including how to return or dispose of the affected food;<br>(2) Notify the public about any hazard presented by the food when appropriate to protect public health;<br>(3) Conduct effectiveness checks to verify that the recall is carried out; and<br>(4) Appropriately dispose of recalled food—e.g., through reprocessing, reworking, diverting to a use that does not present a safety concern, or destroying the food. 117.139(b) |
| A-39 | Preventive Control Management Components - Supply Chain Program. The supply-chain program established in Table A-IV of these General Provisions is subject to the following preventive control management components as appropriate to ensure the effectiveness of the supply-chain program, taking into account the nature of the hazard controlled before receipt of the raw material or other ingredient:<br>(1) Corrective actions and corrections in accordance with §117.150, taking into account the nature of any supplier non-conformance;<br>(2) Review of records in accordance with §117.165(a)(4); and<br>(3) Reanalysis in accordance with §117.170.<br>The recall plan established in §117.139 is not subject to the requirements of paragraph (a) of this section. 117.140(b)MOD |
| A-40 | Reanalysis - Food Safety Plan - whole. The establishment must conduct a reanalysis of the food safety plan as a whole at least once every 3 years. 117.170(a) |
| A-41 | Reanalysis - Food Safety Plan - whole or partial. The establishment must conduct a reanalysis of the food safety plan as a whole, or the applicable portion of the food safety plan:<br>(1) Whenever a significant change in the activities conducted at the facility creates a reasonable potential for a new hazard or creates a significant increase in a previously identified hazard;<br>(2) Whenever the establishment becomes aware of new information about potential hazards associated with the food;<br>(3) Whenever appropriate after an unanticipated food safety problem in accordance with §117.150(b); and<br>(4) Whenever the establishment finds that a preventive control, combination of preventive controls, or the food safety plan as a whole is ineffective. 117.170(b) |



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TABLE A-II. Food Safety Plan - 21 CFR 117 Subpart C - Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-42 | The establishment must revise the written food safety plan if a significant change in the activities conducted at the facility creates a reasonable potential for a new hazard or a significant increase in a previously identified hazard or document the basis for the conclusion that no revisions are needed. 117.170(d) |
| A-43 | The establishment must conduct a reanalysis of the food safety plan when the DoW determines it is necessary to respond to new hazards and developments in scientific understanding. 117.170(f) |
| A-44 | Requirements Applicable to a Preventive Controls Qualified Individual and a Qualified Auditor. (Adopted with Modifications as stated below). To be a preventive controls qualified individual, the individual shall have successfully completed training in the development and application of risk-based preventive controls at least equivalent to that received under a standardized curriculum available through an accredited institution of higher learning or international trade organization, or be otherwise qualified through job experience to develop and apply a food safety system. Job experience may qualify an individual to perform these functions if such experience has provided an individual with knowledge at least equivalent to that provided through such a standardized curriculum. This individual may be, but is not required to be, an employee of the facility. 117.180(c)(1)MOD |
| A-45 | Implementation records required for this subpart (Food Safety plan). The establishment must establish and maintain the following records documenting implementation of the food safety plan:<br>(1) Documentation, as required by §117.136(b), of the basis for not establishing a preventive control in accordance with §117.136(a);<br>(2) Records that document the monitoring of preventive controls;<br>(3) Records that document corrective actions;<br>(4) Records that document verification, including, as applicable, those related to:<br>(i) Validation;<br>(ii) Verification of monitoring;<br>(iii) Verification of corrective actions;<br>(iv) Calibration of process monitoring and verification instruments;<br>(v) Product testing;<br>(vi) Environmental monitoring;<br>(vii) Records review; and<br>(viii) Reanalysis;<br>(5) Records that document the supply-chain program; and<br>(6) Records that document applicable training for the preventive controls qualified individual and the qualified auditor. 117.190(a) |
| A-46 | The records that must be established and maintained are subject to the requirements of subpart F (Records) of this part. 117.190(b) |



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TABLE A-III. Requirements Applying to Records that Must be Established and Maintained – 21 CFR 117 Subpart F.

| ITEM | REQUIREMENT |
|---|---|
| A-47 | General Requirements Applying to Records - Actual VGalues. Contain the actual values and observations obtained during monitoring and, as appropriate, during verification activities. 117.305(b) |
| A-48 | General Requirements Applying to Records - Accuracy. Records must be accurate, indelible, and legible. 117.305(c) |
| A-49 | General Requirements Applying to Records - Created Concurrently. Records must be created concurrently with performance of the activity documented. 117.305(d) |
| A-50 | General Requirements Applying to Records - Detailed History. Records must be as detailed as necessary to provide history of work performed. 117.305(e) |
| A-51 | General Requirements Applying to Records - Contents. Records must include:<br>(1) Information adequate to identify the plant or facility (e.g., the name, and when necessary, the location of the plant or facility);<br>(2) The date and, when appropriate, the time of the activity documented;<br>(3) The signature or initials of the person performing the activity; and<br>(4) Where appropriate, the identity of the product and the lot code, if any. 117.305(f) |
| A-52 | Additional requirements applying to the food safety plan. The owner, operator, or agent in charge of the facility must sign and date the food safety plan upon initial completion and also upon any modification. 117.310 |
| A-53 | Requirements for Record Retention - Retention Period. All records required by this part must be retained at the plant or facility for at least 2 years after the date they were prepared. 117.315 (a) |
| A-54 | Requirements for Record Retention – Availability. All records required by this part shall be made available to DOW auditors for official review during the audit upon request. 117.320 (modified) |



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TABLE A-IV. Supply-Chain Program - 21 CFR 117 Subpart G.

| ITEM | REQUIREMENT |
|---|---|
| A-55 | Requirement to Establish and Implement a Supply - Chain Program.<br>The manufacturing facility must establish and implement a risk-based supply-chain or supplier certification program. 117.405(a) |
| A-56 | Written Program. The supply-chain or supplier certification program must be written and include active measures appropriate to the identified risks. 117.405(b) & 117.410 |
| A-57 | Requirements and Verification Activities for Raw Materials. The supply chain program must include:<br>(1) Using approved suppliers as required by §117.420;<br>(2) Determining appropriate supplier verification activities (including determining the frequency of conducting the activity) as required by §117.425;<br>(3) Conducting supplier verification activities as required by §§117.430 and 117.435;<br>(4) Documenting supplier verification activities as required by §117.475; and<br>(5) When applicable, verifying a supply-chain-applied control applied by an entity other than the receiving facility's supplier and documenting that verification as required by §117.475, or obtaining documentation of an appropriate verification activity from another entity, reviewing and assessing that documentation, and documenting the review and assessment as required by §117.475. 117.410(a) |
| A-58 | Verification Activities for Raw Materials and Other Ingredients. Appropriate supplier verification activities for raw materials and other ingredients are performed. 117.410(b) |
| A-59 | Required Action by the Receiver if Hazards are not Controlled by the Supplier. If the owner, operator, or agent in charge of a receiving facility determines through auditing, verification testing, document review, relevant consumer, customer or other complaints, or otherwise that the supplier is not controlling hazards that the receiving facility has identified as requiring a supply-chain-applied control, the receiving facility must take and document prompt action to ensure that raw materials or other ingredients from the supplier do not cause food that is manufactured or processed by the receiving facility to be adulterated. 117.410(e) |
| A-60 | Records Documenting the Supply-Chain Program.<br>Records documenting the supply-chain program are subject to the requirements of Table A-III (Requirements applying to records that must be established and maintained) of this document. 117.475 (a) |



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TABLE A-IV. Supply-Chain Program - 21 CFR 117 Subpart G - Continued.

| ITEM | REQUIREMENT |
|---|---|
| A-61 | Receiving Facility Documenting Requirements. The receiving facility must document and review the following in records:<br>(1) The written supply-chain program;<br>(2) Documentation of the approval of an importer supplier;<br>(3) Written procedures for receiving raw materials and other ingredients;<br>(4) Documentation demonstrating use of written procedures for receiving raw materials and other ingredients;<br>(5) Documentation of sampling and testing when determined necessary as a supplier verification activity. This documentation must include:<br>(i) Identification of the raw material or other ingredient tested (including lot number, as appropriate) and the number of samples tested;<br>(ii) Identification of the test(s) conducted, including the analytical method(s) used;<br>(iii) The date(s) on which the test(s) were conducted and the date of the report;<br>(iv) The results of the testing;<br>(v) Corrective actions taken in response to detection of hazards; and<br>(vi) Information identifying the laboratory conducting the testing;<br>(6) Documentation of the review of the supplier's relevant food safety records. This documentation must include:<br>(i) The name of the supplier whose records were reviewed and date of review;<br>(ii) The general nature of the records reviewed;<br>(iii) The conclusions of the review; and<br>(iv) Corrective actions taken in response to significant deficiencies identified during the review;<br>(7) Documentation of other appropriate supplier verification activities based on the supplier performance and the risk associated with the raw material or other ingredient;<br>(8) Documentation of actions taken with respect to supplier non-conformance.<br>117.165(a)(4) & 117.475 |



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# FOOD DEFENSE REQUIREMENTS

## B.1 SCOPE

B.1.1 Scope. This appendix establishes the general food defense requirements for auditing commercial food establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## B.2 GENERAL REQUIREMENTS

B.2.1 General. Commercial food establishments will be audited in accordance with 21 CFR 121. Some CFR Parts and Subparts will apply without an associated scored item; some are not adopted entirely. Applicable parts of 21 CFR 121, Subpart A shall also apply.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 121 | - | Mitigation Strategies to Protect Food Against Intentional Adulteration |

(Copies of these are available online at: https://www.govinfo.gov/.)

## B.3 CHECKLISTS

B.3.1 Checklists. Guidance for auditing food production establishments for food defense is contained in the following Tables B-I and B-II. There may be additional requirements located in the applicable product or process Appendix.



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TABLE B-I. Mitigation Strategies to Protect Food Against Intentional Adulteration (Food Defense Measures) - 21 CFR 121 Subpart C.

| ITEM | REQUIREMENT |
|---|---|
| B-1 | Requirement for a Food Defense Plan. The establishment must prepare, or have prepared, and implement a written food defense plan. 121.126(a) |
| B-2 | Contents of a Food Defense Plan. The written food defense plan must include:<br>(1) The written vulnerability assessment, including required explanations, to identify significant vulnerabilities and actionable process steps as required by §121.130(c); (2) The written mitigation strategies, including required explanations, as required by §121.135(b);<br>(3) The written procedures for the food defense monitoring of the implementation of the mitigation strategies as required by §121.140(a);<br>(4) The written procedures for food defense corrective actions as required by §121.145(a)(1); and<br>(5) The written procedures for food defense verification as required by §121.150(b). 121.126(b) |
| B-3 | Records. The food defense plan required by this section is a record that is subject to the requirements of Table B-II of this document. 121.126(c) |
| B-4 | Vulnerability Assessment to Identify Significant Vulnerabilities and Actionable Process Steps. The establishment must conduct or have conducted a vulnerability assessment for each type of food manufactured, processed, packed, or held at the facility using appropriate methods to evaluate each point, step, or procedure in the food operation to identify significant vulnerabilities and actionable process steps. Appropriate methods must include, at a minimum, an evaluation of:<br>(1) The potential public health impact (e.g., severity and scale) if a contaminant were added;<br>(2) The degree of physical access to the product; and<br>(3) The ability of an attacker to successfully contaminate the product. 121.130(a) |
| B-5 | Inside Attacker. The assessment must consider the possibility of an inside attacker. 121.130(b) |
| B-6 | Written Document with Explanations. Regardless of the outcome, the vulnerability assessment must be written and must include an explanation as to why each point, step, or procedure either was or was not identified as an actionable process step. 121.130(c) |
| B-7 | Mitigation Strategies for Actionable Process Steps. The establishment must identify and implement mitigation strategies at each actionable process step to provide assurances that the significant vulnerability at each step will be significantly minimized or prevented and the food manufactured, processed, packed, or held by the establishment will not be adulterated under section 402 of the Federal Food, Drug, and Cosmetic Act. For each mitigation strategy implemented at each actionable process step, the establishment must include a written explanation of how the mitigation strategy sufficiently minimizes or prevents the significant vulnerability associated with the actionable process step. 121.135(a) |



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TABLE B-I. Mitigation Strategies to Protect Food Against Intentional Adulteration (Food Defense Measures) - 21 CFR 121 Subpart C - Continued.

| ITEM | REQUIREMENT |
|---|---|
| B-8 | Written Strategies. Mitigation strategies and accompanying explanations must be written. 121.135(b) |
| B-9 | Mitigation Strategies Management Components.<br>Mitigation strategies required under§121.135 are subject to the following mitigation strategies management components as appropriate to ensure the proper implementation of the mitigation strategies, taking into account the nature of each such mitigation strategy and its role in the facility's food defense system:<br>(a) Food defense monitoring in accordance with §121.140;<br>(b) Food defense corrective actions in accordance with §121.145; and<br>(c) Food defense verification in accordance with §121.150. 121.138 |
| B-10 | Food Defense Monitoring - Written Procedures and Frequencies.<br>As appropriate to the nature of the mitigation strategy and its role in the facility's food defense system, the establishment must establish and implement written procedures, including the frequency with which they are to be performed, for food defense monitoring of the mitigation strategies. 121.140(a) |
| B-11 | Food Defense Monitoring - Monitoring Mitigation. As appropriate to the nature of the mitigation strategy and its role in the facility's food defense system, the establishment must monitor the mitigation strategies with adequate frequency to provide assurances that they are consistently performed. 121.140(b) |
| B-12 | Food Defense Monitoring - Documenting Monitoring. As appropriate to the nature of the mitigation strategy and its role in the facility's food defense system:<br>(1) The establishment must document the monitoring of mitigation strategies in accordance with this section in records that are subject to verification in accordance with §121.150(a)(1) and records review in accordance with §121.150(a)(3)(i).<br>(2) Records may be affirmative records demonstrating the mitigation strategy is functioning as intended. Exception records demonstrating the mitigation strategy is not functioning as intended may be adequate in some circumstances. 121.140(c) |



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TABLE B-I. Mitigation Strategies to Protect Food Against Intentional Adulteration (Food Defense Measures) - 21 CFR 121 Subpart C - Continued.

| ITEM | REQUIREMENT |
|---|---|
| B-13 | Food Defense Corrective Actions - procedures. As appropriate to the nature of the actionable process step and the nature of the mitigation strategy:<br>(1) The establishment must establish and implement written food defense corrective action procedures that must be taken if mitigation strategies are not properly implemented.<br>(2) The food defense corrective action procedures must describe the steps to be taken to ensure that:<br>(i) Appropriate action is taken to identify and correct a problem that has occurred with implementation of a mitigation strategy; and<br>(ii) Appropriate action is taken, when necessary, to reduce the likelihood that the problem will recur. 121.145(a) |
| B-14 | Records. All food defense corrective actions taken in accordance with this section must be documented in records that are subject to food defense verification in accordance with §121.150(a)(2) and records review in accordance with §121.150(a)(3)(i). 121.145(b) |
| B-15 | Food Defense Verification. Food defense verification activities must include, as appropriate to the nature of the mitigation strategy and its role in the facility's food defense system:<br>(1) Verification that food defense monitoring is being conducted as required by §121.138 (and in accordance with §121.140);<br>(2) Verification that appropriate decisions about food defense corrective actions are being made as required by §121.138 (and in accordance with §121.145);<br>(3) Verification that mitigation strategies are properly implemented and are significantly minimizing or preventing the significant vulnerabilities. To do so, the establishment must conduct activities that include the following, as appropriate to the facility, the food, and the nature of the mitigation strategy and its role in the facility's food defense system:<br>(i) Review of the food defense monitoring and food defense corrective actions records within appropriate timeframes to ensure that the records are complete, the activities reflected in the records occurred in accordance with the food defense plan, the mitigation strategies are properly implemented, and appropriate decisions were made about food defense corrective actions; and<br>(ii) Other activities appropriate for verification of proper implementation of mitigation strategies; and<br>(4) Verification of reanalysis in accordance with §121.157. 121.150(a) |
| B-16 | Written Procedures. The establishment must establish and implement written procedures, including the frequency for which they are to be performed, for verification activities conducted according to §121.150(a)(3)(ii). 121.150(b) |



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TABLE B-I. Mitigation Strategies to Protect Food Against Intentional Adulteration (Food Defense Measures) - 21 CFR 121 Subpart C - Continued.

| ITEM | REQUIREMENT |
|---|---|
| B-17 | Documentation. All verification activities conducted in accordance with this section must be documented in records. 121.150(c) |
| B-18 | Reanalysis Every Three Years. The establishment must conduct a reanalysis of the food defense plan, as a whole at least once every three years. 121.157(a) |
| B-19 | Reanalysis Under Other Conditions. The establishment must conduct a reanalysis of the food defense plan as a whole, or the applicable portion of the food defense plan:<br>(1) Whenever a significant change made in the activities conducted at the facility creates a reasonable potential for a new vulnerability or a significant increase in a previously identified vulnerability;<br>(2) Whenever the establishment becomes aware of new information about potential vulnerabilities associated with the food operation or facility;<br>(3) Whenever the establishment finds that a mitigation strategy, a combination of mitigation strategies, or the food defense plan as a whole is not properly implemented; and<br>(4) Whenever FDA requires reanalysis to respond to new vulnerabilities, credible threats to the food supply, and developments in scientific understanding including, as appropriate, results from the Department of Homeland Security biological, chemical, radiological, or other terrorism risk assessment. 121.157(b) |
| B-20 | Completion of Reanalysis. The establishment must complete such reanalysis required by paragraphs (a) and (b) of this section and implement any additional mitigation strategies needed to address the significant vulnerabilities identified, if any:<br>(1) Before any change in activities (including any change in mitigation strategy) at the facility is operative;<br>(2) When necessary within 90-calendar days after production; and<br>(3) Within a reasonable timeframe, providing a written justification is prepared for a timeframe that exceeds 90 days after production of the applicable food first begins. 121.157(c) |
| B-21 | Revising the Food Defense Plan. The establishment must revise the written food defense plan if a significant change in the activities conducted at the facility creates a reasonable potential for a new vulnerability or a significant increase in a previously identified vulnerability or document the basis for the conclusion that no revisions are needed. 121.157(d) |



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TABLE B-II. Requirements Applying to Records that Must be Established and Maintained – 21 CFR 121 Subpart D.

| ITEM | REQUIREMENT |
|---|---|
| B-22 | Records Subject to the Requirements of this Subpart.<br>(a) Except as provided by paragraph (b) of this section, all records required by subpart C of this part are subject to all requirements of this subpart.<br>(b) The requirements of §121.310 apply only to the written food defense plan. 121.301 |
| B-23 | Records - Originals or Copies. Records must be kept as original records, true copies (such as photocopies, pictures, scanned copies, microfilm, microfiche, or other accurate reproductions of the original records), or electronic records. 121.305(a) |
| B-24 | Records - Actual Values. Records must contain the actual values and observations obtained during food defense monitoring. 121.305(b) |
| B-25 | Records - Accuracy. Records must be accurate, indelible, and legible. 121.305(c) |
| B-26 | Records - Documented Activity Performance. Records must be created concurrently with performance of the activity documented. 121.305(d) |
| B-27 | Records - Details. Records must be as detailed as necessary to provide history of work performed. 121.305(e) |
| B-28 | Records - Additional Content Requirements. Records must include:<br>(1) Information adequate to identify the facility (e.g., the name, and when necessary, the location of the facility);<br>(2) The date and, when appropriate, the time of the activity documented;<br>(3) The signature or initials of the person performing the activity; and<br>(4) Where appropriate, the identity of the product and lot code, if any. 121.305(f) |
| B-29 | Records - Exemption. Records that are established or maintained to satisfy the requirements of this part and that meet the definition of electronic records in §11.3(b)(6) of this chapter are exempt from the requirements of part 11 of this chapter. Records that satisfy the requirements of this part, but that also are required under other applicable statutory provisions or regulations, remain subject to part 11 of this chapter. 121.305(g) |
| B-30 | Additional Requirements Applying to the Food Defense Plan.<br>The owner, operator, or agent in charge of the facility must sign and date the food defense plan: (a) Upon initial completion; and (b) Upon any modification. 121.310 |
| B-31 | Requirements for Record Retention.<br>(1) All records required by this part must be retained at the facility for at least 2 years after the date they were prepared.<br>(2) Records that a facility relies on during the 3-year period preceding the applicable calendar year to support its status as exempt as a very small business must be retained at the facility as long as necessary to support the status of a facility as a very small business during the applicable calendar year. 121.315(a) |



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TABLE B-II. Requirements Applying to Records that Must be Established and Maintained - 21 CFR 121 Subpart D - Continued.

| ITEM | REQUIREMENT |
|---|---|
| B-32 | Retention of Food Defense Plan. The food defense plan must be retained for at least two years after its use is discontinued. 121.315(b) |
| B-33 | Offsite Storage of Records. Except for the food defense plan, offsite storage of records is permitted if such records can be retrieved and provided onsite within 24 hours of request for official review. The food defense plan must remain onsite. Electronic records are considered to be onsite if they are accessible from an onsite location. 121.315(c) |
| B-34 | Facility Closure. If the facility is closed for a prolonged period, the food defense plan may be transferred to some other reasonably accessible location but must be returned to the facility within 24 hours for official review upon request. 121.315(d) |



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# MANUFACTURED CHEESE PRODUCTS

## C.1 SCOPE

C.1.1 Scope. This appendix establishes the food safety and related requirements for manufactured cheese establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## C.2 APPLICABLE DOCUMENTS

C.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

C.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 7, Part 58 | - | Grading and Inspection, General Specifications for Approved Plants and Standards for Grades of Dairy Products |
| Title 21, Part 133 | - | Cheeses and Related Cheese Products |

(Copies of these documents are available online at: https://www.govinfo.gov/.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

U.S. Public Health Service (USPHS)/Food and Drug Administration (FDA) Pasteurized Milk Ordinance (PMO), current version.

(Copies of this document are available online at: https://www.fda.gov/food/guidance-documents-regulatory-information-topic-food-and-dietary-supplements/milk-guidance-documents-regulatory-information.)



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## C.3 DEFINITIONS

C.3.1 Definitions. The definitions in section 3 of this standard, as well as those contained in all reference and associated documents apply to this appendix.

C.3.2 Milk. In manufacturing non-grade “A” dairy products, milk is used and is defined as follows.

a. Milk is the lacteal secretion, practically free from colostrum, obtained by the complete milking of one or more healthy cows.

b. Goat milk is the lacteal secretion, practically free from colostrum, obtained by the complete milking of one or more healthy goats.

## C.4 REQUIREMENTS

C.4.1 Animal health requirements. In terms of animal health, milk shall be obtained from healthy animals. Cows, goats, or any other hoofed mammal used for obtaining milk will meet the animal health requirements as specified in Section 8 (Animal Health) of the PMO or equivalent program as determined by the Directory listing authority.

C.4.2 Goat milk. Goat milk will only be used to manufacture dairy products that are legally provided for in 21 CFR 133 or recognized as non-standardized traditional products normally manufactured from goat’s milk.

C.4.3 Cheese manufactured from unpasteurized milk. Cheese varieties may be manufactured from pasteurized or unpasteurized dairy products. If manufactured from un-pasteurized dairy products, the cheese shall be aged for no less than 60 days and at a temperature of no less than 35°F in accordance with 7 CFR Part 58.439.

## C.5 CHECKLIST

C.5.1 Checklist. Guidelines for the auditing of shell egg establishments are contained in the following Table D-I to D-V checklists.



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TABLE C-I. Manufactured Cheese Products Requirements as Specified in the USPHS/FDA Pasteurized Milk Ordinance (PMO), 7 CFR Part 58 and 21 CFR Part 133.

| ITEM | REQUIREMENT |
|---|---|
| C-1 | Milk originates from farms that meet the requirements or intent of the PMO and herds that meet the animal health requirements as specified in Section 8 (Animal Health) of the PMO or equivalent program as determined by the DoW policy. PMO, Sec. 8 |
| C-2 | Where applicable, every particle of product was heated and held continuously, in properly operated equipment, at or above the specified time/temperature (or other time/temperature relationship for equivalent microbial destruction). 133.3(d) and (e) |
| C-3 | Product contact surfaces of utensils and equipment such as holding tanks, pasteurizers, coolers, vats, agitators, pumps, sanitary piping and fittings or any specialized equipment must be constructed of stainless steel, or other materials which under conditions of intended use are equally corrosion resistant. Non-metallic parts, other than glass, having product contact surfaces shall comply with 3-A Sanitary Standards for Plastic or Rubber and Rubber-Like Materials. All Clean In Place (CIP) systems must comply with 3-A Acceptable Practices for Permanently Installed Sanitary Product, Pipelines and Cleaning Systems. 58.128(a) |
| C-4 | If applicable, all can washers and their associated water and steam lines are equipped with proper temperature and pressure controls and properly maintained. The steam pressure to the can washer is not less than 80 pounds and the final rinse temperature should be automatically controlled and does not exceed 140ºF (60°C). 58.128(c) |
| C-5 | Raw milk conforms to basic quality and classification specifications of 58.132 – 58.138 and is tested at the frequencies required, and records are maintained. 58.132 – 58.138 |
| C-6 | Receiving, holding, and processing of milk and cream and the manufacturing, handling, packaging, storing, and delivery of dairy products is in accordance with methods consistent with good commercial practices; milk is not more than three days old when picked up from the producer and delivered to the plant, receiving station or transfer. 58.142 |
| C-7 | Bulk milk in storage tanks within the dairy plant shall be handled in such a manner as to minimize bacterial increase and shall be maintained at 45°F or lower until processing begins, unless holding at a higher temperature is applicable to the particular manufacturing or process practice. 58.143(a) |
| C-8 | The bacteriological quality of commingled raw milk in storage tanks shall not exceed 1,000,000/ml. 58.143(b) |
| C-9 | Sanitary seal assemblies are removable on all agitators, pumps, and vats, and are inspected at regular intervals and kept clean. 58.146(a) |
| C-10 | Records are maintained for all required tests and analyses. Such records are available for examination. 58.148 |
| C-11 | A separate starter room or properly designed starter tanks and satisfactory air movement is provided. All necessary precautions are taken to prevent contamination of the facility, equipment, and air therein. 58.406 |



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TABLE C-II. Manufactured Cheese Products Requirements as Specified in the USPHS/FDA Pasteurized Milk Ordinance (PMO), 7 CFR Part 58 and 21 CFR Part 133 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| C-12 | When applicable, mold counts for make rooms are not more than 15 colonies per plate during a 15 minute exposure. 58.407 |
| C-13 | When applicable to rind cheese, a separate room or container of adequate size is provided for paraffining and boxing; maintained at proper temperature to prevent sweating prior to paraffining. 58.410. |
| C-14 | Separate rooms are provided for preparation of bulk cheese to be cut and wrapped into smaller packages. Air movement is moving outward. 58.413 |
| C-15 | Bulk starter vats are of applicable 3-A Sanitary Standards, in good repair, and equipped with tight fitting lids and have adequate temperature controls and indicating/recording devices. 58.415 |
| C-16 | Vats, tanks, and drain tables are of metal construction. The inner liner is a minimum 16-gauge steel or equally corrosion resistant metal, properly pitched, welded, and fitted with sanitary outlets and valves for maintenance of heat to the lines. 58.416(a) |
| C-17 | Automatic salter is properly designed to ensure proper and equal distribution of salt and provide for adequate salt absorption in the curd. 58.418(c) |
| C-18 | Reuse of single service press cloths is prohibited. 58.421 |
| C-19 | Cheese wax is kept clean. Paraffin tanks are of adequate size, fitted with wooden racks, and have heat controls and an indicating thermometer. 58.427 |
| C-20 | Hydrogen peroxide, catalase, cheese cultures, calcium chloride, and other authorized ingredients comply with requirements. 58.431 through 58.437; 133 |
| C-21 | Enzyme preparations used in the manufacture of cheese shall be safe and suitable. 58.436 |
| C-22 | Cheese from unpasteurized milk is cured for at least 60 days at a temperature of no less than 35°F. When held more than 2 hours from between the time of receipt and setting, the milk must be cooled to 45°F or lower until time of setting. 58.439 |
| C-23 | Laboratory and quality control tests are conducted for each vat of finished cheese. Appropriate methods for the chemical analysis are utilized. 58.442 |
| C-24 | Based on the variety of products produced, the stated quality, identity, and analytical requirements of 7 CFR 58 are met. |
| C-25 | Milkfat from whey is not more than four days old at time of shipping. 58.443(c) |
| C-26 | Nonfat dry milk and whey shall be of USDA Extra grade (or equivalent) except for moisture (Processed cheese). 58.716, and 58.717 |
| C-27 | Color additives shall be Annatto or any other cheese color approved by U.S. FDA. 58.329, 58.435, and 58.719 |
| C-28 | Hydrogenated vegetable fats or oils used as coating of the rind. Several sections of 21 CFR 133 |



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# PASTEURIZED DAIRY PRODUCTS AND FROZEN DESSERTS

## D.1 SCOPE

D.1.1 Scope. This appendix establishes the food safety and related requirements for pasteurized milk establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## D.2 APPLICABLE DOCUMENTS

D.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

D.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 7, Part 58 | - | Grading and Inspection, General Specifications for Dairy Plants Approved USDA Inspection and Grading Service. |
| Title 21, Part 131 | - | Milk and Cream |
| Title 21, Part 135 | - | Frozen Desserts |
| Title 40, Part 180 | - | Tolerance exemptions for active and inert Ingredients for use in antimicrobial formulations (Food-contact surface sanitizing solutions) |

(Copies of these are available online at: https://www.govinfo.gov/.)



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U.S. DEPARTMENT OF AGRICULTURE (USDA) AGRICULTURAL MARKETING SERVICE (AMS)

Dairy Plants Surveyed and Approved for USDA Grading Service.

(Available on-line at: https://apps.ams.usda.gov/dairy/ApprovedPlantList/.)

USDA Guidelines for the Sanitary Design and Fabrication of Dairy Processing Equipment

(Available online at: https://www.ams.usda.gov/sites/default/files/media/DairyEquipmentReviewGuidelines.pdf.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

U.S. Public Health Service (USPHS)/Food and Drug Administration (FDA) Pasteurized Milk Ordinance (PMO), current version.

(Copies of this document are available online at: https://www.fda.gov/food/guidance-documents-regulatory-information-topic-food-and-dietary-supplements/milk-guidance-documents-regulatory-information.)

U.S. Food and Drug Administration (FDA), Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers, Interstate Milk Shippers List

(Available on-line at: https://www.fda.gov/Food/GuidanceRegulation/FederalStateFoodPrograms/ucm2007965.htm.)

FDA CFSAN, Hazards and Controls Guide for Dairy Foods HACCP, Guidance for Processors.

(Available on-line at: http://ncims.org/wp-content/uploads/2015/12/Hazard-GuideJune-16-2006-revised-October-2007.doc.)

FDA Milk Pasteurization Controls and Tests Handbook.

(Available from FDA State Training Branch, or online at: https://wifoodprotection.org/wp-content/uploads/2021/07/Milk_Pasteurization_Training_Manual_8th_Edition.pdf.)

FDA Dairy Farm Sanitation & Inspection Handbook.

FDA Milk Plant Sanitation & Inspection Handbook.

(Publications are available from FDA State Training Branch.)



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FDA, Milk and Milk Product Equipment, A Guideline for Evaluating Construction.

(Available online at: https://agriculture.ny.gov/system/files/documents/2021/05/milkproductequipment.pdf.)

U.S. ENVIRONMENTAL PROTECTION AGENCY

Selected EPA Registered Disinfectants

(Available on-line at: https://www.epa.gov/pesticide-registration/selected-epa-registered-disinfectants.)

D.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein.

3-A SANITARY STANDARDS INC

3-A Standards and Practices

(Available on-line: at http://www.3-a.org/.)

D.3 DEFINITIONS

D.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all references and associated documents apply to this appendix.

D.4 CHECKLIST

D.4.1 Checklist. Guidelines for the auditing of shell egg establishments are contained in the following Table D-I to D-V checklists.



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TABLE D-I. Pasteurized Milk Products and Frozen Desserts General Requirements as Specified in the USPHS/FDA Pasteurized Milk Ordinance (PMO).

| ITEM | REQUIREMENT |
|---|---|
| D-1 | Milk originates from farms that meet the requirements or intent of the PMO and herds that meet the animal health requirements as specified in Section 8 (Animal Health) of the PMO or equivalent program as determined by the DHA policy. PMO, Sec. 8 |
| D-2 | Milking is done in a milking barn, stable or parlor. Brushing is completed prior to milking. Flanks, bellies, tails and udders are clipped as often as necessary to facilitate cleaning of these areas and are free from dirt. Udders and teats of all milking animals are clean and dry before milking. Provided that the sanitizing of teats shall not be required if the udder is dry and the teats have been thoroughly cleaned (not dry wiped) and dried (manually wiped dry) prior to milking. Sec. 7, Item 13r, 1 through 4 |
| D-3 | Milk from lactating animals being treated with medicinal agents, which are capable of being secreted in the milk, is not offered for sale for such a period as is recommended by the attending veterinarian or as indicated on the package label of the medicinal agent. Sec. 7, Item 1r, 1 |
| D-4 | Milk from lactating animals treated with or exposed to insecticides, not approved for use on dairy animals by the EPA is not offered for sale. Sec. 7, Item 1r, 2 |
| D-5 | Bloody, stringy, off-colored milk, or milk that is abnormal to sight or odor, is so handled and disposed of as to preclude the infection of other lactating animals and the contamination of milk utensils. Sec. 7, Item 1r, 4 |
| D-6 | Lactating animals secreting milk with abnormalities are milked last or in separate equipment, which effectively prevents the contamination of the wholesome supply. Sec. 7, Item 1r, 5 |
| D-7 | Equipment, utensils and containers used for the handling of milk with abnormalities are not used for the handling of milk to be offered for sale, unless they are first cleaned and effectively sanitized. Sec. 7, Item 1r, 6 |
| D-8 | Processed animal waste derivatives, used as a feed ingredient for any portion of the total ration of the lactating dairy animal shall not contain levels of deleterious substances, harmful pathogenic organisms or other toxic substances, which are secreted in the milk at any level, which may be deleterious to human health. Sec. 7, Item 1r, 7 |
| D-9 | Animal drugs shall be properly labeled, to include directions for use and prescribed withholding times, active ingredient(s), name and address of the manufacturer or distributor for over-the-counter drugs, or veterinary practitioner dispensing the product for prescription (Rx) and extra label use drugs. Sec. 7, Item 15r, 4 and 5 |
| D-10 | Drugs intended for the treatment of non-lactating dairy animals are segregated from those drugs used for lactating dairy animals. Unapproved and/or improperly labeled drugs are not used to treat dairy animals and are not stored in the milk-house, milking barn, stable or parlor. Sec. 7, Item 15r, 3 and 6 |
| D-11 | Drugs are stored in such a manner that they cannot contaminate the milk or milk product- contact surfaces of the containers, utensils or equipment. Sec. 7, Item 15r, 7 |



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TABLE D-I. Pasteurized Milk Products and Frozen Desserts General Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-12 | Raw milk for pasteurization, ultra-pasteurization, aseptic processing and packaging or retort processed after packaging shall be cooled to 10ºC (50ºF) or less within four (4) hours or less, of the commencement of the first milking, and to 7ºC (45ºF) or less, within two (2) hours after the completion of milking. Provided, that the blend temperature after the first milking and subsequent milkings does not exceed 10ºC (50ºF). Sec. 7, Item 18r, 1 |
| D-13 | Recirculated cooling water, which is used in plate or tubular coolers and/or heat exchangers, including those systems in which a freezing point depressant is used, is from a safe source, protected from contamination, and tested semiannually to comply with the Bacteriological Standards for reclaimed and recirculated water in Appendix G of the PMO. Sec. 7, Item 18r, 2 |
| D-14 | Freezing point depressants and other chemical additives, when used in recirculating systems, shall be non-toxic under conditions of use. Propylene glycol and all additives shall be either USP Grade, Food Grade or generally-recognized-as-safe (GRAS). Sec. 7, Item 18r, 3 |
| D-15 | In milk-houses, an accurate, accessible temperature-recording device shall be installed in the milk line downstream from an effective cooling device, which cools the milk to 7°C (45°F) or less. Sec. 7, Item 5r |
| D-16 | An indicating thermometer shall be installed as close as possible to the recording device. This thermometer shall be used to check the temperature-recording device and the results recorded on the recording record or into the electronic data collection, storage and reporting system. Sec. 7, Item 5r |
| D-17 | Temperature-recording charts or electronic storage of required temperature records shall be maintained on the premises for a period of a minimum of six (6) months and readily available for review. Sec. 7, Item 5r |
| D-18 | After the completion of milk transfer, the milk lines and transfer hose shall be properly CIP cleaned. Sec. 7, Item 5r |
| D-19 | The sanitary piping outside the milk-house shall be capped at all times, except when transferring milk or being CIP cleaned. When the caps are not being used, they shall be properly cleaned and sanitized after each use and stored in the milk-house to protect them from contamination. Sec. 7, Item 5r |
| D-20 | All multi-use containers, utensils and equipment, which are exposed to milk or milk products, or from which liquids may drip, drain or be drawn into milk or milk products, are made of smooth impervious, nonabsorbent, safe materials of the following types: (a) Stainless steel of the AISI 300 series; or (b) Equally corrosion-resistant metal which is non-toxic and nonabsorbent; or (c) Heat resistant glass; or (d) Plastic, or rubber and rubber-like materials which are relatively inert and resistant to decomposition under |



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TABLE D-I. Pasteurized Milk Products and Frozen Desserts General Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-20 cont. | normal use conditions, are non-toxic, fat resistant, relatively nonabsorbent, which do not impart flavor or odor to the product, and which maintain their original properties under repeated use conditions. Sec. 7, Item 9r, 1 |
| D-21 | Milking systems shall not have components incorporated in the return solution lines, which by design do not comply with the criteria for product-contact surfaces. Some examples of these are: (a) Ball type plastic valves; (b). Plastic tees with barbed ridges to better grip the plastic or rubber hoses; and (c). The use of polyvinyl chloride (PVC) water type piping for return solution lines. Sec. 7, Item 9r, 9 |
| D-22 | All milking machines, including heads, milk claws, milk tubing and other milk-contact surfaces can be easily cleaned and inspected. Sec. 7, Item 9r, 9 |
| D-23 | All containers, utensils and equipment are free of breaks and corrosion. All joints in such containers, utensils and equipment are smooth and free from pits, cracks or inclusions. Sec. 7, Item 9r, 4 and 5 |
| D-24 | CIP cleaned milk pipelines and return-solution lines are self-draining. Sec. 7, Item 9r, 6 |
| D-25 | Detailed plans for CIP cleaned pipeline systems have written approval prior to installation. No alteration or addition shall be made without approval. Sec. 7, Item 9r, 7 |
| D-26 | The product-contact surface of all multi-use containers, equipment and utensils used in the handling, storage or transportation of milk are cleaned after each milking or once every twenty-four (24) hours for continuous operations. Sec. 7, Item 10r, 2 |
| D-27 | All product-contact surfaces of multi-use containers, utensils and equipment used in the handling, storage or transportation of milk are sanitized before each usage by using 77ºC (170ºF) water for at least 5 minutes or chemical compounds effective for the sanitization of milk contact surfaces as described in 40 CFR 180.940, or by any method which has been demonstrated to be equally effective. Sec. 7, Item 11r |
| D-28 | All milk containers, utensils and equipment, including milking machine vacuum hoses, are stored in the milk-house in a sanitizing solution, or on racks, until used. Pipeline equipment which is designed for CIP cleaning may be CIP cleaned, sanitized and stored in the milking barn or parlor, provided this equipment is designed, installed and operated to protect the product and solution contact surfaces from contamination at all times. Sec. 7, Item 12r, 1 |
| D-29 | Means are provided to effect complete drainage of equipment when such equipment cannot be stored to drain freely. Sec. 7, Item 12r, 3 |
| D-30 | All product-contact surfaces of containers, utensils and equipment are covered or otherwise protected to prevent the access of insects, dust, condensation and other contamination. All openings, including valves and piping attached to milk storage tanks and milk tank trucks, pumps or vats, shall be capped or otherwise properly protected. Sec. 7, Item 14r, 4 |



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TABLE D-I. Pasteurized Milk Products and Frozen Desserts General Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-31 | During milking and milk-house operations, pipelines and equipment, used to contain or conduct milk, shall be effectively separated from tanks/silos and/or circuits containing cleaning and/or sanitizing solutions. Sec. 7, Item 14r, 2 |
| D-32 | Whenever air under pressure is used for the agitation or movement of milk, or is directed at a milk-contact surface, it is free of oil, dust, rust, excessive moisture, extraneous materials and odor. Sec. 7, Item 14r, 8 |
| D-33 | Sanitized product-contact surfaces, including bulk milk tank openings and outlets, are protected against contact with unsanitized utensils and equipment, hands, clothing, splash, condensation and other sources of contamination. Sec. 7, Item 14r, 9 |
| D-34 | Any sanitized product-contact surface, which has been otherwise exposed to contamination, is again cleaned and sanitized before being used. Sec. 7, Item 14r, 10 |
| D-35 | There shall not be any partial removal of milk from milk storage/holding tanks by the bulk milk hauler/sampler, except partial pickups may be permitted when the milk storage/holding tank is equipped with a seven (7) day recording device, provided the milk storage/holding tank shall be clean and sanitized when empty and shall be emptied at least every seventy-two (72) hours. Sec. 7, Item 10r, 3 |
| D-36 | Means shall be provided to sanitize the milk-contact surfaces of the transfer hose and bulk milk pickup tanker fittings prior to the connection of the transfer hose to the bulk milk pickup tanker. Sec. 7, Item 5r |
| D-37 | At all times, the bulk milk pickup tanker manhole openings(s) shall remain closed, except for brief periods for sampling and examination when environmental conditions permit. Sec. 7, Item 5r |
| D-38 | The milk shall be sampled in a manner so as to preclude contaminating the milk tank truck or sample, by a permitted milk sample collector. Sec. 7, Item 5r |
| D-39 | The milk tank truck shall be effectively agitated in order to collect a representative sample. Sec. 7, Item 5r |
| D-40 | The milk tank truck shall be washed and sanitized at the permitted milk plant, receiving station or transfer station receiving the milk or at a permitted milk tank truck cleaning facility. Sec. 7, Item 5r |
| D-41 | Pasteurized milk and milk products are not strained or filtered, except through a perforated metal strainer. Provided, that pasteurized milk and milk products that are concentrated (condensed) in membrane processing systems may be filtered provided that a single-service in-line filter that is sanitized after assembly, may be allowed if it is a part of the membrane processing system. Sec. 7, Item 15p.(A).10 |
| D-42 | Except as permitted in properly designed and operated pasteurizers, there shall be no physical connection between unpasteurized products, dairy, non-dairy, or water, and pasteurized milk or milk products. Sec. 7, Item 15p.(B).2 |
| D-43 | All milk and/or milk products that have overflowed, leaked, been spilled or improperly handled are discarded. Sec. 7, Item 15p.(B).4 |



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TABLE D-I. Pasteurized Milk Products and Frozen Desserts General Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-44 | In no case shall pasteurized milk or milk products, be standardized with unpasteurized milk or milk products, unless the standardized milk or milk product is subsequently pasteurized. Sec 7, Item 15p.(B).8 |
| D-45 | Reconstituted or recombined milk and milk products shall be pasteurized after reconstitution or recombining of all ingredients. Sec. 7, Item 15p.(B).9 |
| D-46 | Pasteurization shall be performed in equipment, which is properly designed and operated and which ensures that every particle of milk or milk product will be held continuously at the proper temperature for the specified period of time. Sec. 7, Item 16p |
| D-47 | Instruments and devices in pasteurizers that have been identified as public health controls shall be tested upon installation; at least once each three (3) months thereafter; and whenever any alteration or replacement is made which may affect the proper operation of the instrument or device. Provided, that the pasteurization holding time tests shall be conducted at least once each six (6) months thereafter. Item 16p.(D).2 |
| D-48 | In the case of milk plants with approved HACCP plans, during an audit, the auditor may conduct any or all of the Pasteurization Equipment Tests. The auditor shall, through a combination of the physical examination of the pasteurization equipment and a records review, satisfy themselves that the pasteurization equipment is properly installed and operated. Item 16p.(D).2.HACCP.f |
| D-49 | All raw milk and milk products shall be maintained at 7ºC (45ºF) or less until processed, except that acid-type whey with a titratable acidity of 0.40% or above, or a pH of 4.6 or below, is exempted from these temperature requirements. Item 17p, Admin, 1 |
| D-50 | All whey and whey products for condensing and/or drying are maintained at a temperature of 7ºC (45ºF) or less; or 57ºC (135ºF) or greater until processed. Item 17p, Admin, 2 |
| D-51 | Storage tanks containing whey and whey product above 7°C (45°F) and below 57ºC (135ºF) shall be emptied, cleaned and sanitized after each four (4) hours of use or less. Other time and temperature relationships, which have been recognized to be equally efficient and approved are also acceptable. Item 17p, Admin, 2 |
| D-52 | Tanks and/or vessels used to blend and hold milk or milk product flavoring slurry shall be completely emptied and cleaned after each four (4) hours of operation or less, unless the slurry is stored at a temperature of 7ºC (45ºF) or less, or at a temperature of 66ºC (150ºF) or greater and maintained thereat. Item 17p, Admin, 3 |



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TABLE D-I. Pasteurized Milk Products and Frozen Desserts General Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-53 | All pasteurized products, except the following, are cooled immediately in approved equipment prior to filling or packaging to a temperature of 7ºC (45ºF) or less, unless drying is commenced immediately after condensing: (1) Those to be cultured; (2) Cultured sour cream with a pH of 4.70 or below; (3) Acidified sour cream with a pH of 4.60 or below; (4) All yogurt products with an initial pH of 4.80 or below at filling; (5) Cultured buttermilk with a pH of 4.60 or below; (6) Cultured cottage cheese with a pH of 5.2 or below and meeting other critical factors. Item 17p, Admin, 4 |
| D-54 | All condensed whey and whey products shall be cooled during the crystallization process to 10°C (50°F) or less within seventy-two (72) hours of condensing, including the filling and emptying time, unless filling occurs above 57ºC (135ºF), in which case, the seventy-two (72) hour time period begins when cooling is started. Item 17p, Admin, 4 |
| D-55 | All pasteurized milk and milk products, except the following, shall be stored at a temperature of 7°C (45ºF) or less and be maintained thereat following filling or until further processed: (1) Cultured sour cream with a pH of 4.70 or below and cooled to 7°C (45°F) or less within one hundred sixty eight (168) hours of filling; (2) Acidified sour cream with a pH of 4.60 or below and cooled to 7°C (45°F) or less within one hundred sixty eight (168) hours of filling; (3) All yogurt products with an initial pH of 4.80 or below at filling, with a pH of 4.60 or below within twenty-four (24) hours of filling and cooled to 7°C (45°F) or less within ninety-six (96) hours of filling; (4) Cultured buttermilk with a pH of 4.60 or below and cooled to 7°C (45°F) or less within twenty-four (24) hours of filling; (5) Cultured cottage cheese with a pH of 5.2 or below and in conformance with other critical factors and specified cooling periods. Item 17p, Admin, 5 |
| D-56 | All pasteurized milk products to be condensed and/or dried, shall be stored at a temperature of 10ºC (50ºF) or less and be maintained thereat until further processed. If storage tanks are used between the condenser and dryer, any such storage tank(s) containing pasteurized milk products stored above 10ºC (50ºF) and below 57ºC (135ºF) shall be completely emptied and cleaned after each six (6) hours of operation or less. Item 17p, Admin, 6 |
| D-57 | Each storage tank shall be equipped with an indicating thermometer, the sensor of which shall be located to permit the registering of the temperature of the contents when the tank contains no more than twenty percent (20%) of its calibrated capacity. Item 17p, Admin, 8 |
| D-58 | All milk products, including concentrated (condensed) milk products, are bottled and packaged at the plant where final pasteurization is performed without undue delay following final pasteurization. Item 18p, 1 |



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TABLE D-II. Vat Pasteurizer Requirements as Specified in the PMO.

| ITEM | REQUIREMENT |
|---|---|
| D-59 | The pasteurizer is designed so that the simultaneous temperature difference between the warmest and coldest product in the vat will not exceed 0.5ºC (1ºF) at any time during the holding period. Sec. 7, Item 16p.(A) |
| D-60 | Product should cover a sufficient area of the agitator to ensure adequate agitation throughout the holding period. Sec. 7, Item 16p.(A) |
| D-61 | Batch pasteurizers shall be equipped with an indicating and recording thermometer and product shall cover the bulbs of both before pasteurization starts. Sec. 7, Item 16p.(A) |
| D-62 | The milk plant operator shall check the temperature shown by the recording thermometer against the temperature shown by the indicating thermometer at the start of the holding period. The comparison shall be noted in the recording thermometer chart. Sec. 7, Item 16p.(A) |
| D-63 | The recording thermometer shall not read higher than the indicating thermometer. Sec. 7, Item 16p.(A) |
| D-64 | The thermometers shall not read less than the required pasteurization temperature throughout the required holding period. Sec. 7, Item 16p.(A) |
| D-65 | Means shall be provided to keep the atmosphere above the product at a temperature no less than 3ºC (5ºF) higher than the minimum required pasteurization temperature, throughout the holding period. Sec. 7, Item 16p.(A) |
| D-66 | Batch pasteurizers shall be equipped with an airspace thermometer. The surface of the product shall be at least 25 mm (1 inch) below the bottom of the thermometer bulb. Sec. 7, Item 16p.(A) |
| D-67 | The milk plant operator shall check the temperature shown by the airspace thermometer and recorded in the recording thermometer chart at the start and at the end of the holding period. Sec. 7, Item 16p.(A) |
| D-68 | The use of the recording thermometer chart shall not exceed the time limit for which it is designed. Overlapping of recorded data is a violation of this item. Sec. 7, Item 16p.(D).1 |
| D-69 | Recording thermometer charts shall include: (1) Date; (2) Number or location of recording thermometer when more than one is used; (3) A continuous record of the product temperature; (4) Extent of holding period, including filling and emptying times when required; (5) Reading of the airspace thermometer, at the start and end of the holding period; (6) Reading of the indicating thermometer, at the start of the holding period; (7) Amount and name of the pasteurized product, represented by each batch or run on the chart; (8) Record of unusual occurrences; (9) Signature or initials of the operator; and (10) Name of the milk plant. Sec. 7, Item 16p.(D).1.a |
| D-70 | The time accuracy of the recording thermometer shall be determined quarterly. Sec. 7, Item 16p.(D).1.a |
| D-71 | Batch pasteurizers shall be operated so that every particle of product will be held at no less than the minimum pasteurization temperature continuously for at least the minimum required holding time. Sec. 7, Item 16p.(A) |



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TABLE D-II. Vat Pasteurizer Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-72 | When product is preheated to pasteurization temperature before entering the vat, the recording chart shall show a holding period of at least the minimum pasteurization time, at no less than the minimum pasteurization temperature, plus the time of filling from the level of the recording thermometer bulb. Sec. 7, Item 16p.(A) |
| D-73 | When the cooling is begun in the batch pasteurizer, after opening the outlet valve, or is done entirely outside the batch pasteurizer, the recording chart shall show at least the minimum required holding time, at no less than the minimum pasteurization temperature, plus the time of emptying to the level of the recording thermometer bulb. Sec. 7, Item 16p.(A) |
| D-74 | When the recording time interval on the recording chart at the pasteurization temperature includes filling and/or emptying time, such intervals shall be indicated on the recording chart, by the operator, by inscribing the holding time on the recording chart. Sec. 7, Item 16p.(A) |
| D-75 | No product shall be added to the batch pasteurizer after the start of the holding period. Sec. 7, Item 16p.(A) |
| D-76 | All sanitary piping, fittings and connections which are exposed to products or from which liquids may drip, drain or be drawn into products shall consist of smooth, impervious, corrosion-resistant, non-toxic, easily cleanable material, which is approved for milk product-contact surfaces. All piping shall be in good repair. Sec. 7, Item 10p |
| D-77 | All sanitary piping, connections and fittings shall consist of: (a) Stainless steel of the AISI 300 series; or (b) Equally corrosion-resistant metal which is non-toxic and nonabsorbent; or (c) Heat resistant glass. Plastic, or rubber and rubber-like materials which are relatively inert and resistant to decomposition under normal use conditions, are non-toxic, fat resistant, relatively nonabsorbent, which do not impart flavor or odor to the product, and which maintain their original properties under repeated use conditions, may be used for gaskets, sealing applications and for short flexible takedown jumpers or connections where flexibility is required for essential functional reasons. Sec. 7, Item 10p |
| D-78 | Pasteurized milk products shall be conducted from one piece of equipment to another only through sanitary piping. Sec. 7, Item 10p |
| D-79 | Sanitary piping, fittings and connections are designed to permit easy cleaning; kept in good repair; free of breaks or corrosion; and contain no dead ends of piping in which product may collect. Sec. 7, Item 10p |
| D-80 | All interior surfaces of demountable piping, including valves, fittings and connections are designed, constructed and installed to permit inspection and drainage. Sec. 7, Item 10p |
| D-81 | All CIP cleaned product pipelines and return-solution lines are rigid, self-draining and so supported to maintain uniform slope and alignment. Sec. 7, Item 10p |
| D-82 | If gaskets are used, they shall be self-positioning and applied to form a smooth, flush interior surface. Sec. 7, Item 10p |



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TABLE D-II. Vat Pasteurizer Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-83 | If gaskets are not used, all fittings shall have self-positioning faces designed to form a smooth, flush interior surface. Sec. 7, Item 10p |
| D-84 | All interior surfaces of welded joints in pipelines shall be smooth and free from pits, cracks or inclusions. In the case of welded lines, all welds shall be inspected as they are made. Sec. 7, Item 10p |
| D-85 | Each cleaning circuit shall have access points located at sufficient intervals that are adequate to determine the general condition of the interior surfaces of the pipeline. Sec. 7, Item 10p |
| D-86 | All pipelines and fittings shall be so constructed and located that leakage shall not occur. Leakage is defined as the entrance of unpasteurized product into a batch pasteurizer during the holding or emptying period, or entrance of unpasteurized product into any pasteurized product line at any time. Sec. 7, Item 16p.(A) |
| D-87 | To prevent clogging, and to promote drainage, all leak-protection grooves in plug-type outlet valves shall be at least 5 mm wide and at least 2.3 mm deep at the center. Mating grooves shall provide these dimensions throughout their combined length, whenever the valve is in, or approximately in, the fully closed position. Sec. 7, Item 16p.(A) |
| D-88 | All single leak grooves, and all mating leak grooves when mated, shall extend throughout the entire depth of the seat, so as to divert leakage occurring at all points throughout the depth of the seat and so as to prevent air binding. Washers or other parts shall not obstruct leak-protector grooves. Sec. 7, Item 16p.(A) |
| D-89 | A stop shall be provided on all plug-type outlet valves in order to guide the operator in closing the valve so that unpasteurized product may not inadvertently be permitted to enter the outlet line. Sec. 7, Item 16p.(A) |
| D-90 | The stop shall be so designed that the plug will be irreversible when the plug is provided with any grooves or their equivalent, unless duplicate, diametrically opposite grooves are also provided. Sec. 7, Item 16p.(A) |
| D-91 | Stops shall be so designed that the operator cannot turn the valve beyond the stop position, either by raising the plug or by any other means. Sec. 7, Item 16p.(A) |
| D-92 | Outlet valves, in addition to previous requirements, shall be so designed as to prevent the accumulation of unpasteurized product in the product passages of the valve in any closed position. Sec. 7, Item 16p.(A) |
| D-93 | All outlets from vat pasteurizers shall be equipped with close-coupled leak-protector valves or be otherwise similarly protected during filling, holding and emptying periods. Sec. 7, Item 16p.(A) |
| D-94 | All leak-protector grooved outlet valves shall be installed in the proper position to ensure the function of the leak-protector groves and the drainage of the leak-detector valve. Sec. 7, Item 16p.(A) |
| D-95 | All outlet valves shall be kept fully closed during filling, heating, and holding periods. Sec. 7, Item 16p.(A) |



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TABLE D-II. Vat Pasteurizer Requirements as Specified in the PMO – Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-96 | Close-coupled vat pasteurizer outlet valve bodies and plugs shall be made of stainless steel or of other materials that have heat transfer properties at least equal to stainless steel. Sec. 7, Item 16p.(A) |
| D-97 | All inlet pipelines are disconnected during the holding and emptying periods, and all outlet pipelines are disconnected during the filling and holding periods. Sec. 7, Item 16p.(A) |

TABLE D-III. HTST and HHST Continuous Flow Pasteurizer Requirements as Specified in the PMO.

| ITEM | REQUIREMENT |
|---|---|
| D-98 | Each continuous-flow pasteurization system shall be equipped with an automatic product flow diversion device (FDD), which complies with the specifications and performance requirements in the PMO. Item 16p.(B).2 |
| D-99 | The FDD shall automatically cause diversion of product in response to a sub-legal pasteurization temperature. Item 16p.(B).2.a |
| D-100 | At sub-legal temperatures, FDDs return product to the raw product side of the heating system continuously until legal pasteurization temperatures are obtained, at which time, the device restores forward-flow through the pasteurizer. Item 16p.(B).2.a |
| D-101 | When the recording/controlling thermometer temperature is sub-legal and the FDD is not in the fully diverted position, as signaled by the position detection devices, all motive pumps capable of causing flow through the FDD shall be de-energized, or any other equally satisfying means of preventing forward flow of product. Item 16p.(B).2.b.(1) |
| D-102 | When a packing gland is used to prevent leakage around the actuating FDD stem, it shall be impossible to tighten the stem-packing nut to such an extent as to prevent the valve from assuming the fully diverted position. Item 16p.(B).2.b.(2) |
| D-103 | A leak-escape shall be installed on the forward-flow side of the FDD valve seat. However, when backpressure is exerted on the forward-flow side of the valve seat, while the milk or milk product-flow is being diverted, the leak-escape should lie between two valve seats or between two portions of the same seat, one upstream and the other downstream from the leak-escape. Item 16p.(B).2.b.(3) |
| D-104 | The leak-escape shall be designed and installed to discharge all leakage to the outside, or to the constant-level tank through a line separate from the diversion line. Provided, that when leakage is discharged to the constant-level tank, a sight glass shall be installed in the leak-escape line to provide a visual means of leak detection. Item 16p.(B).2.b.(3) |



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TABLE D-III. HTST and HHST Continuous Flow Pasteurizer Requirements as Specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-105 | The closure of the forward-flow seat shall be sufficiently tight so that leakage past it shall not exceed the capacity of the leak-escape device, as evidenced when the forward-flow line is disconnected; and, in order that proper seating may not be disturbed, the length of the connecting rod shall not be adjustable by the user. Item 16p.(B).2.b.(4) |
| D-106 | The FDD shall be so designed and installed that failure of the primary motivating power shall automatically divert the flow of milk or milk product. Item 16p.(B).2.b.(5) |
| D-107 | The FDD shall be located downstream from the holding tube. Item 16p.(B).2.b.(6) |
| D-108 | The recording/controlling thermometer shall be located in the product holding tube not more than 46 centimeters (18 inches) upstream from the FDD. Unless a sufficient delay is in place to prevent forward-flow after product temperature becomes legal following divert-flow. Item 16p.(B).2.b.(6) |
| D-109 | The use of the recording thermometer chart shall not exceed the time limit for which it is designed. Overlapping of recorded data is a violation of this item. Sec. 7, Item 16p.(D).1 |
| D-110 | Recording/controlling thermometer charts shall include: (1) Date; (2) Number or location of recording thermometer when more than one is used; (3) A continuous record of the product temperature; (4) Reading of indicating thermometer, at the start of the holding period, at a given time or reference point as indicated on the chart, and immediately after a chart has been changed; (5) The cut-in and cut-out product temperatures, recorded daily by the operator, at the beginning of the run (HTST only); (6) A record of the time during which the FDD is in the forward-flow position; (7) Amount and name of the pasteurized product, represented by each run on the chart; (8) Record of unusual occurrences; (9) Signature or initials of the operator; and (10) Name of the milk plant. Sec. 7, Item 16p.(D).1.b |
| D-111 | The time accuracy of the recording thermometer shall be determined quarterly. Sec. 7, Item 16p.(D).1.b |
| D-112 | The pipeline from the diversion port of the FDD shall be self-draining and shall be free of restrictions or valves; unless such restrictions are noticeable and valves are so designed that stoppage of the diversion line cannot occur. Item 16p.(B).2.b.(8) |
| D-113 | When it is used, the pipeline from the leak-detector port of the FDD shall be self-draining and shall be free of restrictions or valves. Item 16p.(B).2.b.(9) |
| D-114 | For the timing pump, a one (1) second maximum “off” time delay is allowed to maintain the flow-promoting device in the “on” position through the travel time of the FDD. Item 16p.(B).2.b.(10) |
| D-115 | If the area between the divert and leak-detect valve seats is not self-draining when the FDD is in the diverted position, a delay of at least one (1) second and not more than five (5) seconds is required between the movement of the divert and leak-detect valves when the FDD assumes the forward-flow position. Except that, the delay may be longer than five (5) seconds if holding time is legal in divert-flow. Item 16p.(B).2.b.(11) |



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TABLE D-III. HTST and HHST Continuous Flow Pasteurizer Requirements as Specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-116 | The control system shall be set and sealed so that forward-flow of product cannot start unless the temperature at the recording/controlling thermometer is above the required pasteurization temperature for the product and the process used, nor continue during descending temperatures when the temperature is below the required pasteurization temperature. Item 16p.(B).2.c.(1) |
| D-117 | The system shall be so designed that no milk and/or milk product can be bypassed around the recording/controlling thermometer that shall not be removed from its proper position during the pasteurization process. Item 16p.(B).2.c.(1) |
| D-118 | The cut-in and cut-out product temperatures, as shown by the indicating thermometer, shall be determined at the beginning of each day's operation and entered upon the recorder chart daily by the milk plant operator. Item 16p.(B).2.c.(1) |
| D-119 | Manual switches for the control of pumps, homogenizers or other devices, which produce flow through the holder, shall be wired so that the circuit is completed only when product is above the required pasteurization temperature for product and the process used, or when the FDD is in the fully-diverted position. Item 16p.(B).2.c.(3) |
| D-120 | Holding tubes shall be designed to provide for the holding of every particle of product for at least the time required in the definition of Pasteurization for the product and the process used. Item 16p.(B).2.d.(1) |
| D-121 | The holding tube shall be designed so that the simultaneous temperature difference between the hottest and coldest product, in any cross section of flow, at any time during the holding period, will not be greater than 0.5ºC (1ºF). This requirement may be assumed to have been satisfied, without testing, in tubular holders of 17.8 centimeters (7 inches) or smaller diameter that are free of any fittings through which the product may not be thoroughly swept. Item 16p.(B).2.d.(2) |
| D-122 | No device shall be permitted for short-circuiting a portion of the holding tube to compensate for changes in rate of product-flow. Holding tubes shall be installed so that sections of pipe cannot be left out, resulting in a shortened holding time. Item 16p.(B).2.d.(3) |
| D-123 | The holding tube shall be arranged to have a continuously upward slope in the direction of flow of not less than 2.1 centimeters per meter (0.25 of an inch per foot). Item 16p.(B).2.d.(4) |
| D-124 | Supports for holding tubes shall be provided to maintain all parts of the holding tubes in a fixed position, free from any lateral or vertical movement. Item 16p.(B).2.d.(5) |
| D-125 | The holding tube shall be so designed that no portion between the inlet and the recorder-controller temperature sensor is heated. Item 16p.(B).2.d.(6) |
| D-126 | Since laminar flow, the fastest flowing particle travels twice as fast as the average flowing particle, can occur in the holding tube during pasteurization of high-viscosity milk or milk products, holding tube lengths shall be calculated as twice the length required to hold the average flow for the time standard. Item 16p.(B).2.d.(7) |



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TABLE D-III. HTST and HHST Continuous Flow Pasteurizer Requirements as Specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-127 | An indicating thermometer shall be located as near as practicable to the temperature sensor of the recording/controlling thermometer, but may be located a short distance upstream where product temperature does not differ significantly between the two. Item 16p.(B).2.e.(1) |
| D-128 | The temperature shown by the recording/controlling thermometer shall be checked daily by the plant operator against the temperature shown by the indicating thermometer. Readings shall be recorded on the recorder chart. The recorder/controller shall be adjusted to read no higher than the indicating thermometer. Item.16p.(B).2.e.(2) |
| D-129 | Pumps and other equipment which may produce flow through the holding tube shall be located upstream from the holding tube. Provided that pumps and other flow-promoting devices may be located downstream from the holding tube, if means are provided to eliminate negative pressure between the holding tube and the inlet to such equipment. Item 16p.(B).2.f.(1) |
| D-130 | The speed of pumps or other flow-promoting devices, governing the rate of flow through the holding tube, shall be so controlled as to ensure that the holding time cannot be shortened without detection. Item 16p.(B).2.f.(2) |
| D-131 | For those systems that utilize a homogenizer by-pass line to by-pass the homogenizer while processing product, the by-pass line shall be connected with valves that are so designed that both lines cannot be open at the same time. Item 16p.(B).2.f.(2) |
| D-132 | Tests for the holding time shall be made when all equipment and devices are operated and adjusted to provide for maximum flow. Item 16p.(B).2.f.(3) |
| D-133 | When a homogenizer is located upstream from the holding tube, the holding time shall be determined with the homogenizer in operation with no pressure on the homogenizer valves. Item 16p.(B).2.f.(3) |
| D-134 | For systems which utilize homogenizer by-pass lines, the holding time shall be tested in both flow patterns and the fastest time used. Item 16p.(B).2.f.(3) |
| D-135 | When vacuum equipment is located downstream from the holding tube, the holding time shall be tested with the timing pump operating at maximum flow and the vacuum equipment adjusted to provide for the maximum vacuum. Item 16p.(B).2.f.(3) |
| D-136 | The holding time shall be tested during both forward and diverted-flow. Item 16p.(B).2.f.(3) |
| D-137 | The holding time shall be tested in both forward-flow and diverted-flow initially; semiannually thereafter; and after any alteration or replacement that may affect the holding time. Item 16p.(B).2.f.(3) |
| D-138 | Product-to-product regenerators shall be constructed, installed and operated so that pasteurized product in the regenerator will automatically be under greater pressure than raw product in the regenerator at all times. Item 16p.(C).1 |



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TABLE D-III. HTST and HHST Continuous Flow Pasteurizer Requirements as Specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-139 | The pasteurized product, between its outlet from the regenerator and the nearest point downstream open to the atmosphere, shall rise to a vertical elevation of 30.5 centimeters (12 inches) above the highest raw product level, downstream from the constant-level tank, and shall be open to the atmosphere at this or a higher elevation. Item 16p.(C).2 |
| D-140 | The overflow of the top rim of the constant-level tank shall always be lower than the lowest product level in the regenerator. Item 16p.(C).3 |
| D-141 | No pump or flow-promoting device which can affect the proper pressure relationships within the regenerator shall be located between the pasteurized product outlet from the regenerator and the nearest downstream point open to the atmosphere. Item 16p.(C).4 |
| D-142 | No pump shall be located between the raw milk or milk product inlet to the regenerator and the constant-level tank, unless it is designed and installed to operate only when product is flowing through the pasteurized product side of the regenerator and when the pressure of the pasteurized product is at least 1 psi (0.069 bar) higher than the maximum pressure produced by the pump. Item 16p.(C).5 |
| D-143 | Pressure gauges shall be installed at the raw product inlet to the regenerator and the pasteurized product outlet of the regenerator or the outlet of the cooler. Item 16p.(C).5 |
| D-144 | The accuracy of these required pressure gauges shall be checked on installation; quarterly thereafter; and following repair or adjustment. Item 16p.(C).5 |
| D-145 | All raw product in the regenerator(s) shall automatically drain freely into the constant-level tank or to the floor when the raw milk or milk product pump(s) are shut down and the raw product connection(s) at the regenerator(s) is disconnected. Item 16p.(C).8 |
| D-146 | When vacuum equipment is located downstream from the FDD, means shall be provided to prevent the lowering of the pasteurized product level in the regenerator during periods of diverted-flow or shutdown. An effective vacuum breaker, plus an automatic means of preventing a negative pressure, shall be installed in the line between the vacuum chamber and the pasteurized product inlet to the regenerator. Item 16p.(C).9 |
| D-147 | Product-to-water-to-product regenerators shall be so designed, installed and operated that the heat-transfer-medium side of the regenerator, in the raw product section, will automatically be under greater pressure than the raw product side at all times (OPTION I). Item 16p.(C), Option I.1 |
| D-148 | The heat-transfer water shall be a safe water and the heat-transfer water shall be in a covered tank, which is open to the atmosphere at an elevation higher, by at least 30.5 centimeters (12 inches), than any raw product level downstream from the constant-level tank. Item 16p.(C), Option I.2 |
| D-149 | The heat-transfer water between its outlet from the regenerator and the nearest point downstream open to the atmosphere shall rise to a vertical elevation of at least 30.5 centimeters (12 inches) above any raw product in the system and shall be open to the atmosphere at this or a higher elevation. Item 16p.(C), Option I.2 |



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TABLE D-III. HTST and HHST Continuous Flow Pasteurizer Requirements as Specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-150 | The heat-transfer water circuit shall be full of water at the beginning of the run and all loss of water from the circuit shall be automatically and immediately replenished whenever raw product is present in the regenerator. Item 16p.(C), Option I.3 |
| D-151 | No pump shall be located between the raw product inlet to the regenerator and the constant-level tank, unless it is designed and installed to operate only when water is flowing through the heat-transfer section of the regenerator and when the pressure of the heat-transfer water is at least 1 psi (0.069 bar) higher than the pressure of the raw product. Item 16p.(C), Option I.5 |
| D-152 | Pressure gauges shall be installed at the raw product inlet to the regenerator and the pasteurized heat-transfer water outlet of the regenerator. Item 16p.(C), Option I.5 |
| D-153 | The accuracy of the required differential pressure gauges shall be checked on installation; quarterly thereafter; and following repair or replacement. Item 16p.(C), Option I.5 |
| D-154 | Product-to-water-to-product regenerators may also be constructed, installed and operated such that the pasteurized product in the regenerator will be under greater pressure than the heat-transfer-medium in the pasteurized product side of the regenerator (OPTION II). Item 16p.(C), Option II |
| D-155 | A differential pressure controller shall be used to monitor pressures of the pasteurized product and the heat-transfer-medium. Item 16p.(C), Option II.1 |
| D-156 | One pressure sensor shall be installed at the pasteurized product outlet of the regenerator and the other pressure sensor shall be installed at the heat-transfer-medium inlet of the pasteurized product side of the regenerator. Item 16p.(C), Option II.1 |
| D-157 | This differential pressure controller shall divert the FDD whenever the lowest pressure of pasteurized product in the regenerator fails to exceed the highest pressure of the heat-transfer-medium in the pasteurized product side of the regenerator by at least 1 psi (0.069 bar). Forward-flow of product shall be automatically prevented until all product-contact surfaces between the holding tube and the FDD have been held at or above the required pasteurization temperature continuously and simultaneously for at least the pasteurization time. Item 16p.(C), Option II.1 |
| D-158 | The heat-transfer-medium pump shall be wired so that it cannot operate unless the timing pump is in operation. Item 16p.(C), Option II.2 |
| D-159 | Pasteurized re-circulation lines, divert lines, and leak-detect lines connecting to the constant-level tank shall be designed so that there is an air gap between the termination of these pipelines and the raw product overflow level. This air gap shall be equivalent to at least two (2) times the diameter of the largest of these pipelines. An overflow is defined as the flood rim of the constant-level tank or any unrestricted opening below the flood rim of the constant-level tank which is large enough that it is at least equivalent to two (2) times the diameter of the largest of these pipelines. Item 15p.(B).3 |



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TABLE D-III. HTST and HHST Continuous Flow Pasteurizer Requirements as Specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-160 | Flow rate recording charts in magnetic flow meter based timing systems (MBTS) with a high flow alarm shall include: (1) Date; (2) Number or location of magnetic flow meter when more than one is used; (3) A continuous record of the product flow rate; (4) A continuous record of the status of the high and low-flow/loss of signal alarms; (5) Amount and name of the pasteurized product, represented by each batch or run on the chart; (6) Record of unusual occurrences; (7) Signature or initials of the operator; and (8) Name of the milk plant. Item 16p.(D).1.c |
| D-161 | Electronic collection, storage and reporting of required pasteurization records, with or without hard copy printouts, may be acceptable, provided, the electronically generated records are readily available for review and meet the same criteria as written records. Item 16p.(D).1.d |
| D-162 | A separator may be located between the outlet of a raw regenerator and the timing pump or between raw regenerator sections if the separator is automatically valved-out of the system, and separator stuffing pump(s) are de-energized, whenever booster pumps are required to de-energize. App. H, Separators, 1 |
| D-163 | A separator may not be located between the timing pump and the FDD. App. H, Separators, 2 |

TABLE D-IV. HTST and HHST Pasteurizers with the Flow Diversion Device (FDD) Downstream of the Pasteurized Regenerator as Specified in the PMO.

| ITEM | REQUIREMENT |
|---|---|
| D-164 | The FDD may be located downstream from the regenerator and/or cooler section, provided that the FDD shall be automatically prevented from assuming the forward-flow position until all product-contact surfaces between the holding tube and FDD have been held at or above the required pasteurization temperature continuously and simultaneously for at least the required pasteurization time. Item 16p.(B).2.b.(7) |
| D-165 | If the FDD is located downstream from the regenerator and/or cooler section and is inter-wired or is computer controlled to thoroughly clean the system, including the divert pipeline before the re-starting of production, a cooling section, which is not self-draining, may be present in the divert pipeline. Item 16p.(B).2.b.(8) |
| D-166 | No leak-detect flush time delay is required in these pasteurization if all forward-flow product-contact surfaces of the FDD are sanitized, or sterilized during the normal start-up process. Item 16p.(B).2.b.(11) |
| D-167 | Additional temperature controllers and timers shall be inter-wired with the control system, and the control system shall be set and sealed so that forward-flow of product cannot start until all product-contact surfaces between the holding tube and FDD have been held at or above the required pasteurization temperature, continuously and simultaneously for at least the required pasteurization time. Item 16p.(B).2.c.(2) |



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TABLE D-IV. HTST and HHST Pasteurizers with the FDD Downstream of the Pasteurized Regenerator as specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-168 | In the case of pasteurization systems, with the FDD located downstream from the regenerator and/or cooler section, the control system shall be set and sealed so that forward-flow cannot continue when the temperature of the product in the holding tube is below the required pasteurization temperature. Item 16p.(B).2.c.(2) |
| D-169 | The holding time for HHST systems shall be determined from the pumping rate rather than by the salt conductivity test, because of the short holding tube. Item 16p.(B).2.d.(7 |
| D-170 | With direct steam heating processes, the holding time is reduced because of the product volume increase as the steam condenses to water during the heating in the injector. The surplus water is evaporated as the product is cooled in the vacuum chamber. A volume increase of up to 12% will occur in the holding tube. Measuring the average flow rate before the injector or after the vacuum chamber will not reflect this volume increase. However, calculations shall consider this 12% holding time decrease. Item 16p.(B).2.d.(8) |
| D-171 | Systems capable of operating at 89ºC or higher must have a pressure limit controller or switch and shall cause the FDD to move into the divert-flow position if pressure in the holding tube drops below the boiling pressure of product at the maximum operating temperature plus a 10 psi (0.69 bar) margin of safety Item 16p.(B).2.d.(9) |
| D-172 | With steam injection processes, a differential pressure limit indicator across the injector is needed to keep the heated product in the liquid phase and to ensure adequate isolation of the injection chamber. The instrument shall have a differential pressure switch, so that the FDD will move to the divert-flow position if the pressure drop across the injector falls below 10 psi (0.69 bar). Item 16p.(B).2.d.(10) |
| mD-173 | When culinary steam is injected directly into product, as the means of terminal heating to achieve pasteurization temperature, the steam injector shall be designed, installed and operated so that (1) product and steam flows are isolated from pressure fluctuations inside the injection chamber, and (2) the process shall be as free as possible of non-condensable gases that may evolve from the product or be carried in the steam supply. Item 16p.(B).2.g |
| D-174 | The steam boiler shall be supplied with a de-aerator in order to keep product in the holding tube as free as possible of non-condensable gases. Item 16p.(B).2.g.(2) |
| D-175 | When culinary steam is introduced directly into the milk or milk product, automatic means shall be provided to maintain a proper temperature differential between incoming and outgoing milk or milk product to preclude dilution with water. Item 16p.(B).2.h.(2) |



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TABLE D-IV. HTST and HHST Pasteurizers with the FDD Downstream of the Pasteurized Regenerator as Specified in the PMO - Continued.

| ITEM | REQUIREMENT |
|---|---|
| D-176 | In the case of pasteurization systems, with the FDD located downstream from the re-generator and/or cooler section, the requirements D-139, -140, -144, and -145 may be eliminated. Provided, that a differential pressure controller is used to monitor the highest pressure in the raw product side of the regenerator and the lowest pressure in the pasteurized side of the regenerator, and the controller is interlocked with the FDD and is set so that whenever improper pressures occur in the regenerator, forward-flow of product is automatically prevented and shall not start again until all product-contact surfaces between the holding tube and FDD have been held at or above the required pasteurization temperature, continuously and simultaneously for at least the required pasteurization time. Item 16p.(C).10 |
| D-177 | When culinary steam is introduced directly into milk or milk product to achieve pasteurization temperature, and vacuum equipment is located downstream from the holding tube, the requirement that a vacuum breaker be installed at the inlet to the pasteurized side of the regenerator may be eliminated. Provided, that the differential pressure controller is installed and wired to control the FDD. Item 16p.(C).10 |
| D-178 | When the differential pressure controller is installed and wired to control the FDD, the raw milk or milk product booster pump may be permitted to run at all times. Provided, that the timing pump is in operation. Appendix H, paragraph 4, page 236. |

TABLE D-V. HHST Ultra-Pasteurizers as Specified in the PMO.

| ITEM | REQUIREMENT |
|---|---|
| D-179 | In the case of HHST pasteurizing systems utilizing temperatures and holding times that meet the definition of ultra-pasteurization (UP) with the FDD located downstream of the regenerator and/or cooler section the FDD may alternatively be a system of the "Steam-Block Type." This FDD system shall allow for the flow of water and/or product to the constant-level tank through appropriate valves and coolers during sterilization and when diverted Item 16p.(B).2.b.(12) |
| D-180 | Systems used for the processing of UP products shall meet all the public health control requirements for HHST systems. Item 16p.(B).2.c.(2) |
| D-181 | For systems used for the processing of UP products, the recorder-controller chart shall show that the UP product has been processed at a minimum temperature of 138ºC (280ºF) and has been verified to have a calculated holding time of at least two (2) seconds, or other time and temperature combinations of equivalent microbial inactivation. Item 16p.(B).2.c.(2) |



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# SHELL EGGS

## E.1 SCOPE

E.1.1 Scope. This appendix establishes food safety and related requirements for shell egg establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## E.2. APPLICABLE DOCUMENTS

E.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

E.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 7, Part 56 | - | Voluntary Grading of Shell Eggs |
| Title 7, Part 57 | - | Inspection of Eggs (Egg Products Inspection Act) |
| Title 21, Part 118 | - | Production, Storage, and Transportation of Shell Eggs |

(Copies of these documents are available online at: https://www.govinfo.gov/.)

U. S. DEPARTMENT OF AGRICULTURE, AGRICULTURAL MARKETING SERVICE

QAD 700 Shell Egg Graders Procedures.

(Available online at: https://www.ams.usda.gov/publications/content/qad-700-shell-egg-graders-handbook.)



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## E.3 DEFINITIONS

E.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

E.3.2 Shell oiling. The generic term of "shell oil" or "shell oiling" refers to any edible protective coating applied to the shell surface immediately following washing. If used, the substance shall be applied immediately after approved washing operations and shall be a product acceptable to the FDA.

## E.4 REQUIREMENTS

E.4.1 Flock health program. The flock health program includes environmental testing of the pullets (approximately 14 to 16 weeks of age) and laying hens (40 to 45 weeks of age) for *Salmonella enteritidis*. If the environmental testing is positive then the eggs shall either undergo a 5-log reduction treatment for the life of the flock, or test 1,000 intact eggs at 2-week intervals for a total of 4,000 eggs.

E.4.2 Removal of damaged eggs. Packaging plants shall provide effective methods of screening eggs so that damaged eggs are removed.

E.4.3 Wash water. Wash water shall be potable and should be added continuously to maintain a constant overflow rate. Chlorine or quaternary ammonium sanitizing compounds may be added to the replacement water provided they are compatible with the detergent. The iron content of the water may influence the growth of bacteria when the egg membrane is penetrated. Consequently, wash water may contain no more than 2 ppm of iron. USDA regulations require that wash water temperature be at 90ºF (32ºC) or higher, and shall be at least 20ºF warmer than the highest egg temperature (whichever is greater). These temperatures shall be maintained throughout the cleaning cycle. Wash water shall be changed every four hours, or more often if needed to maintain sanitary conditions. According to USDA regulations, the eggs cannot be immersed at any time.

E.4.4 Egg oiling. Oiling of the washed eggs is not required, but if performed shall adhere to 7 CFR 56.76.

## E.5 CHECKLIST

E.5.1 Checklist. Guidelines for the auditing of shell egg establishments are contained in the following Table E-I checklist.



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TABLE E-I. Shell Egg Requirements as Specified in 7 CFR Part 56 and 21 CFR Part 118.

| ITEM | REQUIREMENT |
|---|---|
| E-1 | The egg grading or candling area is adequately darkened to make possible the accurate quality determination of the candled appearance of eggs. There are no other light sources or reflections of light that interfere with, or prohibit accurate quality determination of eggs in the grading or candling area. Other light sources and equipment or facilities are provided to permit the detection and removal of stained or dirty eggs, or other under grade eggs. 56.76(c)(1) |
| E-2 | Cooler rooms holding shell eggs that are identified with a consumer grade shall be refrigerated and capable of maintaining an ambient temperature no greater than 45°F (7.2°C). 56.76(d) |
| E-3 | Storage facility shall be equipped with humidifying equipment capable of maintaining a relative humidity which will minimize shrinkage. Accurate thermometers and hygrometers are provided. 56.76(d) |
| E-4 | Eggs must be held and transported at or below 45°F ambient temperature beginning 36 hours after time of lay. If the eggs are to be processed as table eggs and are not processed for the ultimate consumer within 36 hours from the time of lay and, therefore, are held and transported as required at or below 45°F ambient temperature, then the eggs may be held at room temperature for no more than 36 hours just prior to processing to allow an equilibration step to temper the eggs. 118.4(e) |
| E-5 | Shell egg protecting operations shall be conducted in a manner to avoid contamination of the product and maximize conservation of its quality. 56.76(e) |
| E-6 | Component equipment within the shell egg protecting system must be sanitarily designed and maintained and provide an adequate amount of oil for shell coverage of the volume of eggs processed. 56.76(e) |
| E-7 | Eggs with excess moisture on the shell are not shell protected (oil processed). 56.76(e) |
| E-8 | Oil having any off odor, or that is obviously contaminated, is not used in shell egg protection. Oil is to be filtered prior to application. 56.76(e) |
| E-9 | The component equipment of the application system shall be washed, rinsed and treated with a bactericidal agent each time the oil is removed. 56.76(e) |
| E-10 | Adequate coverage and protection against dust and dirt shall be provided when the equipment is not in use. 56.76(e) |
| E-11 | Shell egg cleaning operations:<br>Shell egg washing equipment must be sanitarily designed, maintained in a clean and sanitary manner, and thoroughly cleaned at the end of each operating day. 56.76(f) |
| E-12 | Shell egg drying equipment must be sanitarily designed and maintained in a clean and sanitary manner. Air used for drying purposes must be filtered. These filters shall be cleaned or replaced as needed to maintain a sanitary process. 56.76(f) |
| E-13 | The temperature of the wash water must be 90°F (32.2°C) or higher, and will be at least 20°F warmer than the temperature of the eggs to be washed. Temperatures will be maintained throughout cleaning. 56.76(f) |



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TABLE E-I. Shell Egg Requirements as Specified in 7 CFR Part 56 and 21 CFR Part 118 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| E-14 | Approved cleaning compounds shall be used in the wash water. 56.76(f) |
| E-15 | Wash water shall be changed approximately every 4 hours or more often if needed to maintain sanitary conditions, and at the end of each shift. Remedial measures shall be taken to prevent excess foaming during the egg washing operation. 56.76(f) |
| E-16 | Replacement water shall be added continuously to the wash water of washers. Chlorine or quaternary sanitizing rinse water may be used as part of the replacement water, provided, they are compatible with the washing compound. Iodine sanitizing rinse water may not be used as part of the replacement water. 56.76(f) |
| E-17 | Only potable water may be used to wash eggs. An analysis of the iron content of the water supply, stated in parts per million, is performed. When the iron content exceeds 2 parts per million (ppm), equipment is provided to correct the excess iron content. If the water source is changed, new tests are performed. 56.76(f) |
| E-18 | Waste water from the egg washing operation shall be piped directly to the drains. 56.76(f) |
| E-19 | The washing, rinsing, and drying operations shall be continuous and shall be completed as rapidly as possible to maximize conservation of the egg's quality and to prevent sweating of eggs. Eggs shall not be allowed to stand or soak in water. Immersion-type washers shall not be used. 56.76(f) |
| E-20 | Pre-wetting shell eggs prior to washing may be accomplished by spraying a continuous flow of water over the eggs in a manner which permits the water to drain away. The temperature of the water shall be the same as prescribed above. 56.76(f) |
| E-21 | Washed eggs shall be spray-rinsed with water having a temperature equal to, or warmer than, the temperature of the wash water. The spray-rinse water shall contain a sanitizer that has been determined acceptable for intended use and of not less than 100 ppm nor more than 200 ppm of available chlorine or its equivalent. 56.76(f) |
| E-22 | Test kits shall be used to determine the strength of the sanitizing solution. 56.76(f) |
| E-23 | During non-processing periods, eggs shall be removed from the washing and rinsing area of the egg washer and from the scanning area whenever there is a buildup of heat that may diminish the quality of the egg. 56.76(f) |
| E-24 | Washed eggs shall be reasonably dry before packaging and packing. 56.76(f) |
| E-25 | Steam, vapors, or odors originating from the washing and rinsing operation shall be continuously and directly exhausted to the outside of the building. 56.76(f) |
| E-26 | Facility must have and implement a written *Salmonella enteritidis* prevention plan. 118.4 |
| E-27 | Facility must perform *Salmonella enteritidis* testing on pullets at 14 to 16 weeks of age and laying hens at 40 to 45 weeks of age. 118.4(a)(2) and 118.5(a) |



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# FRESH-CUT FRUITS AND VEGETABLES

## F.1 SCOPE

F.1.1 Scope. This appendix establishes the food safety and related requirements for fresh-cut fruit and vegetable establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## F.2 APPLICABLE DOCUMENTS

F.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

F.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Guidance for Industry: Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables, latest revision, U.S. Department of Health and Human Services, Food and Drug Administration, CFSAN.

(Available online at: https://www.fda.gov/Food/GuidanceRegulation/GuidanceDocumentsRegulatoryInformation/ucm064458.htm.)

## F.3 DEFINITIONS

F.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.



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## F.4 REQUIREMENTS

F.4.1 Water quality. Potable water shall be used for all post-harvest operations including grading, flume transports, cooling, washing, and final rinse and equipment sanitization. The quality of water may vary from one point in the process to another (e.g., cooling water versus final rinse water). All water shall meet the microbial standards for drinking water consistent with EPA standards. Reused water in a series of processes shall flow counter to the movement of produce. All chemical substances that disinfect wash water and contact food shall be used in accordance with FDA and Environmental Protection Agency (EPA) regulations. Approved water disinfectants and their amount allowed during processing (see checklist) and the pH of the product determine the final efficacy of the disinfectants on fresh-cut-produce.

F.4.2 Temperature control. Temperature control of minimally processed products is important for controlling the growth of *Clostridium botulinum*. All product packages shall display a “KEEP REFRIGERATED” label. Packages for food service or retail should have a clear code dating system. A “USE BY” date is recommended. Product labeling should not interfere with transparency in order to detect signs of spoilage.

## F.5 CHECKLIST

F.5.1 Checklist. Guidelines for auditing fresh-cut fruits and vegetables are contained in the following Table F-I checklist.



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TABLE F-I. Fresh-Cut Fruits and Vegetables Checklist Requirements as Specified in 21 CFR Part 117 and 123.

| ITEM | REQUIREMENT |
|---|---|
| F-1 | Wash water disinfectant level is established and pH (if applicable) is monitored. Disinfection methods include, but are not limited to, chlorine, hydrogen peroxide, peroxyacetic acid, ozone, chlorine dioxide, and ultraviolet (UV) light. All chemical substances that disinfect wash water and contact food shall be used in accordance with FDA and Environmental Protection Agency (EPA) regulations. Guide to Minimize Microbial Food Safety Hazards for Fresh Fruits and Vegetables and 21 CFR 173. |
| F-2 | Product contact time is established and monitored IAW established FSP (dump tank, submersion, sprayer, flume, hydrocooler method). Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-3 | Water recirculation method is established and monitored (filtration, displacement, replacement). Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-4 | Only approved treatment process water additive(s) or water additive(s) such as surfactants to increase chlorine performance are used. Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-5 | Dewatering, centrifugation, or drying methods established (when applicable) to ensure removal of free surface moisture after washing. Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-6 | Packaging materials are made of approved material, gas-permeable, and preclude packaging migration, the entrance of foreign materials, spoilage prior to toxin production and avoid anaerobic respiration. Gas permeable packaging material must be transparent over an area of at least 50% of the surface to permit the visual detection of spoiled product. Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-7 | Parameters for modified atmosphere(s) packaging are established and monitored (e.g., 1-5% oxygen). Microbiological Safety of Controlled and Modified Atmosphere Packaging of Fresh and Fresh-Cut Produce. Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-8 | Each retail and food service package has a “USE BY DATE” or distinguishable date coding and a “KEEP REFRIGERATED” label or recommended storage conditions / temperature range. Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-9 | Fresh-cut products requiring temperature control for food safety must be maintained at 5º C (41º F) or below during storage and distribution. Guide to Minimize Microbial Food Safety Hazards of Fresh-Cut Fruits and Vegetables |
| F-10 | All product contact water has established and monitored disinfectant parameters to include filtration and recirculation methods. Guide to Minimize Microbial Food Safety Hazards for Fresh Fruits and Vegetables. Note: Recycled water should run counter flow to produce (i.e., final rinse water may be used as cooling water). |



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# ICE

## G.1 SCOPE

G.1.1 Scope. This appendix establishes the food safety and related requirements for ice establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## G.2 APPLICABLE DOCUMENTS

G.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

G.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 40, Part 141 - National Primary Drinking Water Regulations

(Copies of these documents are available online at: https://www.govinfo.gov/.)

G.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL PACKAGED ICE ASSOCIATION (IPIA)

The Packaged Ice Quality Control Standards (PIQCS) Manual (current version)

(Application for copies should be addressed to the International Packaged Ice Association, P.O. 238 East Davis Blvd., Suite 213, Tampa, FL 33606).

## G.3 DEFINITIONS

G.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.



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## G.4 CHECKLIST

G.4.1 Checklist. Guidelines for auditing ice establishments are contained in the following Table G-I checklist.

TABLE G-I. Ice Requirements as Specified in the PIQCS Manual.

| ITEM | REQUIREMENT |
|---|---|
| G-1 | Ice manufacturing, processing, packaging, and storage operations shall be conducted in an enclosed building maintained in a sanitary condition and in a state of good repair. The building shall protect the ice, equipment, and utensils from dust, dirt, rodents, insects, and other sources of contamination. PIQCS Manual, page 81 |
| G-2 | Ice for human consumption shall be processed and packaged only in rooms used solely for those operations. The floors, walls, and ceilings of all rooms, in which ice is manufactured, processed, packaged, and stored shall be of such impervious material and so constructed that it can be maintained in a clean, sanitary condition. PIQCS Manual, page 81 |
| G-3 | Floor drains shall be functional and properly trapped. Floor drainage shall be provided in all areas where floors are subject to flood-type cleaning, or where normal operations discharge, or release water or other liquid waste onto the floor. Drainage and sewage disposal systems must be adequately maintained and controlled to meet sanitary and regulatory standards. PIQCS Manual, page 102 |
| G-4 | No live animals, including dogs, cats, or birds shall be allowed in any area of the plant. PIQCS Manual, page 83 |
| G-5 | Bacteriological tests of the finished product shall be conducted at least monthly. Ice shall be tested for Heterotrophic Plate Count (HPC) (NTE 500 cfu/ml) and total coliform (<1 cfu/100 ml). If sample is positive for total coliforms, sample must be tested for fecal coliforms (NEG / 100 mL). PIQCS Manual, page 119 and DoW Ice Testing Matrix, Appendix O |
| G-6 | Adequate provisions (such as sanitary gloves) must be made so that hands do not come in direct contact with the ice at any time during manufacturing, processing, packaging, and storage. PIQCS Manual, page 119. |
| G-7 | Packaging shall be done with non-toxic materials and in a sanitary manner. The bags shall be of sound strength and quality to prevent fracture or tearing during handling and be constructed of FDA approved materials. PIQCS Manual, page 119. |
| G-8 | Ice product must be in a tightly sealed bag or container. Use of Kwik Lock closures are acceptable. Bags that have small standard vent holes in the bag are deemed to be sealed. The label must show (at a minimum) the product name, net weight, and manufacturer or distributor name and contact information. PIQCS Manual, page 120. |
| G-9 | Air lines, vacuum lines, and water lines must be regularly cleaned, and sanitized. They shall be made of food grade material when coming into contact with the product, water, or ice. PIQCS Manual, page 194 |



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TABLE G-I. Ice Requirements as Specified in the PIQCS Manual - Continued

| ITEM | REQUIREMENT |
|---|---|
| G-10 | Potable water shall be used in any operation which comes into contact with the product ice; this includes, but is not limited to, sprays, thaw tanks, or dipping wells, used to remove the ice from containers (and/or molds) used to freeze them. PIQCS Manual, page 195 |
| G-11 | Ice cans (applicable to product subject to the IPIA Block Ice Addendum) must be leak-proof and the inner surfaces of such containers shall be free of corrosion and made of material approved for food contact. PIQCS Manual, page 114 |



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# FISH AND FISHERY PRODUCTS

## H.1 SCOPE

H.1.1 Scope. This appendix establishes the food safety and related requirements for fresh or frozen processed fish, retorted seafood and specialty fishery product establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## H.2 APPLICABLE DOCUMENTS

H.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed. Items A-30 through A-84 are not applicable to this appendix.

H.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 101 | - | Food Labeling |
| Title 21, Part 123 | - | Fish and Fishery Products |
| Title 21, Part 172 | - | Food Additives Permitted For Direct Addition to Food for Human Consumption |

(Copies of these are available online at: https://www.govinfo.gov/.)

U.S. FOOD AND DRUG ADMINISTRATION (FDA)

FDA Fish and Fisheries Products Hazards & Control Guide, current version.

(Copies of this document are available online at: https://www.fda.gov/downloads/Food/GuidanceRegulation/UCM251970.pdf .)

Hazard Analysis and Critical Control Point Principles and Application Guidelines.

(Available on-line at: https://www.fda.gov/food/hazard-analysis-critical-control-point-haccp/haccp-principles-application-guidelines.)



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National Shellfish Sanitation Program (NSSP), Guide for the Control of Molluscan Shellfish (2015 Revision).

(Copies of this document are available online at: .https://www.fda.gov/food/federal-state-local-tribal-and-territorial-cooperative-human-food-programs/national-shellfish-sanitation-program-nssp).)

Regulatory Fish Encyclopedia (RFE)

(Available on-line at: https://www.fda.gov/food/reference-databases-and-monitoring-programs-food/regulatory-fish-encyclopedia-rfe.)

H.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Compendium of Fish and Fishery Product Processes, Hazards and Controls. National Seafood HACCP Alliance for Training and Education.

(Available on-line at https://seafood.oregonstate.edu/seafood-network-information-center/compendium-fish-and-fishery-product-processes-hazards-and)

H.3 DEFINITIONS

H.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

H.4 REQUIREMENTS

H.4.1 *Clostridium botulinum*. *Clostridium botulinum* are known to be present in the viscera of fish; any product that will be preserved by salting, drying, pickling, or fermentation will be eviscerated prior to processing. Evisceration shall be thorough and performed to minimize contamination of the fish flesh. If even a portion of the viscera or its contents is left behind, the risk of toxin formation by *C. botulinum* remains.

H.4.1.1 Small fish, less than 5 inches in length (e.g., anchovies and herring sprats), that are processed in a manner that prevents toxin formation, and that reach a water phase salt content of 10 percent in refrigerated products, or a water activity of below 0.85, or a pH of 4.6 or less in shelf-stable products, are exempt from the evisceration requirement. Note: this value is based upon the minimum water activity for growth of *Staphylococcus aureus*.

H.4.2 Reduced oxygen packaging and refrigeration. In reduced oxygen packaged products in which refrigeration is the sole barrier to outgrowth of non-proteolytic *C. botulinum* and the spores have not been destroyed (e.g. vacuum packaged raw fish, unpasteurized crayfish



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meat), the temperature shall be maintained at 38ºF (3.3ºC) or below from packing to consumption.

## H.5 CHECKLIST

H.5.1 Checklist. Guidelines for auditing fish and fishery products establishments are contained in the following Table H-I through H-V checklists.



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TABLE H-I. Seafood General Requirements as Specified in 21 CFR Part 123 and Part 172

| ITEM | REQUIREMENT |
|---|---|
| H-1 | Fish or fishery products that are obtained from foreign sources are safe and acceptable as required. 123.12 |
| H-2 | Fresh fish, except those immediately processed or as otherwise required, are iced or otherwise refrigerated to an internal temperature of 38°F (3.3°C) or below upon receipt and are maintained at that temperature until fish are to be processed. FDA Guide, Chapter 12 and 13 |
| H-3 | All fish received in a frozen state are thawed promptly and processed, or stored at a temperature which will maintain it in a frozen state. Compendium, Chapter 7 |
| H-4 | After thawing, fish are washed thoroughly with a vigorous potable water spray or a continuous water flow system. When thawing and brining occur concurrently, the fish are washed in this same fashion following the thawing and brining. Compendium, Chapter 7 |
| H-5 | The evisceration of fish is conducted in a segregated or separate processing room. The evisceration is performed with minimal disturbance of the intestinal tract contents, and the fish, including the body cavity, is washed thoroughly with a vigorous spray or a continuous water flow system following evisceration. Compendium, Chapter 7 |
| H-6 | Sodium nitrite content is used for authorized species only (salmon, sable, shad, chub and tuna) and meets regulatory requirements for use; product is labeled declaring such use. 172.175 and 172.177 |
| H-7 | Incoming shrimp or lobsters are tested for the presence of sulfite residues at 10 ppm or below or a supplier's certificate stating the lack of sulfiting agents is presented upon receipt. 101.100 |
| H-8 | Lots received are accompanied by harvest vessel records that show sensory examination (i.e., a representative sample of scombrotoxin-forming fish shows decomposition [persistent and readily perceptible] in less than 2.5% of the fish in the sample) and internal temperature measurements (e.g., maximum refrigerated brine or seawater temperature, maximum fish size, maximum fish to brine/seawater/ice ratio, maximum initial temperature of fish) necessary to achieve a cooling rate that will prevent development of an unsafe level of histamine in the specific species, or analysis of a representative sample with less than 35 ppm histamine, as established through a scientific study or as provided in the Fish and Fishery Products Hazards and Controls Guidance, Chapter 7: Scombrotoxin (Histamine) Formation. FDA Guide, Chapter 7 |
| H-9 | Shipping containers, retail packages and shipping records relating to processed fish are appropriately labeled in accordance with the perishable nature of the product. Compendium, Chapter 7 |
| H-10 | The finished products are properly cooled to 70ºF (21°C) within 2 hours and further cooled to 40ºF (4.4°C) within an additional 4 hours. Finished products are then maintained at 40ºF (4.4°C). Where the control of nonproteolytic *Clostridium botulinum* is a factor during storage, products are stored at 38ºF (3.3°C). FDA Guide and Compendium, Chapter 7 |



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TABLE H-I. Seafood General Requirements as Specified in 21 CFR Part 123 and Part 172 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| H-11 | Fish processing records include the name and location of the processor or importer; the date and time of the activity that the record reflects; the signature or initials of the person performing the operation; and where appropriate, the identity of the product and / or the production code, if any. Processing and other information shall be entered on records at the time that it is observed. 123.9(a) |
| H-12 | Records are maintained for the chemical examination of finished product for the purpose of validating the water phased salt and sodium nitrite requirements. Compendium, Chapter 7 |
| H-13 | Records that relate to the general adequacy of equipment or process being used by a processor, including the results of scientific studies and evaluations, shall be retained at the processing facility. 123.9(b) |

TABLE H-II. Smoked, Dried, and Brined Product Requirements as Specified in 21 CFR Part 123.

| ITEM | REQUIREMENT |
|---|---|
| H-14 | Each smoking chamber is equipped with a temperature monitoring device so installed as to indicate accurately at all times the internal temperature of the fish within the smoking chamber. Compendium, Chapter 7<br><br>DoW exempts small fish, less than 5 inches in length (e.g., anchovies and herring sprats), that are processed in a manner that prevents toxin formation, and that reach a water phase salt content of 10 percent in refrigerated products, or a water activity of below 0.85, or a pH of 4.6 or less in shelf-stable products, are exempt from the evisceration requirement. This value is based upon the minimum water activity for growth of Staphylococcus aureus. |
| H-15 | All fish are free of viscera prior to processing, except as otherwise exempt. Compendium, Chapter 7 |
| H-16 | All processed fish except smoked fish are processed pursuant to a scheduled process established by a competent processing authority. The scheduled process includes processing methods, procedures, and controls for each product, as well as packaging and labeling requirements. The scheduled process is filed with the appropriate regulatory authority prior to the commencement of processing and is available for examination in each processing establishment. Compendium, Chapter 7 |
| H-17 | All processed fish are distributed and sold in a manner that ensures that the internal temperature is maintained at 38°F (3.3°C) or below except as otherwise exempt. Compendium, Chapter 7 |



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TABLE H-II. Smoked, Dried, and Brined Product Requirements as Specified in 21 CFR Part 123 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| H-18 | Fish are of relatively uniform size and weight and arranged without overcrowding or touching each other within the smokehouse oven. Compendium, Chapter 7 |
| H-19 | Liquid smoke, generated smoke, or a combination of liquid smoke and generated smoke are applied to all surfaces of the product at the appropriate times. Compendium, Chapter 7 |
| H-20 | For hot processed smoked fish to be air packaged, a controlled process is used to heat fish to a continuous temperature of at least 145ºF (62.8ºC) throughout each fish for a minimum of 30 minutes for fish brined to contain not less than 2.5% water phase salt in the loin muscle of the finished product. Compendium, Chapter 7 |
| H-21 | For hot processed smoked fish to be vacuum or modified atmosphere packaged, a controlled process is used to heat the fish to a continuous temperature of at least 145ºF (62.8ºC) throughout each fish for a minimum of 30 minutes for fish brined to contain not less than 3.5% water phase salt in the loin muscle of the finished product or the combination of 3.0% water phase salt in the loin muscle of the finished product and not less than 100 ppm or more than 200 ppm sodium nitrite. Compendium, Chapter 7 |
| H-22 | Brining operations are performed so that the temperature of the brine does not exceed 60ºF (15.5ºC) at the start of brining. If the brining time exceeds 4 hours, the brining is performed in a refrigerated area at 38ºF (3.3ºC) or lower. Compendium, Chapter 7 |
| H-23 | For dry salting, the fish are returned to a refrigerated area of 38°F (3.3°C) or lower immediately after the application of the salt. Compendium, Chapter 7 |
| H-24 | Different species of fish are not mixed in the same brine tank. Compendium, Chapter 7 |
| H-25 | Brines are not reused without an adequate process available to return the brine to an acceptable microbiological level. Compendium, Chapter 7 |
| H-26 | Cold processed smoked fish are produced by a controlled process that utilizes a temperature monitoring system positioned within the smoking chamber to assure all products do not exceed process temperatures or exceed the representative drying and smoking period in accordance with one of the authorized methods (90ºF [32.2ºC] for 20 hours; 50ºF [10.0ºC] for 24 hours; 120ºF [48.9ºC] for 6 hours [sablefish only]). Compendium, Chapter 7 |
| H-27 | For cold smoked fish to be air packaged, fish that have brine contain not less than 2.5 percent water phase salt in the loin muscle of the finished product. Compendium, Chapter 7 |



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TABLE H-II. Smoked, Dried, and Brined Product Requirements as Specified in 21 CFR Part 123 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| H-28 | For cold smoked fish to be vacuum or modified atmosphere packaged, fish that have been brined contain not less than 3.5 percent water phase salt in the loin muscle of the finished product, or a combination of 3.0 percent water phase salt in the loin muscle of the finished product and not less than 100 ppm nor more than 200 ppm of sodium nitrite. Compendium, Chapter 7 |
| H-29 | The vacuum packaging or modified atmosphere packaging of processed fish is conducted only within the facilities of the manufacturer. Compendium, Chapter 7 |

TABLE H-III. MAP Requirements as Specified in Compendium of Fish and Fishery Product Processes, Hazards, and Controls (Compendium).

| ITEM | REQUIREMENT |
|---|---|
| H-30 | Processed fish to be vacuum packaged or modified atmosphere packaged are chemically analyzed for water phase, salt, and for nitrate and other additives when used, with sufficient frequency to ensure conformance with finished product specification requirements. Compendium, Chapter 7 |

TABLE H-IV. Shellfish Requirements as Specified in Good Manufacturing Practices 21 CFR Part 123: National Shellfish Sanitation Program (NSSP).

| ITEM | REQUIREMENT |
|---|---|
| H-31 | Molluscan shellfish will be derived from shellstock received from harvesters or processors that shuck, ship, or reship or pack shellfish are certified by a shellfish control authority. 123.28 |
| H-32 | Identification tags on in-shell and shucked molluscan shellfish, containing all required information are affixed to each container. NSSP CH X.05 and 21 CFR 123.28 |
| H-33 | Chilled or iced shucked shellstock maintained at 45°F during storage and transport. NSSP CH XI.01 |
| H-34 | Shucked shellfish from different lots are not commingled. NSSP CH VII.02 |
| H-35 | Mandatory Critical Control Points Identified by the NSSP Model Ordinance are addressed in the HACCP Plan. NSSP CH XI.01, XII.01, XIII.01, XIV.01 |
| H-36 | Critical limits will include those listed in Chapter XI. 01, Chapter XII. 01, Chapter XIII. 01, and Chapter XIV. 01, as applicable. As an alternative the dealer may establish other critical limits, which the dealer has demonstrated, provide equivalent public health protection with the exception of receiving which will always be considered as a critical control point. In any case, the critical limits identified in Chapter XI. 01, Chapter XII. 01, Chapter XIII. 01, and Chapter XIV. 01, will be met as components of good manufacturing practices. NSSP |



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TABLE H-IV. Shellfish Requirements as Specified in Good Manufacturing Practices 21 CFR Part 123: National Shellfish Sanitation Program (NSSP) - Continued.

| ITEM | REQUIREMENT |
|---|---|
| H-37 | Transaction and shipping records will be sufficient to: Document that the shellfish are from a source authorized under this Ordinance (the NSSP); Permit a container of shellfish to be traced back to the specific incoming lot of shucked shellfish from which it was taken; Permit a lot (or commingled lots) of shucked shellfish or a lot of shellstock to be traced back to the growing area(s), date(s) of harvest, date and location of wet storage, if applicable and if possible, the harvester or group of harvesters; trace the wet storage history of the shellstock including, original harvest site, original harvest date, wet storage site(s) and dates. NSSP CH X.08.B.2 |

TABLE H-V. Hazard Analysis and Critical Control Point (HACCP) Plan Requirements as Specified in 21 CFR Part 123.

| ITEM | REQUIREMENT |
|---|---|
| H-38 | Every processor shall conduct, or have conducted for it, a hazard analysis to determine whether there are food safety hazards that are reasonable likely to occur for each kind of fish and fishery product processed by the processor and to identify the preventive measures that the processor can apply to control those hazards. 123.6(a) |
| H-39 | Every processor shall have and implement a written HACCP plan whenever a hazard analysis reveals one or more food safety hazards that are reasonable likely to occur as required. 123.6(b) |
| H-40 | The contents of the HACCP plan shall, at a minimum, include the following: Identify food safety hazards that are reasonable likely to occur; identify critical control points (CCP) for hazards that may be introduced both inside AND outside of the processing establishment; list the critical limits for each CCP; identify procedures AND frequencies to monitor critical limits; include corrective actions; list the verification procedures and frequencies used; and provide a recordkeeping system that documents the monitoring of the critical control points, as required. Consideration should be given to whether any food safety hazards are reasonably likely to occur as a result of the following: natural toxins; microbiological contamination; chemical contamination; pesticides; drug residues; decomposition in scombroid toxin-forming species (or in any other species where a food safety hazard has been associated with decomposition); parasites; unapproved use of direct or indirect food or color additives; and physical hazards. 123.6(c) |
| H-41 | The HACCP plan shall be signed and dated as required. 123.6(d) |
| H-42 | Failure of a processor to have and implement a HACCP plan that complies as otherwise required shall render the fish or fishery products of that processor adulterated under section 402(a) (4) of the act. 123.6(g) |



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# WAREHOUSES AND DISTRIBUTORS

## I.1 SCOPE:

I.1.1 Scope. This appendix establishes food safety and related requirements for food warehouses and distributors. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## I.2 APPLICABLE DOCUMENTS

I.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

I.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 117 | - | Current Good Manufacturing Practice, Hazard Analysis, and Risk-Based Preventive Controls for Human Food |

(Copies of these are available online at: https://www.govinfo.gov/.)

I.3 DEFINITIONS: The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

I.4 CHECKLIST: Guidelines for auditing commercial food warehouses are contained in the following Table I-I checklist.



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TABLE I-I. Warehouse Requirements as Specified in 21 CFR 117 Subpart B.

| ITEM | REQUIREMENT |
|---|---|
| I-1 | Storage and transportation of food must be under conditions that will protect against allergen cross-contact and against biological, chemical (including radiological), and physical contamination of food, as well as against deterioration of the food and the container. 117.93 |
| I-2 | Establish and implement temperature controls for refrigerated foods that adequately minimize or prevent the growth of pathogens. 117.206 |
| I-3 | Monitor the temperature controls for refrigerated food with adequate frequency to provide assurance that the temperature controls are consistently performed. 117.206 |
| I-4 | If there is a loss of temperature control that may impact the safety of refrigerated food, take appropriate corrective actions to:<br>(i) Correct the problem and reduce the likelihood that the problem will recur;<br>(ii) Evaluate all affected food for safety; and<br>(iii) Prevent the food from entering commerce, if they cannot ensure the affected food is not adulterated under section 402 of the Federal Food, Drug, and Cosmetic Act. 117.206 |
| I-5 | Verify that temperature controls are consistently implemented for refrigerated food by:<br>(i) Calibrating temperature monitoring and recording devices (or checking them for accuracy);<br>(ii) Reviewing records of calibration within a reasonable time after the records are created; and<br>(iii) Reviewing records of monitoring and corrective actions taken to correct a problem with the control of temperature within 7 working days after the records are created or within a reasonable timeframe, provided that the preventive controls qualified individual prepares (or oversees the preparation of) a written justification for a timeframe that exceeds 7 working days. 117.206 |
| I-6 | Establish and maintain the following records for refrigerated foods:<br>(i) Records (whether affirmative records demonstrating temperature is controlled or exception records demonstrating loss of temperature control) documenting the monitoring of temperature controls for any such refrigerated packaged food;<br>(ii) Records of corrective actions taken when there is a loss of temperature control that may impact the safety of any such refrigerated packaged food; and<br>(iii) Records documenting verification activities. 117.206 |



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# PASTEURIZED, REFRIGERATED JUICES

## J.1 SCOPE

J.1.1 Scope. This appendix establishes the food safety and related requirements for pasteurized, refrigerated juice establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## J.2 APPLICABLE DOCUMENTS

J.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed. Items A-30 through A-84 are not applicable to this appendix.

J.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 113 | - | Thermally Processed Low-Acid Canned Foods Packaged in Hermetically Sealed Containers |
| Title 21, Part 114 | - | Acidified Foods |
| Title 21, Part 120 | - | Hazard Analysis and Critical Control Point (HACCP) Systems |

(Copies of these documents are available online at: https://www.govinfo.gov/.)

FOOD AND DRUG ADMINISTRATION (FDA)

FDA Guidance for Industry, “Juice HACCP Hazards and Controls Guidance, First Edition”, Mar 2004.

(Available on-line at: https://www.fda.gov/food/guidanceregulation/guidancedocumentsregulatoryinformation/juice/ucm072557.htm.)



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U.S. DEPARTMENT OF AGRICULTURE

The National Advisory Committee on Microbiological Criteria for Foods (NACMCF), “Requisite Scientific parameters for Establishing the Equivalence of Alternative Methods of Pasteurization”, Aug 2006.

(Available on-line at: https://downloads.regulations.gov/FDA-2011-N-0920-0001/attachment_177.pdf)

J.3 DEFINITIONS

J.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

J.4 REQUIREMENTS

J.4.1 DoW procurement of juice. All juice produced for DoW procurement will be pasteurized by a system that is validated to achieve a 5-log reduction in microbiological load.

J.4.2 HACCP. A validated HACCP plan is compulsory for all pasteurized, refrigerated juice manufacturers. Each establishment is allowed to evaluate its own process.

J.4.3 Control of pathogens. Processors of juice products will include in their Hazard Analysis and Critical Control Point (HACCP) plans control measures that will consistently produce, at a minimum, a five log reduction of the pertinent microorganism, for a period at least as long as the shelf life of the product when stored under normal and moderate abuse conditions. For the purposes of this regulation, the "pertinent microorganism" is the most resistant microorganism of public health significance that is likely to occur in the juice. The most commonly found pathogens in unpasteurized juices are *Escherichia coli* O157:H7 (bacteria), *Salmonella* (bacteria), and *Cryptosporidium* (parasite). The following juice processors are exempt from this requirement:

a. A juice processor that is subject to the requirements of part 21 CFR Parts 113 or 114.

b. A juice processor using a single thermal processing step sufficient to achieve shelf-stability of the juice or a thermal concentration process that includes thermal treatment of all ingredients, provided that the processor includes a copy of the thermal process used to achieve shelf-stability or concentration in its written hazard analysis required by 21 CFR 120.7.

J.5 CHECKLIST

J.5.1 Checklist. Guidelines for auditing pasteurized, refrigerated juice processing establishments are contained in the following Table J-I checklist.



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TABLE J-I. Pasteurized, Refrigerated Juices Checklist Requirements as Specified in 21 CFR Part 120

| ITEM | REQUIREMENT |
|---|---|
| J-1 | Except as provided by 117.5(c), parts 110 and 117 of this chapter apply in determining whether the facilities, methods, practices, and controls used to process juice are safe, and whether the food has been processed under sanitary conditions. 120.5 |
| J-2 | Each processor shall have and implement a sanitation standard operating procedure (SSOP) that addresses sanitation conditions and practices before, during, and after processing. 120.6 |
| J-3 | Each processor shall develop, or have developed for it, a written hazard analysis to determine whether there are food hazards that are reasonably likely to occur for each type of juice processed by that processor and to identify control measures that the processor can apply to control those hazards. The written hazard analysis shall consist of at least the topics outlined in 21 CFR 120.7 (a) through (e). 120.7 |
| J-4 | Each processor shall have and implement a written HACCP plan whenever a hazard analysis reveals one or more food hazards that are reasonably likely to occur during processing, as described in 120.7. The HACCP plan shall be developed by an individual or individuals who have been trained in accordance with 120.13 and shall be subject to the recordkeeping requirements of 120.12. 120.8 |
| J-5 | Whenever a deviation from a critical limit occurs, a processor shall take predetermined corrective actions from written corrective action plans. 120.10 |
| J-6 | Processors will take corrective action to ensure no product enters commerce that is either injurious to health or is otherwise adulterated as a result of the deviation and the cause of the deviation is corrected. 120.10 |
| J-7 | Each processor shall verify the HACCP system is being implemented according to design and validate the HACCP plan by a trained individual to ensure adequate control of food hazards that are reasonably likely to occur at least once within 12 months after implementation and at least annually or whenever changes occur. 120.11 |
| J-8 | Whenever a juice processor has no HACCP plan because a hazard analysis has revealed no food hazards that are reasonably likely to occur, the processor shall reassess the adequacy of that hazard analysis whenever there are any changes in the process that could reasonably affect whether a food hazard exists. 120.11 |
| J-9 | Each processor shall maintain records documenting the processor's HACCP system, SSOPs, written hazard analysis, written HACCP plan, monitoring of CCPs and critical limits, including recording times, temperatures, or other measurements as prescribed in the HACCP plan. 120.12 |
| J-10 | All records required by this part shall be retained at the processing facility, in the case of perishable or refrigerated juices, at least 1 year after the date that such products were prepared, and for, in the case of frozen, preserved, or shelf stable products, 2 years or the shelf life of the product, whichever is greater, after the date that the products were prepared. 120.12 |



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TABLE J-I. Pasteurized, Refrigerated Juices Checklist Requirements as Specified in 21 CFR Part 120 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| J-11 | Only an individual who has successfully completed training in the application of HACCP principles to juice processing or is qualified through job experience shall develop the hazard analysis, develop a HACCP plan, verify, modify, validate the HACCP plan, and perform record review. 120.13 |
| J-12 | Imported juice will be accompanied by a Certificate of Conformance (COC). 120.14 |
| J-13 | HACCP plans shall include control measures that will consistently produce, at a minimum, a 5-log reduction in the product, for a period at least as long as the shelf life of the product when stored under normal and moderate abuse conditions, in the pertinent microorganism. 120.24 |
| J-14 | Juice processors that rely on treatments that do not come into direct contact with all parts of the juice to achieve the requirements of 21 CFR 120.24 will analyze the finished product for biotype I Escherichia coli. 120.25 |



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# BOTTLED WATER AND BEVERAGES

## K.1 SCOPE

K.1.1 Scope. This appendix establishes the food safety and related requirements for bottled water and beverage establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## K.2 APPLICABLE DOCUMENTS

K.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

K.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 129 | - | Processing and Bottling of Bottled Drinking Water |
| Title 21, Part 165 | - | Beverages |

(Copies of these documents are available online at: https://www.govinfo.gov/.)

## K.3 DEFINITIONS

K.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

## K.4 REQUIREMENTS

K.4.1 Microbiological Criteria. Source water must meet the DoW requirements stated in paragraph 5.6

## K.5 CHECKLIST

K.5.1 Checklist. Guidelines for auditing bottled water establishments are contained in the following Table K-I checklist.



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TABLE K-I. Bottled Water Requirements as Specified in 21 CFR Part 129 and 165.

| ITEM | REQUIREMENT |
|---|---|
| K-1 | The bottling room is separated from other plant operations or storage areas by tight walls, ceilings, self-closing doors, and size of conveyor opening. Bottle washing and sanitizing are in an enclosed area and are positioned to minimize post-sanitization contamination. 129.20 |
| K-2 | The product water supply for each plant shall be from an approved source properly located, protected, and operated and shall be easily accessible, adequate, and of a safe, sanitary quality which shall be in conformance at all times with the applicable laws and regulations of the government agency or agencies having jurisdiction. 129.35(a) |
| K-3 | If spring water is collected with the use of an external force, water shall continue to flow naturally to the surface of the earth through the spring's natural orifice. 165.110(a)(2)(vi) |
| K-4 | Representative bacteriological samples tested weekly for each type of finished product produced during a day's production. 129.80 (g) and 165.110 |
| K-5 | Product and operations water samples of source water from each source in use by the plant are to be taken and analyzed by the plant at a minimum frequency of once each year for chemical contaminants and once every four years for radiological contaminants. Additionally, source water obtained from other than a public water system is to be sampled and analyzed for total coliform at least once each week. If any coliform organisms are detected, follow-up testing must be conducted to determine whether any of the coliform organisms are Escherichia coli. 129.35(a)(3)(i) |
| K-6 | Source and finished product test results meet the compliance of 21 CFR Part 165.110(b). Source water that has been treated with a chlorine-based disinfectant or ozone will be tested for the residual disinfectants and disinfection by-products (DBPs) listed in 21 CFR 165.110(b)(4)(iii)(H). 129.80(g) and 129.35(a)(4) |
| K-7 | Containers, caps, or seals are purchased and stored in sanitary closures (original containers) in a clean, dry place. They are examined before use and are handled, dispensed and used in a sanitary manner. They are washed, rinsed, and sanitized as needed. 129.37(c) |
| K-8 | Treatment methods accomplish their intended purpose. Records are maintained to show type and date of treatments and physical inspections of equipment. Conditions found, performance and effectiveness are noted. 129.80(a) |
| K-9 | Product water samples are taken after processing, prior to bottling, to assure uniformity and effectiveness of the treatment process. Methods of analysis are approved by the government agency having jurisdiction. 129.80(a) |
| K-10 | All unsanitary or defective containers are reprocessed or rendered unusable and discarded. Multi-service primary containers are cleaned, sanitized, and inspected immediately prior to being filled, capped, and sealed. 129.80(b) |
| K-11 | Mechanical washers are inspected. Records of physical maintenance, inspections, conditions found, and performance of the mechanical washer, are maintained by the plant. 129.80(b) |
| K-12 | Sanitizing operations meet requirements contained in 129.80(d). |



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TABLE K-I. Bottled Water Requirements as Specified in 21 CFR Part 129 and 165 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| K-13 | Records of the intensity of the sanitizing agent and the contact time duration will be maintained by the plant. 129.80(c) and (d) |
| K-14 | Each unit package is identified by a production code. The code identifies the particular batch or segment of a continuous run, and the production date. 129.80(e) |
| K-15 | Records are maintained of product type, volume produced, date produced, lot code used, and distribution to wholesale and retail outlets. 129.80(e) |
| K-16 | Containers and closures are nontoxic and comply with applicable standards. 129.80(f) |
| K-17 | Filling, capping, and sealing are monitored. Filled containers are visually or electronically inspected. 129.80(f) |
| K-18 | A swab and/or rinse bacterial count performed quarterly on four containers and closures immediately prior to filling the containers and meets the microbiological criteria in accordance with 21 CFR 129.80(f). |
| K-19 | Records are maintained of sampling date, type of product, production code, and results of each analysis. 129.80(h) |
| K-20 | Current certificates or notifications of approval authority for source and supply of product and operations water are on file. 129.80(h) |



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# CATERERS AND READY-TO-EAT MANUFACTURED PRODUCTS

## L.1 SCOPE

L.1.1 Scope. This appendix establishes the food safety and related requirements for caterers and other Ready-to-Eat (or Heat and Eat) establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## L.2 APPLICABLE DOCUMENTS

L.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that processors of Ready-to-Eat products must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed. Caterers will be audited only against the requirements listed in this appendix.

L.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

U.S. Public Health Service (USPHS) / Food and Drug Administration (FDA) Food Code, current version.

(Copies of this document are available online at: https://www.fda.gov/food/retail-food-protection/fda-food-code

## L.3 DEFINITIONS

L.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix. Caterers are not required to have a food defense plan that meets the definition listed in paragraph 3.16.

## L.4 CHECKLIST

L.4.1 Checklist. Guidelines for auditing caterers and RTE manufactured product establishments are contained in the following Table L-I checklist.



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TABLE L-I. Caterers and Ready-to-Eat Manufactured Products Requirements as Specified in the Food Code.

| | |
|---|---|
| L-1 | Supervision responsibility, knowledge, and duty requirements are met. All debitable provisions under Part 2-1. |
| L-2 | Employee health requirements are met. All debitable provisions under Part 2-2. |
| L-3 | Personal cleanliness (hands and arms, fingernails, jewelry, and outer clothing) requirements are met. All debitable provisions under Part 2-3. |
| L-4 | Hygienic requirements are met. All debitable provisions under Part 2-4. |
| L-5 | A written procedure is in place to respond to contamination events (vomiting and diarrhea). All debitable provisions under Part 2-5. |
| L-6 | Food shall be safe, unadulterated, and honestly presented. All debitable provisions under Part 3-1. |
| L-7 | Food source, receiving, original container, and food donation requirements are met. All debitable provisions under Part 3-2. |
| L-8 | Food is protected from contamination after receiving. All debitable provisions under Part 3-3. |
| L-9 | Organisms of public health concern are destroyed through cooking, freezing, reheating, or other methods. All debitable provisions under Part 3-4. |
| L-10 | Growth of organisms of public health concern are limited through temperature and time control and/or specialized processing methods. All debitable provisions under Part 3-5. |
| L-11 | Food identity, presentation, and labeling requirements are met. All debitable provisions under Part 3-6. |
| L-12 | Disposition of contaminated food is handled appropriately. All debitable provisions under Part 3-7. |
| L-13 | Special requirements are in place when providing food for highly susceptible populations. All debitable provisions under Part 3-8. |
| L-14 | Material requirements are met for surfaces, equipment, and utensils within the facility. All debitable provisions under Part 4-1. |
| L-15 | Design and construction requirements are met for surfaces, equipment, and utensils within the facility regarding durability, strength, cleanability, accuracy, functionality, and acceptability. All debitable provisions under Part 4-2. |
| L-16 | Availability and capacity requirements are met for equipment, utensils, temperature measuring devices, testing devices, cleaning agents, and sanitizers. All debitable provisions under Part 4-3. |
| L-17 | Location requirements are met for cabinets (storing food, sanitized equipment, laundered linens, single-service/single-use articles) and clothes washers and dryers. Installation requirements are met for fixed equipment. All debitable provisions under Part 4-4. |
| L-18 | Maintenance and operation requirements are met for equipment, utensils, and temperature and pressure measuring devices. All debitable provisions under Part 4-5. |



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TABLE L-I. Caterers and Ready-to-Eat Manufactured Products Requirements as Specified in the Food Code – Continued.

| | |
|---|---|
| L-19 | Cleaning requirements are met for equipment and utensils to include adequacy of cleaning, frequency, and methods for cleaning. All debitable provisions under Part 4-6. |
| L-20 | Sanitization requirements are met for equipment and utensils to include adequacy of sanitization, frequency, and methods for sanitization. All debitable provisions under Part 4-7. |
| L-21 | Laundering requirements are met . All debitable provisions under Part 4-8. |
| L-22 | Clean items (equipment, utensils, wiping clothes, food contact surfaces) are protected from contamination. All debitable provisions under Part 4-9. |
| L-23 | Source water requirements are met to include quality, quantity, availability, distribution. All debitable provisions under Part 5-1. |
| L-24 | Plumbing system requirements are met to include approved materials, design, construction, installation, numbers and capacity, location and placement, operation, and maintenance. All debitable provisions under Part 5-2. |
| L-25 | Mobile water tank and mobile food establishment water tank requirements are met. All debitable provisions under Part 5-3. |
| L-26 | Sewage, other liquid waste, and rainwater requirements are met. All debitable provisions under Part 5-4. |
| L-27 | Refuse, recyclable, returnable requirements are met. All debitable provisions under Part 5-5. |
| L-28 | Surface characteristic requirements are met for indoor and outdoor areas. All debitable provisions under Part 6-1. |
| L-29 | Physical facility design, construction, and installation requirements are met to include cleanability, and functionality. All debitable provisions under Part 6-2 . |
| L-30 | Quantity and capacity requirements are met for handwashing sinks, toilets and urinals, lighting, ventilation, dressing areas and lockers, and service sinks. All debitable provisions under Part 6-3. |
| L-31 | Location and placement requirements are met for handwashing sinks, toilet rooms, employee accommodations, distressed merchandise, refuse, recyclables, and returnables. All debitable provisions under Part 6-4. |
| L-32 | Maintenance and operation requirements are met for the premises, structures, attachments, and fixtures. All debitable provisions under Part 6-5. |
| L-33 | Labeling and identification requirements are met for poisonous and toxic materials. All debitable provisions under Part 7-1. |
| L-34 | Storage and application requirements are met for poisonous and toxic materials. All debitable provisions under Part 7-2. |



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# SLAUGHTER AND FABRICATION OF MEAT PRODUCTS OCONUS

## M.1 SCOPE

M.1.1 Scope. This appendix establishes the food safety and related requirements for meat slaughter and fabrication establishments OCONUS. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## M.2 APPLICABLE DOCUMENTS

M.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed. Items A-30 through A-84 are not applicable to this appendix.

M.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 9, Part 94 | - | Foot-and-Mouth Disease, Newcastle Disease, Highly Pathogenic Avian Influenza, African Swine Fever, Classical Swine Fever, Swine Vesicular Disease, and Bovine Spongiform Encephalopathy: Prohibited and Restricted Importations |
| Title 9, Part 307 | - | Facilities for Inspection |
| Title 9, Part 309 | - | Ante-Mortem Inspection |
| Title 9, Part 310 | - | Post-Mortem Inspection |
| Title 9, Part 311 | - | Disposal of Diseased or Otherwise Adulterated Carcasses and Parts |
| Title 9, Part 313 | - | Humane Slaughter of Livestock |
| Title 9, Part 416 | - | Sanitation |
| Title 9. Part 417 | - | Hazard Analysis and Critical Control Point (HACCP) Systems |



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M.3 DEFINITIONS

M.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

M.4 REQUIREMENTS

M.4.1 HACCP. A validated HACCP plan is required for meat processing establishments in accordance with 9 CFR Part 417.

M.4.2 Bleeding. Kosher/Halal animals are bled without previous stunning, simultaneously by severing the trachea and major blood vessels.

M.4.3 Meat fabrication. Meat fabrication only includes meats that have been cut (even to individual cuts) and chilled. Comminuted meats, hearts, livers, and kidneys will be included in the category of fabrication. All fabrication rooms shall be equipped with the appropriate number of disinfection units (sometimes referred to as sanitizers). The temperature of the product should never exceed 45°F (7°C). Meat that is not boned shall be fabricated in a properly chilled room and shall immediately be chilled. Prior to each day's production, a pre-operational sanitation inspection shall be performed (by either an auditor or acceptable company representative).

M.5 CHECKLIST

M.5.1 Checklists. Guidelines for auditing of meat slaughter and fabrication establishments OCONUS are contained in the following Tables M-I and M-II checklists.



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TABLE M-I. Slaughter and Fabrication of Meat Products OCONUS Requirements as Specified in 9 CFR Part 309, 310, 311, 313 and 416.

| ITEM | REQUIREMENT |
|---|---|
| M-1 | Livestock entering the establishment receive an adequate ante-mortem inspection on the day of and before slaughter and are properly segregated when required. 309.1 and 309.2 |
| M-2 | When holding pens of an establishment are located in a public stockyard, such pens are regarded as part of the premises of that establishment. 309.1(b) |
| M-3 | Animals have access to water in all holding pens. If held longer than 24 hours, feed is provided. 313.2(e) |
| M-4 | Floors of livestock pens, ramps, and driveways are constructed and maintained as to provide good footing for livestock. 313.1 |
| M-5 | Handling of livestock from the unloading ramps to the stunning area is done in a humane manner. 313.2(a) |
| M-6 | Seriously crippled animals and non-ambulatory disabled beef, “downers," are not slaughtered for human consumption (BSE requirement). 309.2(b) |
| M-7 | Livestock found to be dead or in a dying condition on the premises of an establishment are identified as condemned and disposed of. 309.3(a) |
| M-8 | Any swine having a temperature of 106°F or higher and any cattle, sheep, goats, horses, mules, or other equines having a temperature of 105°F or higher are identified as condemned. 309.3(c) |
| M-9 | All livestock showing, on ante-mortem inspection, symptoms of anaplasmosis, ketosis, leptospirosis, listeriosis, parturient paresis, pseudorabies, rabies, scrapie, tetanus, grass tetany, transport tetany, strangles, purpura hemorrhagica, azoturia, infectious equine encephalomyelitis, toxic encephalomyelitis (forage poisoning), dourine, acute influenza, generalized osteoporosis, glanders (farcy), acute inflammatory lameness or extensive fistula shall be condemned. 309.4 |
| M-10 | Humane methods of slaughter are applied within an appropriate stunning area. 313 |
| M-11 | Animals are adequately stunned prior to being shackled, hoisted, thrown, cast, or cut (bleeding). 313.2 |
| M-12 | A careful post-mortem examination and inspection is made of carcasses and parts of all livestock slaughtered. 310.1(a) |
| M-13 | The head, tail, tongue, thymus gland, and all viscera of each slaughtered animal are handled in such a manner as to identify them with the rest of the carcass and as being derived from the particular animal involved, until the post-mortem examination of the carcass and parts thereof has been completed. 310.2(a) |
| M-14 | Identification devices (i.e., ear tags) are removed from the animal's hide or ear by an establishment's employee and are properly affixed to the corresponding carcass. Supervising veterinarian may allow alternate methods of tracing carcasses back to live animal records. 310.2(b) |



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TABLE M-I. Slaughter and Fabrication of Meat Products OCONUS Requirements as Specified in 9 CFR Part 309, 310, 311, 313 and 416 – Continued.

| ITEM | REQUIREMENT |
|---|---|
| M-15 | Each carcass, including all detached organs and other parts, in which any lesion or condition is found that, might render the meat or any part unfit for food purposes, or otherwise adulterated, and which, for that reason, would require a subsequent inspection, is retained. All parts are retained until an approved veterinary final inspection has been completed. Retained carcasses are not washed or trimmed unless authorized by veterinary official. 310.3 |
| M-16 | Each carcass or part which is found on final inspection to be unsound, unhealthful, unwholesome, or otherwise adulterated is conspicuously marked. 310.5 |
| M-17 | Spermatic cords and pizzles are removed from all carcasses. Preputial diverticuli are removed from hog carcasses. 310.7 |
| M-18 | Carcasses found before evisceration to be affected with anthrax are not eviscerated but are retained, condemned, and immediately tanked and the complete working area is cleaned and disinfected immediately and disinfected. 310.9 |
| M-19 | When a carcass is to be dressed with the skin left on, the skin is thoroughly washed and cleaned before any incision is made for the purpose of removing any part thereof or evisceration. 310.10 |
| M-20 | The sternum of each carcass is split and abdominal and thoracic viscera are removed at the time of slaughter in order to allow proper inspection. 310.12 |
| M-21 | The kidney capsule is opened to expose the kidneys for the purpose of inspection. 310.19 |
| M-22 | When only a portion of a carcass is to be condemned on account of slight bruises, either the bruised portion is removed immediately and disposed of, or the carcass is promptly placed in a retaining room and kept until chilled, and the bruised portion is then removed and disposed of. 310.14 |
| M-23 | The entire carcass of swine, cattle, sheep, goats, and equines shall be condemned when the lesions of tuberculosis are generalized, or a lesion is found in any muscle or intermuscular tissue, bone, joint, abdominal organ or in any lymph node as a result of draining a muscle, bone, joint, or abdominal organ. 311.2 |



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TABLE M-I. Slaughter and Fabrication of Meat Products OCONUS Requirements as Specified in 9 CFR Part 309, 310, 311, 313 and 416 – Continued.

| ITEM | REQUIREMENT |
|---|---|
| M-24 | a. Livestock carcasses will be condemned on post-mortem inspection for: anthrax, blackleg, unhealed vaccine lesions, infectious anemia (swamp fever), strangles, purpura hemorrhagica, azoturia, infectious equine encephalomyelitis, toxic encephalomyelitis (forage poisoning), dourine, acute influenza, generalized osteoporosis, glanders (farcy), acute inflammatory lameness or extensive fistula.<br>b. Livestock carcasses showing lesions of the following named diseases or conditions on post-mortem examination shall be condemned, except when recovery has occurred to the extent that only localized lesions remain and the affected organs and parts have been removed. Anaplasmosis, bacillary hemoglobinuria in cattle, babesiosis, bluetongue, hemorrhagic septicemia, icterohematuria in sheep, infection bovine rhinotracheitis, leptospirosis, and malignant epizootic catarrhal fever. 311.10 |
| M-25 | The equipment used in the dressing of such carcasses, such as viscera trucks or inspection tables, shall be sanitized with hot water having a minimum temperature of 180°F. Carcasses or parts of carcasses contaminated by contact with such diseased carcasses shall be condemned unless all contaminated tissues are removed within 2 hours. 311.16(b) |
| M-26 | Carcasses, organs, and other parts are handled in a sanitary manner to prevent contamination (adulteration) with fecal material, urine, bile, hair, dirt, or foreign matter; however, if contamination occurs, it is promptly removed in a manner satisfactory to prevent further contamination and that removes all traces of the adulterant. 310.18(a) |
| M-27 | A supply of potable, running water at a suitable temperature and under pressure as needed, must be provided in all areas when required (for cleaning rooms and equipment, utensils, and packaging facilities and for employee sanitary facilities, etc. 416.2(g)(1) |
| M-28 | Non-potable water may not be used in areas where edible product is handled or prepared or in any manner that would allow it to adulterate edible product or create insanitary conditions. 416.12 (g)(6) |
| M-29 | All food contact surfaces must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions and the adulteration of product. 416.4(a) |
| M-30 | Non-food contact surfaces of facilities, equipment, and utensils used in the operation of the establishment must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions and the adulteration of product. 416.4(a) |
| M-31 | Any person who has or appears to have an infectious disease, open lesion, including boils, sores, or infected wounds, or any other abnormal source of microbial contamination, must be excluded from any operations which could result in product adulteration and the creation of insanitary conditions until the condition is corrected. 416.5(c) |



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TABLE M-I. Slaughter and Fabrication of Meat Products OCONUS Requirements as Specified in 9 CFR Part 309, 310, 311, 313 and 416 – Continued.

| | |
|---|---|
| M-32 | Aprons, frocks, and other outer clothing worn by persons who handle product are clean and are changed each day. 416.5(b) |
| M-33 | Livestock, except swine, must be tested for *Escherichia coli* Biotype 1 (*E. coli*) at a minimum of one sample during each week of operation. 310.25 (a)(2) |
| M-34 | Official swine slaughter establishments must develop, implement, and maintain written procedures to prevent contamination of carcasses and parts with visible fecal material, ingesta, and milk contamination throughout the entire slaughter and dressing operation. Establishments must incorporate these procedures into their HACCP plans, or sanitation SOPs, or other prerequisite programs. These procedures must include sampling and analysis for microbial organisms IAW the sampling location and frequency requirements in (c)(1) and (2) of this section to monitor their ability to maintain process control. 310.18(c) |
| M-35 | Swine slaughter establishments must collect and analyze samples for microbial organisms at the pre-chill and post-chill points in the process. 310.18(c)(1) |
| M-36 | Swine slaughter establishments must, at a minimum, collect and analyze samples at a frequency proportional to the establishment's volume of production once per 1,000 carcasses, but a minimum of once during each week of operation. 310.18(c)(2) |
| M-37 | Swine slaughter establishments must maintain daily records sufficient to document the implementation and monitoring of all test results and retain them for at least one year and must be accessible to audit personnel. 310.18(c)(2) and (d) |



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TABLE M-II. Hazard Analysis and Critical Control Point Development and Application in OCONUS Meat Product Slaughter and Fabrication Plants.

| ITEM | REQUIREMENT |
|---|---|
| M-38 | Every plant subject to this Appendix shall conduct, or have conducted for it, a hazard analysis to determine the food safety hazards reasonably likely to occur in the production process and identify the preventive measures the establishment can apply to control those hazards. 417.2(a)(1) |
| M-39 | Hazard analysis shall include food safety hazards that can be reasonably expected to occur before, during, and after entry into the establishment. 417.2(a)(1) |
| M-40 | A flow chart describing the steps of each process and product flow in the establishment shall be prepared, and the intended use or consumers of the finished product shall be identified. 417.2(a)(2) |
| M-41 | At a minimum, food safety hazards that must be addressed (as they apply to the product or associated processes) include: natural toxins, microbiological contamination, chemical contamination, pesticides, drug residues, zoonotic diseases, decomposition, parasites, unapproved use of direct or indirect food or color additives, and physical hazards. 417.2(a)(3) |
| M-42 | Every plant subject to this Appendix develop and implement a written HACCP plan covering each product produced by that establishment whenever a hazard analysis reveals one or more food safety hazards that are reasonably likely to occur, based on the hazard analysis conducted IAW this section, in each of the following processing categories:<br>(1) Slaughter – All Species.<br>(2) Raw product – Ground.<br>(3) Raw product – Not ground.<br>(4) Thermally processed – Commercially sterile.<br>(5) Not heat treated – Shelf stable.<br>(6) Heat treated – Shelf stable.<br>(7) Fully cooked – Not shelf stable.<br>(8) Heat treated but not fully cooked – Not shelf stable.<br>(9) Product with secondary inhibitors – Not shelf stable. 417.2(b) |
| M-43 | A single HACCP plan may encompass multiple products within a single processing category identified in this paragraph, if the food safety hazards, critical control points, critical limits, and procedures required to be identified and performed are essentially the same, provided that any required features of the plan that are unique to a specific product are clearly delineated in the plan and are observed in practice. 417.2(b)(2) |
| M-44 | HACCP plans for thermally processed/commercially sterile products do not have to address food safety hazards associated with microbiological contamination if the raw materials are produced in a Directory listed or exempt canned food plant. 417.2(b)3 |



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TABLE M-II. Hazard Analysis and Critical Control Point Development and Application in OCONUS Meat Product Slaughter and Fabrication Plants - Continued.

| ITEM | REQUIREMENT |
|---|---|
| M-45 | The HACCP plan shall, at a minimum:<br>(1) List the food safety hazards identified which must be controlled.<br>(2) List the critical control points (CCPs) for each of the identified food safety hazards, including, (as appropriate), CCPs designed to control food safety hazards that could be introduced in the establishment and CCPs designed to control food safety hazards introduced outside the establishment.<br>(3) List the Critical Limits (CLs) that must be met at each of the CCPs to adequately control the identified hazard.<br>(4) List the procedures, and the frequency with which those procedures will be performed, that will be used monitor each of the CCPs to ensure compliance with the CLs.<br>(5) Include all corrective actions that have been developed to be followed in response to any deviation from a CL at a CCP.<br>(6) Provide for a recordkeeping system that documents the monitoring of the CCPs which contains the actual values and observations obtained during monitoring.<br>(7) List the verification procedures and the frequency with which those procedures will be performed. 417.2(c) |
| M-46 | The HACCP plan shall be signed and dated:<br>(1) By the responsible establishment individual, signifying that the establishment accepts and will implement the HACCP plan,<br>(2) Upon initial acceptance, any modification, and annually. 417.2(d) |
| M-47 | The written HACCP plan shall identify the corrective action to be followed in response to a deviation from a critical limit. The HACCP plan shall describe the corrective action to be taken, and assign responsibility for taking corrective action, to ensure:<br>(1) The cause of the deviation is identified and eliminated,<br>(2) The CCP will be under control after the corrective action is taken,<br>(3) Measures to prevent recurrence are established, and<br>(4) No product that is injurious to health or otherwise adulterated as a result of the deviation enters commerce. 417.3 (a) |



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TABLE M-II. Hazard Analysis and Critical Control Point Development and Application in OCONUS Meat Product Slaughter and Fabrication Plants - Continued.

| ITEM | REQUIREMENT |
|---|---|
| M-48 | If a deviation not covered by a specified corrective action occurs, or if another unforeseen hazard arises, the establishment shall:<br>(1) Segregate and hold the affected product, at least until the CCP is under control and measures to prevent recurrence are established,<br>(2) Perform a review to determine the acceptability of the affected product for distribution,<br>(3) Take action, when necessary, with respect to the affected product to ensure that no product that is injurious to health or otherwise adulterated, as a result of the deviation, enters commerce,<br>(4) Perform or obtain reassessment by an individual trained in accordance with § 417.7 of this part, to determine whether the newly identified deviation or other unforeseen hazard should be incorporated into the HACCP plan. 417.3 (b) |
| M-49 | Upon completion of the hazard analysis and development of the HACCP plan, the establishment shall conduct activities designed to determine that the HACCP plan is functioning as intended. During this validation period, the establishment shall repeatedly test the adequacy of the CCP's, CLs, monitoring and recordkeeping procedures, and corrective actions set forth in the HACCP plan. 417.4 (a)(1) |
| M-50 | Ongoing verification activities include, but are not limited to:<br>(1) The calibration of process-monitoring instruments,<br>(2) Direct observations of monitoring activities and corrective actions, and<br>(3) The review of records generated and maintained IAW CCP monitoring and Corrective actions. 417.4(a)(2) |
| M-51 | (1) Each plant shall reassess the adequacy of the HACCP plan at least annually and whenever any changes occur that could affect the hazard analysis or alter the HACCP plan. Such changes may include, but are not limited to, changes in: raw materials or source of raw materials, product formulation, slaughter or processing methods or systems, production volume, personnel, packaging, finished product distribution systems, or the intended use or consumers of the finished product. This shall include plants that do not have a HACCP plan because a hazard analysis has revealed no food safety hazards that are reasonably likely to occur.<br>(2) The reassessment shall be performed by an individual who has successfully completed a course of instruction in the application of the seven HACCP principles to meat or poultry processing, including a segment on the development of a HACCP plan for a specific product and on record review. 417.4(a)(3), 417.4(b) |



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TABLE M-II. Hazard Analysis and Critical Control Point Development and Application in OCONUS Meat Product Slaughter and Fabrication Plants - Continued.

| ITEM | REQUIREMENT |
|---|---|
| M-52 | Each plant shall maintain the following records documenting the establishment's HACCP plan:<br>(1) The written hazard analysis, including all supporting documentation,<br>(2) The written HACCP plan, including decision-making documents associated with the selection and development of CCP's and critical limits, and documents supporting both the monitoring and verification procedures selected and the frequency of those procedures,<br>(3) Records documenting the monitoring of CCP's and their critical limits, including the recording of actual times, temperatures, or other quantifiable values, as prescribed in the establishment's HACCP plan,<br>(4) Calibration of process-monitoring instruments,<br>(5) Corrective actions, including all actions taken in response to a deviation,<br>(6) Verification procedures and results,<br>(7) Product codes, product name or identity, or slaughter production lot. 417.5 |
| M-53 | Individuals involved in the development or reassessment and modification of the HACCP plan, to include adaptation of generic models, shall have successfully completed a course of instruction in the application of the seven HACCP principles to meat or poultry processing, including a segment on the development of a HACCP plan for a specific product and on record review. 417.7 |



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# ASEPTIC PROCESSING AND PACKAGING OF THERMALLY PROCESSED LOW-ACID FOODS PACKAGED IN HERMETICALLY SEALED CONTAINERS

## N.1 SCOPE

N.1.1 Scope. This appendix establishes the food safety and related requirements for food establishments performing aseptic processing and packaging of thermally processed low-acid foods packaged in hermetically sealed containers. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## N.2 APPLICABLE DOCUMENTS

N.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

N.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 21, Part 113 - Thermally Processed Low-Acid Canned Foods Packaged in Hermetically Sealed Containers

(Copy of this document is available online at: https://www.govinfo.gov/.)

U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Guide to Inspections of Aseptic Processing and Packaging for the Food Industry.

(Available on-line at: https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/inspection-guides/aseptic-processing-and-packaging-food-industry)

U.S. Public Health Service (USPHS) / Food and Drug Administration (FDA) Pasteurized Milk Ordinance (PMO), current version.

(Available online at: https://www.fda.gov/food/guidance-documents-regulatory-information-topic-food-and-dietary-supplements/milk-guidance-documents-regulatory-information.)



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N.2.3 Other Government publications. The following other Government publications form a part of this document to the extent specified therein.

GMA RESEARCH AND EDUCATION FOUNDATION

Canned Foods: Principles of Thermal Process Control, Acidification and Container Closure Evaluation (ISBN 978-0-937774-58-8).

(Available for purchase online and upon attendance of the Better Process Control School. Designated by the US FDA as the basic text for use at approved schools of instruction. Emphasis for this appendix should be on the “Aseptic Processing and Packaging Systems” chapter).

N.3 DEFINITIONS

N.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all references and associated documents apply to this appendix.

N.4 CHECKLIST

N.4.1 Checklist. Guidelines for auditing establishments performing aseptic processing and packaging of thermally processed low-acid foods packaged in hermetically sealed containers are contained in the following Table N-I checklist.



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TABLE N-I. Aseptic Processing and Packaging of Thermally Processed Low-Acid Foods Packaged in Hermetically Sealed Containers Requirements as Specified in 21 CFR Part 113 and 173.

| ITEM | REQUIREMENT |
|---|---|
| N-1 | Trained personnel identified by an approved school as having satisfactorily completed a prescribed course of instructions shall supervise the operators of aseptic processing and packaging systems. 113.10 |
| N-2 | Scheduled processes for low-acid foods have been established by qualified persons having expert knowledge of thermal process requirements and having adequate facilities for making such determinations. Critical factors (e.g., aW; pH; target F0 and/or minimum sterilization time and temperature; etc.) shall be specified in the scheduled process. 113.83 |
| N-3 | Processing and production information is entered at the time it is observed by the processing system operator on forms mandatory for the aseptic processing and packaging system used and includes: company name, date, product, production code, sterilizer number, container size, number of containers produced per coding interval, product temperature and pressure in the holding tube outlet; differential pressure, if a product-to-product regenerator is used; product flow rate, as determined by the metering pump or by filling and closing rates; retention time of containers, and closures when applicable, in the sterilizing environment; sterilization media flow rate or temperature or both; retention time of containers, and closures when applicable, in the sterilizing environment; and, when a batch system is used for container and/or closure sterilization, sterilization cycle times and temperatures. The processing system operator or other designated person shall sign or initial each record form. 113.100(a) – (e) |
| N-4 | A representative of plant management who is qualified by suitable training or experience shall review, sign and date all processing and production records for completeness and to ensure that the product received the scheduled process no later than 1 working day after the actual process, and before shipment or release for distribution. 113.100(b) |
| N-5 | Written records of all container closure examinations shall specify the product code, the date and time of container closure inspections, the measurements obtained, and all corrective actions taken. Records shall be signed or initialed by the container closure inspector and reviewed by management with sufficient frequency to ensure that the containers are hermetically sealed. 113.100(c) |
| N-6 | Records shall be maintained to identify the initial distribution of the finished product to facilitate, when necessary, the segregation of specific food lots that may have become contaminated or otherwise rendered unfit for their intended use. 113.100(d) |
| N-7 | Copies of all records, except those required under Section 113.83 establishing scheduled processes, shall be retained at the processing plant for a period of not less than 1 year from the date of manufacture, and at the processing plant or other reasonably accessible location for an additional 2 years. 113.100(e) |



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TABLE N-I. Aseptic Processing and Packaging of Thermally Processed Low-Acid Foods Packaged in Hermetically Sealed Containers Requirements as Specified in 21 CFR Part 113 and 173 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| N-8 | Incubation tests should be conducted on a representative sample of containers of product from each code; records of the test results should be maintained. 113.40.(g).(3) |
| N-9 | Critical factors specified in the scheduled process shall be measured and recorded on the processing record at intervals of sufficient frequency to ensure that the factors are within the limits specified in the scheduled process. Such measurements and recordings should be done at intervals not to exceed 15 minutes. 113.40.(g).(4) |
| N-10 | Each product sterilizer shall be equipped with at least one mercury-in-glass thermometer or an equivalent temperature-indicating device, such as a thermocouple-recorder. Tested for accuracy against a known accurate standard thermometer upon installation and at least once a year thereafter, or more frequently if necessary, to ensure their accuracy. Installed where it can be accurately and easily read. The temperature-indicating device shall be the reference instrument for indicating the processing temperature. 113.40.(g).(1).(i).(a) |
| N-11 | There shall be an accurate temperature recording device on each product sterilizer. Installed in the product at the holding-tube outlet between the holding tube and the inlet to the cooler. The temperature chart shall be adjusted to agree as nearly as possible with, but to be in no event higher than, the reference instrument for indicating the processing temperature. A means of preventing unauthorized changes in adjustment shall be provided. 113.40.(g).(1).(i).(b) |
| N-12 | An accurate temperature recorder-controller shall be located in the product sterilizer at the final heater outlet. It shall be capable of ensuring that the desired product sterilization temperature is maintained. 113.40.(g).(1).(i).(c) |
| N-13 | When a product-to-product regenerator is used to heat the cold unsterilized product entering the sterilizer by means of a heat exchange system, it shall be designed, operated, and controlled so that the pressure of the sterilized product in the regenerator is greater than the pressure of any unsterilized product in the regenerator to ensure that any leakage in the regenerator is from the sterilized product into the unsterilized product. 113.40.(g).(1).(i).(d) |
| N-14 | When a product-to-product regenerator is used, there shall be an accurate differential pressure recorder-controller installed on the regenerator. The controller shall be tested for accuracy against a known accurate standard pressure indicator upon installation and at least once every 3 months of operation thereafter, or more frequently if necessary, to ensure its accuracy. One pressure sensor shall be installed at the sterilized product regenerator outlet and the other pressure sensor shall be installed at the unsterilized product regenerator inlet. 113.40.(g).(1).(i).(e) |



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TABLE N-I. Aseptic Processing and Packaging of Thermally Processed Low-Acid Foods Packaged in Hermetically Sealed Containers Requirements as Specified in 21 CFR Part 113 and 173 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| N-15 | A metering pump shall be located upstream from the holding tube and shall be operated to maintain the required rate of product flow. A means of preventing unauthorized speed changes shall be provided. 113.40.(g).(1).(i).(f) |
| N-16 | The product-sterilizing holding tube shall be designed to give continuous holding of every particle of food for at least the minimum holding time specified in the scheduled process. The holding tube shall be designed so that no portion of the tube between the product inlet and the product outlet can be heated, and it must be sloped upward at least 0.25 inch per foot. 113.40.(g).(1).(i).(g) |
| N-17 | If a processor elects to install a flow-diversion system, it should be installed in the product piping located between the product cooler and the product filler or aseptic surge tank and should be designed to divert flow away from the filler or aseptic surge tank automatically. Controls and/or warning systems should be designed and installed with necessary sensors and actuators to operate whenever the sterilizing temperature in the holding tube, pressure in the holding tube, or pressure differential in the product regenerator drops below specified limits. Flow-diversion systems should be designed and operated in accordance with recommendations of an aseptic processing and packaging authority. 113.40.(g).(1).(i).(h) |
| N-18 | Product coolers, aseptic surge tanks, or any other equipment downstream from the holding tube, with rotating or reciprocating shafts, valve stems, instrument connections, or other such points, are subject to potential entry of microorganisms into the product. Such locations in the system should be equipped with steam seals or other effective barriers at the potential access points. Appropriate means should be provided to permit the operator to monitor the performance of the seals or barriers during operations. 113.40.(g).(1).(i).(i) |
| N-19 | Before the start of aseptic processing operations, the product sterilizer holding tube and all product-contact surfaces downstream all the way to the aseptic surge tank or filler shall be brought to a condition of commercial sterility. 113.40.(g).(1).(ii).(a) |
| N-20 | When product temperature in the holding tube drops below the temperature specified in the scheduled process, product flow should be diverted away from the filler or aseptic surge tank by means of a flow-diversion system. If for any reason product subjected to a temperature drop below the scheduled process is filled into containers, the product shall be segregated from product that received the scheduled process. The product holding tube and any further system portions affected shall be returned to a condition of commercial sterility before product flow is resumed to the filler or to the aseptic surge tank. 113.40.(g).(1).(ii).(b) |



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TABLE N-I. Aseptic Processing and Packaging of Thermally Processed Low-Acid Foods Packaged in Hermetically Sealed Containers Requirements as Specified in 21 CFR Part 113 and 173 – Continued.

| ITEM | REQUIREMENT |
|---|---|
| N-21 | When a regenerator is used, the product may lose sterility whenever the pressure of sterilized product in the regenerator is less than 1 pound per square inch greater than the pressure of unsterilized product in the regenerator. In this case, product flow should be diverted away from the filler or aseptic surge tank by means of the flow-diversion system. If for any reason the product is filled into containers, the product shall be segregated from product that received the scheduled process and shall be reprocessed or destroyed. Product flow to the filler or to the aseptic surge tank shall not be resumed until the cause of the improper pressure relationships in the regenerator has been corrected and the affected system(s) has been returned to a condition of commercial sterility. 113.40.(g).(1).(ii).(c) |
| N-22 | When an aseptic surge tank is used, conditions of commercial sterility may be lost when the sterile air overpressure or other means of protection drops below the scheduled process value. Product flow to and/or from the aseptic surge tank shall not be resumed until the potentially contaminated product in the tank is removed, and the aseptic surge tank has been returned to a condition of commercial sterility. 113.40.(g).(1).(ii).(d) |
| N-23 | Readings at the following points shall be observed and recorded at the start of aseptic packaging operations and at intervals of sufficient frequency to ensure that these values are as specified in the scheduled process: Temperature-indicating device in holding tube outlet; temperature recorder in holding tube outlet; temperature recorder-controller at final heater outlet; differential pressure recorder-controller, if a product-to-product regenerator is used; product flow rate as established by the metering pump or as determined by filling and closing rates and, if an aseptic surge tank is used, sterile air pressure or other protection means; and proper performance of seam seals or other similar devices. The measurements and recordings should be made at intervals not to exceed 1 hour. 113.40.(g).(1).(ii).(e) |
| N-24 | The container and closure sterilization system and product filling and closing system shall be instrumented to demonstrate that the required sterilization is being accomplished continuously. Automatic recording devices shall be used to record, when applicable, the sterilization media flow rates, temperature, concentration, or other factors. When a batch system is used for container sterilization, the sterilization conditions shall be recorded. 113.40.(g).(2).(i).(a) |



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TABLE N-I. Aseptic Processing and Packaging of Thermally Processed Low-Acid Foods Packaged in Hermetically Sealed Containers Requirements as Specified in 21 CFR Part 113 and 173 – Continued.

| ITEM | REQUIREMENT |
|---|---|
| N-25 | A method(s) shall be used either to give the retention time of containers, and closures if applicable, in the sterilizing environment specified in the scheduled process, or to control the sterilization cycle at the rate specified in the scheduled process. A means of preventing unauthorized speed changes must be provided. 113.40.(g).(2).(i).(b) |
| N-26 | Before the start of packaging operations, both the container and closure sterilizing system and the product filling and closing system shall be brought to a condition of commercial sterility. 113.40.(g).(2).(ii).(a) |
| N-27 | A system shall be provided to stop packaging operations, or alternatively to ensure segregation of any product packaged when the packaging conditions fall below scheduled processes. Compliance with this requirement may be accomplished by diverting product away from the filler, by preventing containers from entering the filler, or by other suitable means. In the event product is packaged under conditions below those specified in the scheduled process, all such product shall be segregated. The system(s) shall be returned to a condition of commercial sterility before resuming packaging operations. 113.40.(g).(2).(ii).(b) |
| N-28 | Observations and measurements of operating conditions shall be made and recorded at intervals of sufficient frequency to ensure that commercial sterility of the food product is being achieved; such measurements shall include the sterilization media flow rates, temperatures, the container and closure rates (if applicable) through the sterilizing system, and the sterilization conditions if a batch system is used for container sterilization. The measurements and recordings should be made at intervals not to exceed 1 hour. 113.40.(g).(2).(ii).(c) |
| N-29 | When steam comes into contact with the product, through direct injection or infusion of steam into the product; during the exhausting of containers in a steam exhaust box; through injection of steam into the headspace of containers to form a vacuum, or through any other means, the boiler additives must be approved for use as a food additive and labeled for that use. 173.310 |
| N-30 | Establishment must demonstrate proper evaluation of containers and closures. 113.60 |



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# SLAUGHTER AND FABRICATION OF POULTRY PRODUCTS OCONUS

## O.1 SCOPE

O.1.1 Scope. This appendix establishes the food safety and related requirements for poultry slaughter and processing in OCONUS establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## O.2 APPLICABLE DOCUMENTS

O.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed. Items A-30 through A-84 are not applicable to this appendix.

O.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 9, Part 381 | - | Poultry Products Inspection Regulations |
| Title 9, Part 416 | - | Sanitation |
| Title 9, Part 417 | - | Hazard Analysis and Critical Control Point (HACCP) Systems |

(Copies of these documents are available online at: https://www.govinfo.gov/.)

## O.3 DEFINITIONS

O.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

## O.4 REQUIREMENTS

O.4.1 HACCP. A validated HACCP plan is required for poultry processing establishments in accordance with 9 CFR Part 417.



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O.4.2 Flock health. Flocks destined for food production shall be free of Tuberculosis, diseases of the leukosis complex, septicemia or toxemia, airsacculitis, inflammatory processes, tumors, and parasites.

O.4.3 Bleeding. Birds are bled immediately after stunning (animals shall die by exsanguination). Birds slaughtered to meet Kosher or Halal requirements are killed by bleeding without previous stunning simultaneously by severing the trachea and major blood vessels.

O.4.4 Scalding and de-feathering. Water in the scald tank shall be circulated to preclude the accumulation of pathogens in the scald tank water, which would contaminate the birds passing through it.

O.4.5 Chilling. Establishments utilizing forced air chilling of the product will ensure the Food Safety Program addresses the environmental as well as the proper carcass chilling temperature and time concerns.

O.5 CHECKLISTS

O.5.1 Checklists. Guidelines for auditing poultry slaughter and processing establishments are contained in the following Table O-I through O-IV checklists.



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TABLE O-I. Slaughter of Poultry OCONUS Requirements as Specified in 9 CFR Part 381, 416, and 417.

| ITEM | REQUIREMENT |
|---|---|
| O-1 | Facilities for ante-mortem and post-mortem inspection will comply with minimum facility requirements for the system of ant-mortem and post-mortem inspection employed. 381.36 and 381.76 |
| O-2 | Birds will receive ante-mortem inspection on the day of slaughter and be properly segregate when required. 381.70 thru 381.75 |
| O-3 | Poultry must be slaughtered in accordance with good commercial practices in a manner that will result in thorough bleeding of the carcasses and ensure that breathing has stopped prior to scalding. Blood from the killing operation must be confined to a relatively small area. 381.65(b) |
| O-4 | When thawing frozen ready-to-cook poultry in water methods must be used that prevent adulteration or net weight gain by the poultry. 381.65(c) |
| O-5 | The water used in washing the poultry must be permitted to drain freely from the body cavity. 381.65(d) |
| O-6 | Detached ova may be collected for human food and handled only in accordance with 9 CFR 590.44 and may leave the establishment only to be moved to an official egg product processing plant for processing. Ova from condemned carcasses must be condemned and treated as required in § 381.95. 381.65(e) |
| O-7 | Official poultry slaughter establishments must develop, implement, and maintain written procedures to ensure that poultry carcasses contaminated with visible fecal material do not enter the chiller. Establishments must incorporate these procedures into their HACCP plans, or sanitation SOPs, or other prerequisite programs. 381.65(f) |
| O-8 | Official poultry slaughter establishments must develop, implement, and maintain written procedures to prevent contamination of carcasses and parts by enteric pathogens and fecal contamination throughout the entire slaughter and dressing operation. Establishments must incorporate these procedures into their HACCP plans, or sanitation SOPs, or other prerequisite programs. 381.65(g) |
| O-9 | Establishments must collect and analyze samples for microbial organisms at the pre-chill and post-chill points in the process. 381.65(g) |
| O-10 | Establishments, must, at a minimum, collect and analyze samples at a frequency proportional to the establishment's volume of production at the following rates:<br>(A) Chickens. Once per 22,000 carcasses, but a minimum of once during each week of operation.<br>(B) Turkeys, ducks, geese, guineas, and squabs. Once per 3,000 carcasses, but at a minimum once each week of operation. 381.65(g) |
| O-11 | Establishments must maintain daily records sufficient to document the implementation and monitoring of all test results and retain them for at least one year and must be accessible to audit personnel. 381.65(g) and (h) |



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TABLE O-I. Slaughter of Poultry OCONUS Requirements as Specified in 9 CFR Part 381, 416, and 417 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| O-12 | Each official poultry slaughter establishment must ensure that all poultry carcasses, parts, and giblets are chilled immediately after slaughter operations so that there is no outgrowth of pathogens, unless such poultry is to be frozen or cooked immediately at the official establishment. 381.66(b) |
| O-13 | Previously chilled poultry carcasses and major portions must be kept chilled so that there is no outgrowth of the pathogens, unless such poultry is to be packed and frozen immediately at the official establishment. 381.66(b) |
| O-14 | After product has been chilled, the establishment must prevent the outgrowth of pathogens on the product as long as the product remains at the establishment. 381.66(b) |
| O-15 | The establishment must develop, implement, and maintain written procedures for chilling that address, at a minimum, the potential for pathogen outgrowth, the conditions affecting carcass chilling, and when its chilling process is completed. The establishment must incorporate these procedures into its HACCP plan, or sanitation SOP, or other prerequisite program. 381.66(b) |
| O-16 | Only ice produced from potable water may be used for ice and water chilling, except that water and ice used for chilling may be reused in accordance with § 416.2(g). The ice must be handled and stored in a sanitary manner. 381.66(c) |
| O-17 | Poultry washing, chilling, and draining practices and procedures must be such as will minimize water absorption and retention at time of packaging. 381.66(d) |
| O-18 | An antimicrobial intervention may be applied with water at the beginning of air chilling processes, provided that its use does not result in any net pick-up of water or moisture during the air chilling process. 381.66(e) |
| O-19 | Ready-to-cook poultry shall be placed into a freezer within 48 hours after initial chilling in accordance with paragraph (b) of this section. During this period, if such poultry is not immediately placed into a freezer after chilling and packaging, it shall be held at 36 °F or lower. 381.66(f) |
| O-20 | Warm packaged ready-to-cook poultry which is to be chilled by immediate entry into a freezer within the official establishment shall within 2 hours from time of slaughter be placed in a plate freezer or a freezer with a functioning circulating air system where a temperature of -10 °F or lower is maintained. 381.66(f) |
| O-21 | A post-mortem inspection will be made on a bird-by-bird basis on all poultry eviscerated. 381.76(a) |



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TABLE O-I. Slaughter of Poultry OCONUS Requirements as Specified in 9 CFR, Part 381, 416, and 417 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| O-22 | Each carcass, including all parts thereof, in which there is any lesion of disease, or other condition which might render such carcass or any part thereof adulterated, shall be held for further examination if a final decision cannot be made on first examination. 381.77 |
| O-23 | The identity of each such carcass, including all parts thereof, shall be maintained until a final examination has been completed. 381.77 |
| O-24 | Each carcass found to be adulterated shall be condemned, except that any such articles which may be made not adulterated by re-processing need not be condemned if reprocessed under the supervision of an inspector and thereafter found to be not adulterated. 381.78(a) |
| O-25 | Carcasses suspected of containing biological residues and any parts of carcasses in such lot which are condemned shall be kept separate from all other condemned carcasses or parts. 381.78(b) |
| O-26 | The carcasses or parts of carcasses of all poultry inspected and found at the time of post-mortem inspection, or at any subsequent inspection, to be affected with any of the diseases or conditions named in this subpart, shall be disposed of in accordance with the section pertaining to the disease or condition. 381.80 thru 381.93 |
| O-27 | Condemned carcasses shall be disposed of in accordance with one of the approved methods described in reference requirements, under the supervision of an inspector. 381.95 |
| O-28 | Each official establishment must be operated and maintained in a manner sufficient to prevent the creation of insanitary conditions and to ensure that product is not adulterated. 416.1 |
| O-29 | A supply of running water that complies with the National Primary Drinking Water regulations (40 CFR part 141), at a suitable temperature and under pressure as needed, must be provided in all areas where required (for processing product, for cleaning rooms and equipment, utensils, and packaging materials, for employee sanitary facilities, etc.). 416.2(g)(1) |
| O-30 | If an establishment uses municipal or a private well for its water supply, it must make available upon request, documentation certifying the potability of the water by a local health agency. Documentation must be renewed semi-annually for private well water supply. 416.2(g)(1) |
| O-31 | Non-potable water may not be used in areas where edible product is handled or prepared or in any manner that would allow it to adulterate edible product or create insanitary conditions. 416.2(g)(6) |
| O-32 | Any water that has never contained human waste and that is free of pathogenic organisms may be used in edible and inedible product areas, provided it does not contact edible product. 416.2(g)(5) |



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TABLE O-I. Slaughter of Poultry OCONUS Requirements as Specified in 9 CFR, Part 381, 416, and 417 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| O-33 | All food contact surfaces must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions and the adulteration of product. 416.4(a) |
| O-34 | Non-food contact surfaces of facilities, equipment, and utensils used in the operation of the establishment must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions and the adulteration of product. 416.4(a) |

TABLE O-II. Contamination with Microorganisms; Process Control Verification Criteria and Testing; Pathogen Reduction Standards for Establishments that Slaughter Ratites Requirements as Specified in 9 CFR Part 381.94, 416, and 417.

| ITEM | REQUIREMENT |
|---|---|
| O-35 | Each official establishment that slaughters ratites shall test for generic *Escherichia coli*. Establishments that slaughter ratites and livestock, shall test the type of ratites or livestock slaughtered in the greatest number. 381.94 |
| O-36 | Each establishment that slaughters ratites shall prepare written specimen collection procedures which shall identify employees designated to collect samples, and shall address location(s) of sampling, how sampling randomness is achieved, and handling of the sample to ensure sample integrity. 381.94(2) |
| O-37 | The establishment must collect samples from whole ratites at the end of the chilling process. 381.94(2) |
| O-38 | Establishments that slaughter ratites, must take samples at a frequency proportional to the establishment's volume of production at the following rate: 1 sample per 3,000 carcasses, but at a minimum one sample each week of operation. 381.94(2) |
| O-39 | Establishments shall evaluate *E. coli* test results using statistical process control techniques. Failure to meet these criteria is an indication that the establishment may not be maintaining process controls sufficient to prevent fecal contamination. 381.94(5) and (6) |
| O-40 | Condemned carcasses will be disposed of in accordance with one of the approved methods of described in reference requirements. 381.95 |



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TABLE O-III. Fabrication of Poultry OCONUS Requirements as Specified in 9 CFR, Part 381, 416, and 417.

| ITEM | REQUIREMENT |
|---|---|
| O-41 | Poultry that is mechanically separated must adhere to the bone solids and calcium content requirement in 381.173(b) and (c). |
| O-42 | A poultry product required to be prepared from a particular kind of poultry (e.g., chicken) shall not contain "Mechanically Separated (Kind of Poultry)" described in § 381.173, that is made from any other kind of poultry (e.g., Mechanically Separated Turkey). 381.174(a) |
| O-43 | "Mechanically Separated (Kind of Poultry)" described in § 381.173 may be used in the formulation of any poultry or meat food product, provided such use conforms with any applicable requirements of the definitions and standards of identity or composition in this subchapter or part 319 of this chapter, and provided that it is identified as "Mechanically Separated (Kind of Poultry)." 381.174(b) |

TABLE O-IV. Hazard Analysis and Critical Control Point Development and Application in OCONUS Poultry Slaughter and Fabrication Plants.

| ITEM | REQUIREMENT |
|---|---|
| O-44 | Every plant subject to this Appendix shall conduct, or have conducted for it, a hazard analysis to determine the food safety hazards reasonably likely to occur in the production process and identify the preventive measures the establishment can apply to control those hazards. 417.2(a)(1) |
| O-45 | Hazard analysis shall include food safety hazards that can be reasonably expected to occur before, during, and after entry into the establishment. 417.2(a)(1) |
| O-46 | A flow chart describing the steps of each process and product flow in the establishment shall be prepared, and the intended use or consumers of the finished product shall be identified. 417.2(a)(2) |
| O-47 | At a minimum, food safety hazards that must be addressed (as they apply to the product or associated processes) include: natural toxins, microbiological contamination, chemical contamination, pesticides, drug residues, zoonotic diseases, decomposition, parasites, unapproved use of direct or indirect food or color additives, and physical hazards. 417.2(a)(3) |



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TABLE O-IV. Hazard Analysis and Critical Control Point Development and Application in OCONUS Poultry Slaughter and Fabrication Plants – Continued.

| ITEM | REQUIREMENT |
|---|---|
| O-48 | Every plant subject to this Appendix develop and implement a written HACCP plan covering each product produced by that establishment whenever a hazard analysis reveals one or more food safety hazards that are reasonably likely to occur, based on the hazard analysis conducted IAW this section, in each of the following processing categories:<br>(1) Slaughter – All Species.<br>(2) Raw product – Ground.<br>(3) Raw product – Not ground.<br>(4) Thermally processed – Commercially sterile.<br>(5) Not heat treated – Shelf stable.<br>(6) Heat treated – Shelf stable.<br>(7) Fully cooked – Not shelf stable.<br>(8) Heat treated but not fully cooked – Not shelf stable.<br>(9) Product with secondary inhibitors – Not shelf stable. 417.2(b) |
| O-49 | A single HACCP plan may encompass multiple products within a single processing category identified in this paragraph, if the food safety hazards, critical control points, critical limits, and procedures required to be identified and performed are essentially the same, provided that any required features of the plan that are unique to a specific product are clearly delineated in the plan and are observed in practice. 417.2(b)(2) |
| O-50 | HACCP plans for thermally processed/commercially sterile products do not have to address food safety hazards associated with microbiological contamination if the raw materials are produced in a Directory listed or exempt canned food plant. 417.2(b)3 |
| O-51 | The HACCP plan shall, at a minimum:<br>(1) List the food safety hazards identified which must be controlled.<br>(2) List the critical control points (CCPs) for each of the identified food safety hazards, including, (as appropriate), CCPs designed to control food safety hazards that could be introduced in the establishment and CCPs designed to control food safety hazards introduced outside the establishment.<br>(3) List the Critical Limits (CLs) that must be met at each of the CCPs to adequately control the identified hazard.<br>(4) List the procedures, and the frequency with which those procedures will be performed, that will be used to monitor each of the CCPs to ensure compliance with the CLs.<br>(5) Include all corrective actions that have been developed to be followed in response to any deviation from a CL at a CCP.<br>(6) Provide for a recordkeeping system that documents the monitoring of the CCPs which contains the actual values and observations obtained during monitoring.<br>(7) List the verification procedures and the frequency with which those procedures will be performed. 417.2(c) |



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TABLE O-IV. Hazard Analysis and Critical Control Point Development and Application in OCONUS Meat Product Slaughter and Fabrication Plants - Continued.

| ITEM | REQUIREMENT |
|---|---|
| O-52 | The HACCP plan shall be signed and dated:<br>(1) By the responsible establishment individual, signifying that the establishment accepts and will implement the HACCP plan,<br>(2) Upon initial acceptance, any modification, and annually. 417.2(d) |
| O-53 | The written HACCP plan shall identify the corrective action to be followed in response to a deviation from a critical limit. The HACCP plan shall describe the corrective action to be taken, and assign responsibility for taking corrective action, to ensure:<br>(1) The cause of the deviation is identified and eliminated,<br>(2) The CCP will be under control after the corrective action is taken,<br>(3) Measures to prevent recurrence are established, and<br>(4) No product that is injurious to health or otherwise adulterated as a result of the deviation enters commerce. 417.3 (a) |
| O-54 | If a deviation not covered by a specified corrective action occurs, or if another unforeseen hazard arises, the establishment shall:<br>(1) Segregate and hold the affected product, at least until the CCP is under control and measures to prevent recurrence are established,<br>(2) Perform a review to determine the acceptability of the affected product for distribution,<br>(3) Take action, when necessary, with respect to the affected product to ensure that no product that is injurious to health or otherwise adulterated, as a result of the deviation, enters commerce,<br>(4) Perform or obtain reassessment by an individual trained in accordance with § 417.7 of this part, to determine whether the newly identified deviation or other unforeseen hazard should be incorporated into the HACCP plan. 417.3 (b) |
| O-55 | Upon completion of the hazard analysis and development of the HACCP plan, the establishment shall conduct activities designed to determine that the HACCP plan is functioning as intended. During this validation period, the establishment shall repeatedly test the adequacy of the CCP's, CLs, monitoring and recordkeeping procedures, and corrective actions set forth in the HACCP plan. 417.4 (a)(1) |
| O-56 | Ongoing verification activities include, but are not limited to:<br>(1) The calibration of process-monitoring instruments,<br>(2) Direct observations of monitoring activities and corrective actions, and<br>(3) The review of records generated and maintained IAW CCP monitoring and Corrective actions. 417.4(a)(2) |



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TABLE O-IV. Hazard Analysis and Critical Control Point Development and Application in OCONUS Meat Product Slaughter and Fabrication Plants - Continued.

| ITEM | REQUIREMENT |
|---|---|
| O-57 | (1) Each plant shall reassess the adequacy of the HACCP plan at least annually and whenever any changes occur that could affect the hazard analysis or alter the HACCP plan. Such changes may include, but are not limited to, changes in: raw materials or source of raw materials, product formulation, slaughter or processing methods or systems, production volume, personnel, packaging, finished product distribution systems, or the intended use or consumers of the finished product. This shall include plants that do not have a HACCP plan because a hazard analysis has revealed no food safety hazards that are reasonably likely to occur.<br>(2) The reassessment shall be performed by an individual who has successfully completed a course of instruction in the application of the seven HACCP principles to US meat or poultry processing, including a segment on the development of a HACCP plan for a specific product and on record review. 417.4(a)(3), 417.4(b) |
| O-58 | Each plant shall maintain the following records documenting the establishment's HACCP plan:<br>(1) The written hazard analysis, including all supporting documentation,<br>(2) The written HACCP plan, including decision-making documents associated with the selection and development of CCP's and critical limits, and documents supporting both the monitoring and verification procedures selected and the frequency of those procedures,<br>(3) Records documenting the monitoring of CCP's and their critical limits, including the recording of actual times, temperatures, or other quantifiable values, as prescribed in the establishment's HACCP plan,<br>(4) Calibration of process-monitoring instruments,<br>(5) Corrective actions, including all actions taken in response to a deviation,<br>(6) Verification procedures and results,<br>(7) Product codes, product name or identity, or slaughter production lot. 417.5 |
| O-59 | Individuals involved in the development or reassessment and modification of the HACCP plan, to include adaptation of generic models, shall have successfully completed a course of instruction in the application of the seven HACCP principles to US meat or poultry processing, including a segment on the development of a HACCP plan for a specific product and on record review. 417.7 |



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# COOK-CHILL, SOUS VIDE, AND OTHER REDUCED OXYGEN PACKAGING PRODUCT PROCESSING

## P.1 SCOPE

P.1.1 Scope. This appendix establishes the food safety and related requirements for Cook-Chill and Sous Vide processing establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## P.2 APPLICABLE DOCUMENTS

P.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

P.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

U.S. Public Health Service (USPHS) / Food and Drug Administration (FDA), Food Code.

(Copies of this are available online at: https://www.fda.gov/food/retail-food-protection/fda-food-code.)

## P.3 DEFINITIONS

P.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

## P.4 REQUIREMENTS

P.4.1 HACCP program. A mandatory HACCP program shall be required in all Sous Vide and Cook-Chill operations to maintain the highest levels of safety and production quality.

P.4.2 HACCP plans. HACCP plans are required for all Sous Vide and Cook-Chill retail production operations, and relevant records shall be maintained for all phases of production. Scheduled process control and HACCP plans will include: time-temperature controls, finished product laboratory analyses, and verification procedures to ensure a pathogen-free product.



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Manufacturers will re-evaluate and validate the process and shelf life of products annually. Any change in recipe, ingredients or significant pieces of equipment will also require re-evaluation and verification.

## P.5 CHECKLIST

P.5.1 Checklist. This Appendix is intended for guidance only and does not contain a checklist. Utilize the applicable portions of 21 CFR 117 and 121 for auditing these types of establishments. Findings will be scored IAW Appendix A and Appendix B of this document.



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# MANUFACTURED EGG PRODUCTS

## Q.1 SCOPE

Q.1.1 Scope. This appendix establishes the food safety and related requirements for manufactured egg product establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## Q.2 APPLICABLE DOCUMENTS

Q.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

Q.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 9, Part 590 | - | Inspection of Eggs and Egg Products (Egg Products Inspection Act) |
| Title 9, Part 416 | - | Sanitation |
| Title 9, Part 417 | - | Hazard Analysis and Critical Control Point (HACCP) Systems |

(Copies of these documents are available online at: https://www.govinfo.gov/.)

U.S. DEPARTMENT OF AGRICULTURE

FSIS Food Safety Guideline for Egg Products (September 9, 2020)

(Copies of these documents are available online at: https://www.fsis.usda.gov/guidelines/2020-0005)



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## Q.3 DEFINITIONS

Q.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

## Q.4 REQUIREMENTS

Q.4.1 Egg breaking operations. Eggs will be broken in a sanitary and satisfactory manner, and reexamined by a qualified person before being released into tank or churns. Small blood spots, small meat spots and shell particles shall be removed with a spoon or other approved instrument. Following breaking, egg contents shall be cooled unless pasteurized immediately. The acceptable temperature varies with the type of product and how long it will be held until pasteurization. These requirements are summarized in Figure Q-2, below.

Q.4.2 Egg packaging operations. Packaging rooms will have a filtered, positive air ventilation system unless product is packaged by an automatic closed packaging system.

Q.4.3 Pasteurization. Every particle of liquid eggs shall be heat treated to the extent necessary to ensure that the eggs are free of *Salmonella* and other pathogens. Egg pasteurization equipment is similar to milk pasteurization equipment. 9 CFR 590.570 prescribes that continuous flow pasteurization equipment include a holding tube, automatic flow diversion valve, thermal controls, and recording devices to ensure proper pasteurization. The establishment shall have a program for ensuring the proper operation of the pasteurization equipment. Proper pasteurization times and temperatures for different types of products are shown on Figure Q-1.

| Liquid Egg Products | Minimum Temperature Requirements (°F) | Minimum Holding Time Requirements (Minutes) |
|---|---|---|
| Whole egg | 140 | 3.5 |
| Whole egg blends (less than 2% added nonegg ingredients) | 142<br>140 | 3.5<br>6.2 |
| Fortified whole egg and blends (24–38% egg solids, 2–12 percent added nonegg ingredients) | 144<br>142 | 3.5<br>6.2 |
| Salt whole egg (with 2% or more salt added) | 146<br>144 | 3.5<br>6.2 |
| Sugar whole egg (2–12% sugar added) | 142<br>140 | 3.5<br>6.2 |
| Plain yolk | 142<br>140 | 3.5<br>6.2 |
| Sugar yolk (2% or more sugar added) | 146<br>144 | 3.5<br>6.2 |
| Salt yolk (2–12% salt added) | 146<br>144 | 3.5<br>6.2 |

FIGURE Q-1. Pasteurization requirements.

Q.4.4 Egg Cooling (after pasteurization). Once broken or pasteurized, eggs shall be cooled to an acceptable temperature within 2 hours. Adding previously processed egg to increase



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the rate of cooling is not an acceptable method. The acceptable temperature varies with the type of product. They are summarized in Figure Q-2 below.

| Product | Non-Salted Liquid Product to be held 8 hours or less | Non-Salted Liquid Product to be held more than 8 hours | Salted Liquid Product | Temperatures within 2 hours after Pasteurization | Temperatures within 3 hours after Stabilization |
|---|---|---|---|---|---|
| White (not to be stabilized) | 55°F or lower | 45°F or lower | _____ | 45°F or lower | _____ |
| Whites (to be stabilized) | 70°F or lower | 55°F or lower | _____ | 55°F or lower | [1] |
| All other product with less than 10% added salt | 45°F or lower | 40°F or lower | _____ | 45°F or lower if to be held 8 hours or less<br><br>40°F or lower if to be held more than 8 hours | 45°F or lower if to be held 8 hours or less<br><br>40°F or lower if to be held more than 8 hours |
| All other product with 10% or more added salt | _____ | _____ | 65°F or lower if to be held for 30 hours or less<br><br>45°F or lower if to be held more than 30 hours | 65°F or lower[2] | |

1 Stabilized liquid whites should be dried as soon as possible after removal of glucose. The storage of stabilized liquid whites should be limited to that necessary to provide a continuous operation.
2 The cooling process should be continued to assure that any salt product to be held more than 24 hours is cooled and maintained at 45°F or lower.

FIGURE Q-2. Egg cooling requirements after pasteurization.

Q.4.5 Egg freezing. Frozen eggs will be frozen solid or reduced to a temperature of 10° F or lower within 60 hours of pasteurization.

Q.4.6 *Salmonella* testing. *Salmonella* is a public health threat for manufactured egg products. Therefore, a comprehensive program for laboratory surveillance of product for the presence of *Salmonella* is required for liquid egg, frozen egg, and dried egg products. All lots shall be *Salmonella* negative. The sampling program for this surveillance should be on a lot-by-lot basis. A lot may be considered a day's production for each category of egg product for a continuous thermal processing operation. Although OCONUS areas may have different legal requirements, the USDA FSIS Food Safety Guideline for Egg Products offers the following guidance for reducing the frequency of *Salmonella* testing.

Q.4.6.1 No history of testing. Every lot shall be tested until 60 consecutive lots are found to be *Salmonella* negative.

Q.4.6.2 Level 1. One lot sampled for every two lots produced.

Q.4.6.3 Level 2. One lot sampled for every four lots produced.

Q.4.6.4 Level 3. One lot sampled for every eight lots produced.

Q.4.6.5 Sample frequency reduction.



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Q.4.6.5.1 From 100 percent to Level 1. To reduce the sampling frequency from 100 percent to Level 1, 60 consecutive lots within a product category shall be *Salmonella* negative.

Q.4.6.5.2 From Level 1 to Level 2. To reduce the sampling frequency from Level 1 to Level 2, 60 consecutive lots within a product category shall be *Salmonella* negative.

Q.4.6.5.3 From Level 2 to Level 3. To reduce the sampling frequency from Level 2 to Level 3, 60 consecutive lots within a product category shall be *Salmonella* negative.

Q.4.6.6 Sample frequency increase. If a *Salmonella* positive lot is found at any reduced sampling level, the plant shall immediately begin sampling the entire product category at 100%.

## Q.5 CHECKLISTS

Q.5.1 Checklists. Guidelines for auditing manufactured egg product establishments are contained in the following Tables Q-I to Q-III checklists.



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TABLE Q-I. Manufactured Egg Products Requirements as Specified in 9 CFR Part 590.

| ITEM | REQUIREMENT |
|---|---|
| Q-1 | Operations involving the processing, storing, and handling of egg products must be done in a sanitary manner. 590.504(a) |
| Q-2 | All loss and inedible eggs or inedible egg products must be placed in a container clearly labeled “inedible” and containing a sufficient amount of denaturant or decharacterant suspended in the product. 590.504(c) |
| Q-3 | Inedible egg products that are not denatured or decharacterized may be shipped for industrial use or animal food, provided that it is properly packaged, labeled, and segregated. 590.504(c) |
| Q-4 | Approved cleaning compound will be used in the wash water. 590.516(a) |
| Q-5 | Shell eggs shall be sufficiently dry at time of breaking to prevent contamination or adulteration of the liquid egg product from free moisture on the shell. 590.516(b) |
| Q-6 | Eggs must be handled in a manner that minimizes sweating prior to breaking or processing. 590.508 |
| Q-7 | Shell eggs having strong odors or eggs received in cases having strong odors will be candled and broken separately to determine their acceptability. 590.510(b) |
| Q-8 | Eggs, presented for breaking will be of edible interior quality and free of exterior dirt and foreign material except that:<br>a. Checks on eggs with a portion of the shell missing may be used so long as the membrane is not ruptured and the shell is free of adhering dirt and foreign material.<br>b. Eggs with clean shells which are damaged during candling or transfer may be used so long as the yolk is not broken and the contents of the egg are not exuding over the outside shell. Such eggs will be placed in leaker trays and broken immediately.<br>c. Eggs with meat or blood spots may be used if the spots are removed in an acceptable manner. 590.510(c) |
| Q-9 | Each egg used in processed egg products must be broken in a sanitary manner and examined to ensure that the contents are acceptable for human consumption. 590.522 |
| Q-10 | Freezing rooms, either on or off the premises, must be capable of solidly freezing, or reducing to a temperature of 10°F or lower, all liquid egg products. 590.534 |
| Q-11 | For pasteurized egg product, every particle of egg must receive a sufficient lethality treatment that renders the product ready to eat without additional preparation. 590.570 |
| Q-12 | To ensure adequate pasteurization, pasteurized egg products and heat-treated dried egg products will be analyzed for the presence of *Salmonella* at a frequency that ensures the system is working. 590.580 |
| Q-13 | If the establishment does not test every lot of liquid egg, frozen egg, or dried egg, then the establishment must have an acceptable reduced sampling plan. 590.580 and FSIS Food Safety Guideline for Egg Products, Appendix 1 |



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TABLE Q-II. Manufactured Egg Products Requirements as Specified in 9 CFR Part 416

| ITEM | REQUIREMENT |
|---|---|
| Q-14 | Each official establishment must be operated and maintained in a manner sufficient to prevent the creation of insanitary conditions and to ensure that product is not adulterated. 416.1 |
| Q-15 | A supply of potable, running water at a suitable temperature and under pressure as needed, must be provided in all areas when required (for cleaning rooms and equipment, utensils, and packaging facilities and for employee sanitary facilities, etc. 416.2(g)(1) |
| Q-16 | If an establishment uses municipal or a private well for its water supply, it must make available upon request, documentation certifying the potability of the water by a local health agency. Documentation must be renewed semi-annually for private well water supply. 416.2(g)(1) |
| Q-17 | Non-potable water may not be used in areas where edible product is handled or prepared or in any manner that would allow it to adulterate edible product or create insanitary conditions. 416.2(g)(6) |
| Q-18 | Equipment and utensils used for processing or otherwise handling edible product or ingredients must be of such material and construction to facilitate thorough cleaning and to ensure that their use will not cause the adulteration of product during processing, handling, or storage. 416.3(a) |
| Q-19 | Equipment and utensils must be maintained in sanitary condition so as not to adulterate the product. 416.3(a) |
| Q-20 | Equipment and utensils must not be constructed, located, or operated in a manner that prevents inspecting equipment or utensils to determine whether they are in sanitary condition. 416.3(b) |
| Q-21 | All food contact surfaces must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions and the adulteration of product. 416.4(a) |
| Q-22 | Non-food contact surfaces of facilities, equipment, and utensils used in the operation of the establishment must be cleaned and sanitized as frequently as necessary to prevent the creation of insanitary conditions and the adulteration of product. 416.4(a) |



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TABLE Q-III. Hazard Analysis and Critical Control Point Development and Application in OCONUS Manufactured Egg Products Plants.

| ITEM | REQUIREMENT |
|---|---|
| Q-23 | Every plant subject to this Appendix shall conduct, or have conducted for it, a hazard analysis to determine the food safety hazards reasonably likely to occur in the production process and identify the preventive measures the establishment can apply to control those hazards. 417.2(a)(1) |
| Q-24 | Hazard analysis shall include food safety hazards that can be reasonably expected to occur before, during, and after entry into the establishment. 417.2(a)(1) |
| Q-25 | A flow chart describing the steps of each process and product flow in the establishment shall be prepared, and the intended use or consumers of the finished product shall be identified. 417.2(a)(2) |
| Q-26 | At a minimum, food safety hazards that must be addressed (as they apply to the product or associated processes) include: natural toxins, microbiological contamination, chemical contamination, pesticides, drug residues, zoonotic diseases, decomposition, parasites, unapproved use of direct or indirect food or color additives, and physical hazards. 417.2(a)(3) |
| Q-27 | A single HACCP plan may encompass multiple products within a single processing category identified in this paragraph, if the food safety hazards, critical control points, critical limits, and procedures required to be identified and performed are essentially the same, provided that any required features of the plan that are unique to a specific product are clearly delineated in the plan and are observed in practice. 417.2(b)(2) |
| Q-28 | HACCP plans for thermally processed/commercially sterile products do not have to address food safety hazards associated with microbiological contamination if the raw materials are produced in a Directory listed or exempt canned food plant. 417.2(b)3 |
| Q-29 | The HACCP plan shall, at a minimum:<br>(1) List the food safety hazards identified which must be controlled.<br>(2) List the critical control points (CCPs) for each of the identified food safety hazards, including, (as appropriate), CCPs designed to control food safety hazards that could be introduced in the establishment and CCPs designed to control food safety hazards introduced outside the establishment.<br>(3) List the Critical Limits (CLs) that must be met at each of the CCPs to adequately control the identified hazard.<br>(4) List the procedures, and the frequency with which those procedures will be performed, that will be used monitor each of the CCPs to ensure compliance with the CLs.<br>(5) Include all corrective actions that have been developed to be followed in response to any deviation from a CL at a CCP.<br>(6) Provide for a recordkeeping system that documents the monitoring of the CCPs which contains the actual values and observations obtained during monitoring.<br>(7) List the verification procedures and the frequency with which those procedures will be performed. 417.2(c) |



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TABLE Q-III. Hazard Analysis and Critical Control Point Development and Application in OCONUS Manufactured Egg Products Plants – Continued.

| ITEM | REQUIREMENT |
|---|---|
| Q-30 | The HACCP plan shall be signed and dated:<br>(1) By the responsible establishment individual, signifying that the establishment accepts and will implement the HACCP plan,<br>(2) Upon initial acceptance, any modification, and annually. 417.2(d) |
| Q-31 | The written HACCP plan shall identify the corrective action to be followed in response to a deviation from a critical limit. The HACCP plan shall describe the corrective action to be taken, and assign responsibility for taking corrective action, to ensure:<br>(1) The cause of the deviation is identified and eliminated,<br>(2) The CCP will be under control after the corrective action is taken,<br>(3) Measures to prevent recurrence are established, and<br>(4) No product that is injurious to health or otherwise adulterated as a result of the deviation enters commerce. 417.3 (a) |
| Q-32 | If a deviation not covered by a specified corrective action occurs, or if another unforeseen hazard arises, the establishment shall:<br>(1) Segregate and hold the affected product, at least until the CCP is under control and measures to prevent recurrence are established,<br>(2) Perform a review to determine the acceptability of the affected product for distribution,<br>(3) Take action, when necessary, with respect to the affected product to ensure that no product that is injurious to health or otherwise adulterated, as a result of the deviation, enters commerce,<br>(4) Perform or obtain reassessment by an individual trained in accordance with § 417.7 of this part, to determine whether the newly identified deviation or other unforeseen hazard should be incorporated into the HACCP plan. 417.3 (b) |
| Q-33 | Upon completion of the hazard analysis and development of the HACCP plan, the establishment shall conduct activities designed to determine that the HACCP plan is functioning as intended. During this validation period, the establishment shall repeatedly test the adequacy of the CCP's, CLs, monitoring and recordkeeping procedures, and corrective actions set forth in the HACCP plan. 417.4 (a)(1) |
| Q-34 | Ongoing verification activities include, but are not limited to:<br>(1) The calibration of process-monitoring instruments,<br>(2) Direct observations of monitoring activities and corrective actions, and<br>(3) The review of records generated and maintained IAW CCP monitoring and Corrective actions. 417.4 (a)(2) |



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TABLE Q-III. Hazard Analysis and Critical Control Point Development and Application in OCONUS Manufactured Egg Products Plants – Continued.

| ITEM | REQUIREMENT |
|---|---|
| Q-35 | (1) Each plant shall reassess the adequacy of the HACCP plan at least annually and whenever any changes occur that could affect the hazard analysis or alter the HACCP plan. Such changes may include, but are not limited to, changes in: raw materials or source of raw materials, product formulation, slaughter or processing methods or systems, production volume, personnel, packaging, finished product distribution systems, or the intended use or consumers of the finished product. This shall include plants that do not have a HACCP plan because a hazard analysis has revealed no food safety hazards that are reasonably likely to occur.<br>(2) The reassessment shall be performed by an individual who has successfully completed a course of instruction in the application of the seven HACCP principles to meat or poultry processing, including a segment on the development of a HACCP plan for a specific product and on record review. 417.4(a)(3), 417.4(b) |
| Q-36 | Each plant shall maintain the following records documenting the establishment's HACCP plan:<br>(1) The written hazard analysis, including all supporting documentation,<br>(2) The written HACCP plan, including decision-making documents associated with the selection and development of CCP's and critical limits, and documents supporting both the monitoring and verification procedures selected and the frequency of those procedures,<br>(3) Records documenting the monitoring of CCP's and their critical limits, including the recording of actual times, temperatures, or other quantifiable values, as prescribed in the establishment's HACCP plan,<br>(4) Calibration of process-monitoring instruments,<br>(5) Corrective actions, including all actions taken in response to a deviation,<br>(6) Verification procedures and results,<br>(7) Product codes, product name or identity, or slaughter production lot. 417.5 |
| Q-37 | Individuals involved in the development or reassessment and modification of the HACCP plan, to include adaptation of generic models, shall have successfully completed a course of instruction in the application of the seven HACCP principles to meat or poultry processing, including a segment on the development of a HACCP plan for a specific product and on record review. 417.7 |



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# MUSHROOMS

## R.1 SCOPE

R.1.1 Scope. This appendix establishes the food safety and related requirements for mushroom growing and processing establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## R.2 APPLICABLE DOCUMENTS

R.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

R.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 112 | - | Standards for the Growing, Harvesting, Packing, and Holding of Produce for Human Consumption |
| Title 40, Part 141 | - | National Primary Drinking Water Regulations |

(Copies of these are available online at: https://www.govinfo.gov/.)

R.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

MUSHROOM GOOD AGRICULTURAL PRACTICES PROGRAM (MGAP), 2016 edition.

Industry-Wide Food Safety Standards for Fresh Mushroom Growing, Harvesting, and Shipping.

(Copies of this document are available online at: https://www.americanmushroom.org/clientuploads/Food%20_Safety/MGAP_All_Species_2016.pdf)



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## R.3 DEFINITIONS

R.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

## R.4 REQUIREMENTS

R.4.1 Agricultural water. Agricultural water is subject to the testing requirements of 112.44 unless water is received from a Public Water System. There is no requirement to test any agricultural water that is subject to the requirements of 112.44 under the following conditions.

a. Water is received from a Public Water System, as defined under the Safe Drinking Water Act (SDWA) regulations, 40 CFR part 141, that furnishes water that meets the microbial requirements under those regulations or under the regulations of the State (as defined in 40 CFR 141.2) approved to administer SDWA public water supply program, and the Water System results or certificates of compliance that demonstrate that the water meets that requirement.

b. Water is received from a public water supply that furnishes water that meets the microbial quality requirements described in 112.44(a), and public water system results or certificates of compliance that demonstrate that water meets that requirement.

c. Treat the water in accordance with the requirements of 112.43.

R.4.2 Sampling untreated ground water. If using untreated ground water for the purposes that are subject to the requirements of 112.44(a). Initial testing of the microbial quality of each source of the untreated ground water shall be conducted four times during the growing season or over a period of one year, using a minimum total of four samples collected to be representative of the intended use(s). If the four initial water sample results meet the microbial quality criteria of 112.44(a), testing may occur once annually thereafter, using a minimum of one sample collected to be representative of the intended use(s). Farms shall resume testing four times per growing season or year if any annual test fails to meet the microbial quality criteria in 112.4(a). Post harvest water shall have no detectable generic *Escherichia coli* (*E. coli*) in 100 milliliters (mL), and only applies to the produce covered by 21 CFR 112.1(b).

## R.5 CHECKLIST

R.5.1 Checklist. Guidelines for auditing mushroom establishments are contained in the following Table R-I checklist.



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TABLE R-I. Mushrooms Requirements as Specified in 21 CFR Part 112.

| ITEM | REQUIREMENT |
|---|---|
| R-1 | The label on all packages for wholesale or retail sale includes all items required by federal law to include the name and complete address of the farm and product code to enable trace back to the point at which the mushrooms were grown and the date they were processed. 112.6 |
| R-2 | All agricultural water must be safe and of adequate sanitary quality for its intended use. 112.41 |
| R-3 | At the beginning of a growing season, as appropriate, but at least once annually, water sources, water distribution systems, facilities, and equipment must be inspected to identify conditions that are reasonably likely to introduce known or foreseeable hazards into or onto covered produce or food contact surfaces to the extent that they are under establishment control. 112.42 |
| R-4 | Farm must prepare a written agricultural water assessment for water that you apply to the mushrooms using a direct application method during growing activities. The farm is exempt if the water meets the requirements for water from a public water system. 112.43 |
| R-5 | When using agricultural water for any one or more of the following purposes, farms must ensure there is no detectable generic *E. coli* in 100 milliliters (mL) of agricultural water, and the establishment must not use untreated surface water for any of the following purposes:<br>(1) Applied in any manner that directly contacts produce, including ice that directly contacts covered product during or after harvest activities.<br>(2) Used on food contact surfaces or used for ice that contacts food contact surfaces.<br>(3) Used for washing hands during and after harvest activities. 112.44(a) |
| R-6 | Test untreated ground water used as irrigation water or for harvesting, packing, or holding covered produce to determine if it meets the microbial quality criterion in paragraph (a) of this section, as follows:<br>(1) Initially test the microbial quality of each source of the untreated ground water at least four times during the growing season or over a period of 1 year, using a minimum total of four samples collected aseptically and representative of the intended use(s).<br>(2) If the four initial sample results meet the microbial quality criterion, the farm may test once annually thereafter, using a minimum of one sample collected aseptically and representative of the intended use(s).<br>(3) If any annual test fails to meet the microbial quality criterion, you must immediately discontinue the use(s) and resume testing at least four times per growing season or year until all results meet the microbial quality criterion.<br>(4) You may meet these testing requirements using test results or data collected by a third party, as provided in §112.47. 112.44(b) |
| R-7 | Any method used to treat agricultural water must be effective to make water safe and of adequate sanitary quality for its intended use(s) and meet the microbial quality criterion. 112.46(b) |



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TABLE R-I. Mushrooms requirements as specified in 21 CFR Part 112 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| R-8 | Areas where raw animal manure, unpasteurized substrate which contains raw manure or other potentially hazardous materials are processed, stored, or transported are clearly separated from areas where mushrooms are grown, harvested, and packed. 112.52 |
| R-9 | Raw materials and unpasteurized substrate are in separate areas from where mushrooms are grown, harvested, packed, and stored. 112.52 |
| R-10 | The following treatment processes are acceptable for a biological soil amendment of animal origin:<br>a. A scientifically valid controlled physical process (e.g., thermal), chemical process (e.g., high alkaline pH), biological process (e.g., composting), or a combination of that has been validated to satisfy the microbial standard in 112.55(a) for *Listeria monocytogenes*, *Salmonella* species, and *E. coli* O157:H7; or<br>b. A scientifically valid controlled physical, chemical, or biological process, or combination thereof, that has been validated to satisfy the microbial standard in 112.55(b) for *Salmonella* species and fecal coliforms. 112.54 |
| R-11 | The following microbial standards apply to treatment processes in 112.54, for *L. monocytogenes*, *Salmonella* species, and *E. coli* O157:H7:<br>a. *L. monocytogenes* not detected using a method that can detect one colony forming unit (CFU) per 5 gram (or milliliter if liquid is sampled) analytical portion.<br>b. *Salmonella* species not detected using a method that can detect three most probable numbers (MPN) per 4 grams (or milliliter if liquid is sampled) of total solids.<br>c. *E. coli* O157:H7 not detected using a method that can detect 0.3 MPN per 1 gram (or milliliter if liquid is sampled) analytical portion.<br>d. *Salmonella* species are not detected using a method that can detect three MPN *Salmonella* species per 4 grams (or milliliter if liquid is sampled) of total solids; and less than 1,000 MPN fecal coliforms per gram (or milliliter if liquid is sampled) of total solids. 112.55 |
| R-12 | Mushrooms are packaged in a manner that prevents the formation of *Clostridium botulinum* toxin. 112.115 |
| R-13 | Packaging materials must be designed for single use and unlikely to support growth or transfer of bacteria. 112.116 |
| R-14 | Equipment for moving, mixing, or otherwise handling unpasteurized substrate is not used for handling pasteurized substrate, casing materials, or mushrooms and equipment is cleaned as needed to protect against contamination of the premises. 112.123 |
| R-15 | Traffic patterns for employees and equipment are established to avoid contamination of pasteurized substrate, casing materials, and mushrooms with raw manure and unpasteurized substrate. Where workers and equipment enter and exit the building between phase one and other plant operations, methods of control exist for employees' feet, hands and portable (personal) equipment (knives). 112.126(a)(1)(ii) |
| R-20 | Water that contacts pasteurized substrate or casing soil shall meet U.S. EPA microbial standards for drinking water. Standard 2.1, MGAP |



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# VEGETABLE SPROUTS

## S.1 SCOPE

S.1.1 Scope. This appendix establishes the food safety and related requirements for vegetable sprout growing and processing establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## S.2 APPLICABLE DOCUMENTS

S.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

S.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | | |
|---|---|---|
| Title 21, Part 112 | - | Sprouts |
| Title 21, Part 179 | - | Irradiation in the Production, Processing, and Handling of Food |

(Copies of these are available online at: https://www.govinfo.gov/.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES, FOOD AND DRUG ADMINISTRATION

Reducing Microbial Food Safety Hazards in the Production of Seed for Sprouting: Guidance to Industry

(Available on-line at: https://www.fda.gov/media/127972/download.)

## S.3 DEFINITIONS

S.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.



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## S.4 REQUIREMENTS

S.4.1 Irradiation. For farms that are receiving irradiated seed, the seeds cannot be used for food if they have been subjected to greater than 8.0 KGy of ionizing radiation (Ref: 21 CFR, Part 179.26). Farms should have documentation from the seed supplier indicating that the seed was treated with irradiation and the dose. additionally, the bags of SEED (not the retail or wholesale packages of sprouted seeds) shall be marked with the “Radura” symbol.

## S.5 CHECKLIST

S.5.1 Checklist. Guidelines for auditing vegetable sprouts establishments are contained in the following Table S-I checklist.



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TABLE S-I. Sprouts Requirements as Specified in 21 CFR Part 112.

| ITEM | REQUIREMENT |
|---|---|
| S-1 | Farms must treat seeds or beans that will be used to grow sprouts using a scientifically valid method to reduce microorganisms of public health significance, or obtain a Certificate of Conformance from the grower, distributor, or supplier indicating treatment was completed. 112.142(e) |
| S-2 | Water, to include irrigation water that directly contact sprouts or surfaces that come in contact with sprouts, is potable and meets federal drinking water standards. 112.41 |
| S-3 | Sprouts must be grown, harvested, packed, and held in a fully enclosed building. 112.143(a) |
| S-4 | Food contact surfaces used in growing, harvesting, packing, or holding sprouts must be cleaned and sanitized before contact with sprouts, seeds, or beans used to grow sprouts. 112.143(b) |
| S-5 | Testing must be conducted during growing, harvesting, packing, and holding sprouts, as specified in 112.144. 112.143(c) |
| S-6 | Establishment must establish and implement a written environmental monitoring plan as specified in 112.145. 112.143(d) |
| S-7 | Establishment must take action if *Listeria* species or *L. monocytogenes* is detected in the growing, harvesting, packing, or holding environment specified in 112.146. 112.143(e) |
| S-8 | A written sampling plan to test spent sprout irrigation water or sprouts for pathogens specified in 112.147. 112.143(f) |
| S-9 | Establishment must take action if samples of spent sprout irrigation water or sprouts test positive for a pathogen as specified in 112.148. 112.143(g) |
| S-10 | Establishment must test the growing, harvesting, packing, and holding environment for *Listeria* species or *L. monocytogenes* in accordance with the requirements of 112.145. 112.144(a) |
| S-11 | Farms must either:<br>(1) Test spent sprout irrigation water from each production batch of sprouts for *E. coli* O157:H7, *Salmonella* species, and any pathogen meeting the criteria in paragraph (c) of this section, in accordance with the requirements of 112.147; or<br>(2) If testing spent sprout irrigation water is not practical (for example, spoil grown sprouts with roots or for hydroponically grown sprouts that use very little water), test each production batch of sprouts at the in-process stage (while sprouts are still growing) for *E. coli* O157:H7, *Salmonella* species, and any pathogen meeting the criteria in paragraph (c). 112.144(b) |



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TABLE S-I. Sprouts Requirements as Specified in 21 CFR Part 112 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| S-12 | In addition to *E. coli* O157:H7 and *Salmonella* species, farms must tests as provided in paragraph (b) of this section for additional pathogens when the following conditions are met:<br>(1) Testing for pathogen is reasonably necessary to minimize the risk of serious adverse health consequences or death from use of, exposure to, sprouts; and<br>(2) A scientifically valid test method for the pathogen is available to detect the pathogen in spent sprout irrigation water or sprouts. 112.144(c) |
| S-13 | A written environmental monitoring plan must be implemented to test for *Listeria* species or *L. monocytogenes*. The plan will identify what test is being conducted, how often (not less than monthly), at what point in the process the samples are collected, and the sample collection sites. Sampling numbers and locations must be sufficient to determine effectiveness of both food contact and non-food-contact surfaces. Testing must be in accordance with 21 CFR 112 Subpart N, Sec. 112.152. 112.145 |
| S-14 | At a minimum, establishments must take the following actions if *Listeria* species or *L. monocytogenes* are detected in the growing, harvesting, packing, or holding environment:<br>(a) Conduct additional testing of surfaces and areas surrounding the area where *Listeria* species or *L. monocytogenes* was detected to evaluate the extent of the problem;<br>(b) Clean and sanitize the affected surfaces and surrounding areas;<br>(c) Conduct additional sampling and testing to determine whether the *Listeria* species or *L. monocytogenes* has been eliminated;<br>(d) Conduct finished product testing;<br>(e) Perform any other actions necessary to prevent recurrence of the contamination; and<br>(f) Take appropriate action to prevent any food that is adulterated from entering into commerce. 112.146 |
| S-15 | A written sampling plan must be established for spent sprout irrigation water (or in-process sprouts) for *E. coli* O157:H7, *Salmonella* species, and any additional pathogen that may require monitoring to minimize risk for which a scientifically valid test method exists. Sampling must be conducted by batch of sprouts. Sampling plan must identify the number and location of samples to be collected ensuring a good representation of the batch is tested for contamination. The sampling plan must include a corrective action for positive results. 112.147(a) |
| S-16 | A production batch of sprouts must not enter commerce unless results from spent sprout irrigation water (or in-process sprouts) are negative for the required test. 112.147(b) |
| S-17 | Action must be taken to prevent a production batch of sprouts from entering commerce if results of spent sprout irrigation water (or in-process sprouts) are positive. Immediate and surrounding areas must be cleaned and sanitized to prevent further contamination and reoccurrence. 112.148 |



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# LOW-ACID CANNED FOODS (LACF)

## T.1 SCOPE

T.1.1 Scope. This appendix establishes the food safety and related requirements for low-acid canned food establishments. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## T.2 APPLICABLE DOCUMENTS

T.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

T.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 21, Part 113 - Thermally Processed Low-Acid Foods Packaged In Hermetically Sealed Containers

(Copy of this document is available online at: https://www.govinfo.gov/.)

NATIONAL ADVISORY COMMITTEE ON MICROBIOLOGICAL CRITERIA FOR FOODS

"Microbiological Safety Evaluations and Recommendations on Sprouted Seeds," May 28, 1999.

(Available on-line at: www.fda.gov/media/127972/download.)

## T.3 DEFINITIONS

T.3.1 Definitions. The definitions in section 3 of this standard apply to this appendix.



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## T.4 CHECKLIST

T.4.1 Checklist. Guidelines for auditing low-acid canned food establishments are contained in the following table T-I checklist.

TABLE T-I. Low Acid Canned Foods Requirements as Specified in 21 CFR Part 113 and 173.

| ITEM | REQUIREMENT |
|---|---|
| T-1 | Trained personnel supervise the operators of processing systems, retorts, systems, and container and closure inspections. 113.10 |
| T-2 | Scheduled processes for low-acid foods have been established by qualified persons having expert knowledge of thermal process requirements and having adequate facilities for making such determinations. Critical factors (e.g., minimum headspace; consistency; maximum fill or drained weight; Aw; pH; initial product temperature; type of filling; specific can size/style; etc.) will be specified in the scheduled process. 113.83 |
| T-3 | Processing and production information is entered on forms/mandatory records specific to the process and retort used and includes: company name, date, product, production code, retort number, container size, number of containers produced per coding interval, initial temp, time steam on, time vent closed, vent temp, time temp up, time steam off, actual processing time, mercury-in-glass (MIG) and recorder chart temperatures and other appropriate processing/critical factor data. Records are reviewed by qualified personnel, reviews are documented when required; and records maintained as required. 113.100 |
| T-4 | Processor ensures raw materials and ingredients are suitable for use before using raw materials and ingredients susceptible to microbiological contamination. Process steps are controlled to ensure critical factors are met, as noted on the scheduled process. 113.81 |
| T-5 | Critical factors specified in the scheduled process should be measured and recorded on the processing record at intervals not exceeding 15 min. 113.40 |
| T-6 | Whenever any process is less than the scheduled process or when critical factors are out of control, that low-acid food is fully reprocessed or food is evaluated and substantiated by a qualified scientific authority as free of any potential hazard to public health prior to release. 113.89 |
| T-7 | Container size and type are the same as those identified on the scheduled process for each item being produced. 113.83 |
| T-8 | When pH is the basis for a scheduled process, processor ensures equilibrium pH of the finished product. 113.81 |
| T-9 | Containers must meet container specifications and be free of gross closure defects. 113.60 |
| T-10 | Visual seam inspections are conducted by a qualified individual as often as necessary, but not less than every 30 minutes, and after each startup, prolonged break, and each seamer jam. Can seam teardowns are conducted by a qualified individual at least every 4 hours. 113.60 |



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Table T-I. Low Acid Canned Foods Requirements as Specified in 21 CFR Part 113 and 173 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| T-11 | Required can seam measurements during can seam teardown are performed using the Micrometer Method (in 3 places about 120ºF apart [Required: cover hook, body hook, width, tightness, and thickness]) or the Seam Scope/Seam Projector Method (in 2 places about 180ºF apart [Required: body hook, overlap, tightness, thickness]). 113.60 |
| T-12 | Specific retort type is properly fitted with indicating thermometers, temperature recorders, pressure gauges, steam controls, spreaders and bleeders. Steam inlets and crate supports are of proper design and location. 113.40 |
| T-13 | Each retort is equipped with at least one MIG thermometer properly installed and tested for accuracy against a known accurate standard at installation and at least once per year thereafter. Calibration records are annotated and thermometers are identified. MIG divisions are readable to 1ºF and temperature range does not exceed 17ºF per inch of graduated scale. 113.40 |
| T-14 | Each retort is equipped with accurate temperature recording device with appropriate degree graduations. The temperature recording device will be adjusted to agree with the MIG thermometer, but in no event higher the MIG thermometer. A means of preventing unauthorized adjustments will be provided (e.g., seals, etc.). 113.40 |
| T-15 | Each retort is equipped with a pressure gauge (graduated in divisions of 2 pounds or less) and an automatic steam controller (to properly control steam temperature). 113.40 |
| T-16 | Retort steam inlets are of proper design and enter the retort at a location opposite of the retort vent. 113.40 |
| T-17 | Crate supports, steam spreaders, bleeders, stacking equipment, air valves, water valves, and vents are of proper design, installed, and maintained as required by the specific type of retort being used. 113.40 |
| T-18 | Dividers between the layers of containers should be perforated approximately the equivalent of 1-inch holes on 2-inch centers (or as otherwise specified in the scheduled process) to promote steam flow. 113.40 |
| T-19 | Retort operating and venting procedures for each product and container size being packed are posted near the processing equipment, or are readily available to the retort operator. 113.87 |
| T-20 | Product traffic control in the retort room and processing area is controlled to prevent unprocessed LACF from circumventing the thermal process, as well as mixing of processed and unprocessed containers. 113.87 |
| T-21 | Trucks, crates, etc. of retorted food product are plainly and conspicuously marked with a heat sensitive indicator, or by other effective means that will indicate visually, to thermal processing personnel, the units that have been retorted. 113.87 |
| T-22 | Initial temperature of the contents of the containers to be processed in the retort is determined and recorded. Product initial temperature (e.g., the temperature of the potentially coldest container in the retort about to begin a process) is not lower than the minimum initial temperature specified in the scheduled process. 113.87 |



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Table T-I. Low acid canned foods requirements as specified in 21 CFR Part 113 and 173 - Continued.

| ITEM | REQUIREMENT |
|---|---|
| T-23 | Timing device used to record processing and venting times is accurate to the extent needed. Pocket or wrist watches are not considered satisfactory for timing purposes. Clock times on recording-temperature charts should reasonable correspond to time of day entered on other processing records. 113.87 |
| T-24 | Water used for cooling must be potable. Container cooling water will be chlorinated or otherwise sanitized, as necessary, for cooling canals and for recirculating water supplies. There should be a measurable residual of the sanitizer employed at the water discharge point of the container cooler. 113.60 |
| T-25 | When the steam comes into contact with the LACF, through direct injection of the steam into the food; during the exhausting of containers in a steam exhaust box; through injection of steam into the headspace of containers to form a vacuum, or through any other means, the boiler additives must be approved for use as a food additive and labeled for that use. 173.310 |



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# FRESH FRUIT AND VEGETABLE SUPPLIERS (UNPROCESSED) IN OCONUS AREAS

## U.1 SCOPE

U.1.1 Scope. This appendix establishes the food safety and related requirements for fresh fruits and vegetable suppliers (unprocessed) in OCONUS areas. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

## U.2 APPLICABLE DOCUMENTS

U.2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in appendices A and B of this standard, whether or not they are listed.

U.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 21, Part 112 - Standards for the Growing, Harvesting, Packing, and Holding of Produce for Human Consumption

(Copies of these are available online at: https://www.govinfo.gov/.)

DEPARTMENTS OF THE ARMY, NAVY, AND AIR FORCE

TB MED 530/NAVMED P-5010/AFMAN 48-147_IP, Tri-Service Food Code

U.S. FOOD AND DRUG ADMINISTRATION (FDA)

The Guide to Minimize Microbial Food Safety Hazards: The Guide at a Glance

(Available online at https://www.fda.gov/Food/GuidanceRegulation/GuidanceDocumentsRegulatoryInformation/ProducePlantProducts/ucm187676.htm.)

U.S. ENVIRONMENTAL PROTECTION AGENCY

Selected EPA-registered Disinfectants



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(Available online at https://www.epa.gov/pesticide-registration/selected-epa-registered-disinfectants.)

Setting Tolerances for Pesticide Residues in Foods

(Available online at https://www.epa.gov/pesticide-tolerances/setting-tolerances-pesticide-residues-foods.)

U.3 DEFINITIONS

U.3.1 Definitions. The definitions in section 3 of this standard as well as those contained in all reference and associated documents apply to this appendix.

U.4 CHECKLIST

U.4.1 Checklist. This Appendix is intended for guidance only and does not contain a checklist. Utilize the applicable portions of 21 CFR 117 and 121 for auditing these types of establishments. Findings will be scored IAW Appendix A and Appendix B of this document.



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CONCLUDING MATERIAL

Custodians:
DHA – MB1
Navy – SA
Air Force – 03

Preparing Activity:
DHA – MB1
(Project No. 89GP-2025-003)

Review activities:
Navy - MS, MC
DLA - SS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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